SGA 6 – English Working Draft

Intersection theory and the Riemann–Roch theorem, through Expose V section 2.3

This reader assembles the currently delivered SGA 6 English working translation into a single checking file. It combines the volume front matter, Expose 0, the Riemann–Roch appendix, Expose I through section 8, and the continuation through Expose V section 2.3. The original French reference PDF and the source TeX/render-check packages are included separately in this record.

SGA 6 – Working Draft, Opening Through Expose I

Intersection Theory and the Riemann–Roch Theorem

Working English LaTeX translation: front matter, Expose 0, the RRR appendix, and Expose I through Theorem 1.5.1

Contents

Preface

The present edition of *Seminaire de Geometrie Algebrique 6* reproduces, up to minor changes, the first edition distributed by the IHES in the form of mimeographed exposes. The corrections concern mainly typographical errors, except in Exposes I and III, where L. Illusie made slight changes to a few statements which were incorrect in the original version. Footnotes have also been added, especially in Expose XIV by A. Grothendieck, in order to point out recent progress on questions that were still open at the time of the seminar.

It should also be noted that Expose XI, by D. Ferrand, on the theory of the determinant for perfect complexes, was not written up and will therefore not be published. To avoid errors in references, the numbering of the following exposes has nevertheless been left unchanged.

June 1971

Introduction

The purpose of this seminar is to develop a global theory of intersections and a Riemann–Roch formula for arbitrary schemes. A description of the programme of the seminar is given in Expose 0. Already by the time of the well-known report of Borel–Serre in 1958, it was clear in principle how to prove a Riemann–Roch formula for general regular schemes, at least for a projective morphism. The extension to the general case is less evident. The programme in which it is placed – and which is partially carried out in this seminar – goes back to 1960. As in the theory of duality for coherent sheaves, one essential tool for formulating a satisfactory theory is Verdier's theory of derived categories, and some familiarity with that theory is indispensable for understanding the seminar.

Apart from this theory, the seminar requires little beyond a general knowledge of the foundations of algebraic geometry as presented in EGA I, II, and III. Occasional references are made to EGA IV, mainly for facts concerning flat and smooth morphisms, and most of these also occur in the early exposes of SGA 1. In order to state some results with the generality required for applications, we sometimes use ringed sites and ringed topoi, for which we refer to SGA 4, Exposes I–IV. A reader who is not using the language of sites and topoi may replace them, at first reading, by ordinary topological spaces and their open sets; nevertheless the language of topoi is strongly recommended, since it gives an extremely convenient principle of unification.

The fundamental notion for the theory presented here is that of a perfect complex of modules, together with several variants developed in Exposes I, II, and III. These notions are also important in other parts of algebraic geometry, notably for formulas of Lefschetz–Verdier type in cohomology with discrete coefficients (SGA 5) or coherent coefficients. They should also play a role outside algebraic geometry, for example in the formulation of a complex analytic variant of the Riemann–Roch theorem treated here, or in suitable variants of the Atiyah–Singer index theorem. Some indications in this direction are given in Expose II. At the time of the seminar, however, there was still no statement, even heuristic, that would simultaneously include both the algebraic Riemann–Roch theorem and the index theorem. A new idea seems to be missing, just as a new idea is missing in algebraic geometry for proving Riemann–Roch beyond projective hypotheses; see Expose XIV, no. 2.

The local theory of intersections, as treated in Serre's *Algebre locale. Multiplicites*, is not used in the seminar. We mention only that Serre's course had an evident influence on the genesis of the theory in 1957. Nor do we use the theory of the Chow ring developed in the Chevalley Seminar of 1958. That theory is essentially tied to nonsingularity hypotheses, whereas the aim here is to develop an intersection theory on general schemes, and even on general locally ringed topoi. Expose XIV, nos. 4 and 8, comments on the relation between the present theory and Chow theory, and raises the question of a generalization of the latter. Thus the seminar presents a coherent and self-contained theory of intersections, but it is not exhaustive with respect to the different viewpoints that are useful, or even indispensable, in intersection theory. It should therefore be complemented by Serre's and Chevalley's treatments.

An appendix to Expose 0 reproduces Grothendieck's 1957 report *Classes de faisceaux et theoreme de Riemann–Roch*, which served as the basis for the Borel–Serre seminar at Princeton in the same year and for their article. That report sketches two proofs of the Riemann–Roch theorem for nonsingular quasi-projective varieties. The first proof, for the time being valid only in characteristic zero, gives a slightly sharper result in the case of an immersion and had not yet appeared in the literature. The report can be read independently of the rest of the seminar and can serve, like Expose 0, as an introduction for readers not yet familiar with Borel–Serre. The proof just mentioned applies essentially unchanged to projective morphisms of complex analytic spaces and should apply in arbitrary characteristic once the problem of exterior-power operations in the derived category is solved. The lack of a study of that question is probably the most serious gap in the seminar from the point of view adopted here.

Two names appearing on the cover do not appear in the table of contents. J.-P. Jouanolou took an active part in the technical elaboration of the first part of the seminar, though he was prevented from participating in the oral lectures. In particular, he is responsible for the linear-independence assertion in the important statement VI 1.1 on the structure of the ring $K^\bullet$ of classes of locally free modules on a projective bundle. J.-P. Serre delivered two oral talks, logically independent of the rest of the seminar, and preferred to publish them separately under the title *Groupes de Grothendieck des schemas en groupes reductifs deployes*.

As in the other parts of the Seminaire de Geometrie Algebrique du Bois Marie, the sigla SGA 1 to SGA 6 refer to the different parts of the seminar. The following Roman numeral indicates the expose number, and the following Arabic numerals refer to the internal numbering of that expose. For references internal to the present seminar the siglum SGA 6 is omitted, so that a reference such as I 4.9 means statement 4.9 of Expose I. The siglum EGA refers to the *Elements de Geometrie Algebrique* of J. Dieudonne and A. Grothendieck.

Table of Contents of SGA 6

1 Expose 0 – Outline of a Programme for a Theory of Intersections on General Schemes

by A. Grothendieck

The present expose is introductory. Its reading is not logically necessary for the study of the seminar. It is addressed especially to readers who know the provisional form of the Riemann–Roch theorem as presented in the report of Borel–Serre, or in Grothendieck's earlier report cited below as RRR and reproduced as the appendix to this expose.

1.1 Reminder on the Riemann–Roch formula

Let $f : X \to Y$ be a proper morphism of smooth quasi-projective schemes over a field k, and let F be a coherent sheaf on X. Denote by $\mathrm{Cl}(F)$ the class of F in the group $K(X)$ of classes of coherent sheaves on X. The Riemann–Roch formula has the form

$$\mathrm{Todd}(T_Y)\ \mathrm{ch}_Y\big(f_!(\mathrm{Cl}(F))\big) = f_*\big(\mathrm{Todd}(T_X)\ \mathrm{ch}_X(F)\big). \tag{1}$$

It holds in $A(Y) \otimes_{\mathbb{Z}} \mathbb{Q}$, where $A(Y)$ is the Chow ring of Y. The f_* on the right is the direct image of cycles, tensored with $\mathbb{Q}$,

$$f_* : A(X) \longrightarrow A(Y),$$

while the term $f_!(\mathrm{Cl}(F))$ on the left is the Euler–Poincare characteristic of F relative to f:

$$f_!(\mathrm{Cl}(F)) = \sum_i (-1)^i\,\mathrm{Cl}(R^i f_*(F)).$$

The symbols ch_X and ch_Y denote the Chern characters, and T_X and T_Y the tangent bundles. Since $\mathrm{Todd}(-)$ and $\mathrm{ch}(-)$ are universal polynomials with rational coefficients in the Chern classes, and the constant term of $\mathrm{Todd}(-)$ is 1, the Todd class is invertible. After multiplying by $\mathrm{Todd}(T_Y)^{-1}$, one obtains the more useful form

$$\mathrm{ch}_Y\big(f_!(\mathrm{Cl}(F))\big) = f_*\big(\mathrm{Todd}(T_f)\ \mathrm{ch}_X(F)\big), \tag{2}$$

where

$$T_f = T_X - f^*T_Y \in K(X). \tag{3}$$

Thus T_f plays the role of the virtual relative tangent bundle of X over Y. If f is smooth, then $T_f = T_{X/Y}$. If $f : X \to Y$ is an immersion, then

$$T_f = -\check{N}_{X/Y},$$

where $\check{N}_{X/Y}$ is the dual of the normal sheaf.

The main purpose of the seminar is to generalize (2) in two directions at once:

(a) remove the hypothesis that a base field k exists;

(b) replace regularity assumptions on X and Y by a local regularity condition on f.

Along the way one also studies a third problem: removing quasi-projectivity hypotheses, which in the absence of a base field are expressed by the existence of ample invertible modules on X and Y.

1.2 Removing the base field

We first examine (a), still retaining regularity and the existence of ample invertible modules on X and on Y. In this situation the definition of $K(X)$, $K(Y)$, and $f_! : K(X) \to K(Y)$ causes no new difficulty, because X and Y are regular. A natural way to give meaning to (2) would be to define Chow rings $A(X)$ and $A(Y)$, a direct image

$$f_* : A(X) \to A(Y),$$

Chern classes

$$c_i : K(X) \to A(X),$$

and the virtual relative tangent bundle $T_f \in K(X)$.

For the last of these a definition is essentially forced. The morphism f, using ample invertible modules on X and Y, can be factored as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y, \tag{4}$$

where i is a closed immersion and f' is smooth; one may take, for example, $X' = \mathbb{P}^r_Y$ for r sufficiently large. Set

$$T_f = i^* T_{X'/Y} - \check{N}_{X/X'}. \tag{5}$$

Here $\Omega^1_{X'/Y}$ is the locally free module of relative differentials, and $N_{X/X'}$ is the conormal module J/J^2 for the ideal J defining X in X'. Since X and X' are regular, this conormal module is locally free. Expose VIII proves that the element T_f defined by (5) is independent of the chosen factorization.

For Chow rings and Chern classes under these hypotheses, a theory analogous to the usual theory for smooth quasi-projective varieties should exist, but this work is not undertaken in the seminar. Instead one introduces another ring which plays the same role as the Chow ring and whose rationalization is isomorphic to $A(X) \otimes \mathbb{Q}$ when X is smooth quasi-projective over a field. This ring is the graded ring associated with a suitable filtration of $K(X)$.

One natural filtration, considered in RRR, is the topological filtration by codimension of support: $\operatorname{Fil}^i K(X)$ is generated by the classes $\operatorname{Cl}(F)$ of coherent sheaves whose support has codimension at least i. One must first know that this filtration is compatible with multiplication. In RRR this was done for X smooth over a field by using Chow's moving lemma. Assuming this, one gets a graded ring $\operatorname{Gr}^*_{\mathrm{top}}(X)$. RRR defines a surjective ring homomorphism

$$\phi : A(X) \to \operatorname{Gr}^*_{\mathrm{top}}(X),$$

sending the class of a prime cycle Z to $\operatorname{Cl}_X(\mathcal{O}_Z)$; Expose XIV 4.2 shows that the kernel is a torsion group. Moreover, RRR explains how the images in $\operatorname{Gr}^*_{\mathrm{top}}(X)$ of the Chern classes $c_i(F)$ can be computed directly from $\operatorname{Cl}(F) \in K(X)$ and exterior-power operations in $K(X)$.

Thus for a noetherian regular scheme X with an ample invertible module, the ring $\operatorname{Gr}^*_{\mathrm{top}}(X) \otimes \mathbb{Q}$ is a convenient substitute for the hypothetical Chow ring. Its advantage is that many computations in RRR naturally take place in it. To fully justify this viewpoint, however, one would have to settle the compatibility of the topological filtration with multiplication, a problem closely tied to Chow's moving lemma – precisely the technical ingredient of Chow theory one hoped to avoid.

1.3 The λ-filtration

To bypass this problem, one equips $K(X)$ with a second filtration, finer than the topological filtration and describable in purely algebraic terms from the structure of $K(X)$ as an augmented λ-ring. The definition is suggested by the formula in RRR expressing the Chern classes of $x \in K(X)$ in $\mathrm{Gr}^*_{\mathrm{top}}(X) \otimes \mathbb{Q}$:

$$c_i(x) = \gamma^i(x - \varepsilon(x)) \pmod{\mathrm{Fil}^{i+1}_{\mathrm{top}} K(X)}. \tag{6}$$

Here γ^i is a certain integral combination of the operations λ^j, $j \le i$, and ε is the augmentation. One proves that $\gamma^i(x - \varepsilon(x))$ has topological filtration at least i, so that (6) makes sense. The λ-filtration is then defined by ideals $\mathrm{Fil}^i_\lambda K(X)$ generated by finite sums of expressions

$$\gamma^{i_1}(x_1 - \varepsilon(x_1)) \cdots \gamma^{i_k}(x_k - \varepsilon(x_k)), \qquad i_1 + \cdots + i_k \ge i.$$

The associated graded ring is denoted $\mathrm{Gr}^\bullet(X)$. It is this ring which replaces the Chow ring in the seminar. In the present regular ample situation the canonical map

$$\mathrm{Gr}^\bullet(X) \to \mathrm{Gr}^*_{\mathrm{top}}(X)$$

is an isomorphism modulo torsion. Thus, after tensoring with $\mathbb{Q}$, $\mathrm{Gr}^\bullet(X)$ plays the same role as $A(X) \otimes \mathbb{Q}$. The definition is also designed so that (6), with the λ-filtration in place of the topological filtration, gives a theory of Chern classes with values in $\mathrm{Gr}^\bullet(X)$. Expose V proves, using Expose VI, that $K(X)$ is a special λ-ring in the sense of RRR, and hence that these Chern classes have all the usual formal properties.

A substantial advantage of $\mathrm{Gr}^\bullet(X)$ over $A(X)$ or $\mathrm{Gr}^*_{\mathrm{top}}(X)$ is that it is defined for every scheme X, and more generally for every locally ringed topos. One takes $K(X)$ to be the group of classes of locally free modules on X; it remains an augmented λ-ring and has the corresponding filtration. The price of this generality is that, unless one tensors with $\mathbb{Q}$, torsion phenomena make the properties of $\mathrm{Gr}^\bullet(X)$ less satisfactory than those of its competitors. For instance, for a proper morphism $f : X \to Y$ of noetherian regular schemes with ample invertible modules, it is not generally true integrally that

$$f_* \operatorname{Fil}^i K(X) \subset \operatorname{Fil}^{i-d} K(Y),$$

where d is the relative dimension. Expose VII proves the corresponding assertion after tensoring with $\mathbb{Q}$, giving a degree $-d$ homomorphism

$$f_* : \operatorname{Gr}^\bullet(X) \otimes \mathbb{Q} \to \operatorname{Gr}^\bullet(Y) \otimes \mathbb{Q}$$

which plays the role of the direct image of cycles.

We therefore have all the structural elements needed to make sense of (2) without assuming a base field, when $f : X \to Y$ is a proper morphism of noetherian regular schemes admitting ample invertible modules.

1.4 General schemes

We next indicate how to give meaning to (2) after abandoning regularity of X and Y, replacing it by a local condition on f.

The first difficulty is that the observation underlying RRR – namely, that the group $K(X)$ may be defined either by coherent sheaves or by locally free sheaves – uses regularity in an essential way. For a general noetherian scheme one must distinguish two groups. We write

$$K_\bullet(X) \quad \text{and} \quad K^\bullet(X),$$

the first defined from coherent sheaves, covariant for proper morphisms, and the second from locally free sheaves, contravariant for arbitrary morphisms. The group $K^\bullet(X)$ is naturally a ring, and even an augmented λ-ring; $K_\bullet(X)$ is generally only a module over it. The map

$$x \longmapsto x \cdot \operatorname{Cl}(\mathcal{O}_X)$$

from $K^\bullet(X)$ to $K_\bullet(X)$ is not an isomorphism in general.

For the generalized Riemann–Roch formula the seminar works with $K^\bullet(X)$ and $K^\bullet(Y)$, not with $K_\bullet(X)$ and $K_\bullet(Y)$. These augmented λ-rings, with their natural filtrations, give graded rings $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}^\bullet(Y)$ and Chern-class maps

$$c_i : K^\bullet(X) \to \mathrm{Gr}^\bullet(X).$$

The problem is the direct image

$$f_* : K^\bullet(X) \to K^\bullet(Y), \tag{7}$$

which cannot be defined naively by

$$f_*(\mathrm{Cl}(F)) = \sum_i (-1)^i \, \mathrm{Cl}(R^i f_*(F)),$$

since even for a locally free F the sheaves $R^i f_*(F)$ need not be locally free, nor even of finite Tor-dimension.

Derived categories provide the solution. Although the individual sheaves $R^i f_*(F)$ may be bad, the derived direct image $\mathrm{R}f_*(F)$ often has good properties as a complex. If F has finite Tor-dimension over $\mathcal{O}_Y$, for example if f has finite Tor-dimension and F is locally free, then Expose III shows that $\mathrm{R}f_*(F)$ is perfect: its cohomology sheaves are coherent and it has globally finite Tor-dimension, or equivalently, on each affine open of Y, it is isomorphic in the derived category to a bounded complex of locally free modules of finite type.

Perfect complexes are as good as locally free modules from the viewpoint of homological algebra. They form a triangulated subcategory $\mathrm{Parf}(Y)$ of $\mathrm{D}(Y)$. For any triangulated category C there is a Grothendieck group $K(C)$, generated by objects modulo the relations

$$\mathrm{Cl}(K) - \mathrm{Cl}(K') - \mathrm{Cl}(K'')$$

attached to distinguished triangles

$$K' \to K \to K'' \to K'[1].$$

When Y has an ample invertible module, Expose II proves that every perfect complex on Y is globally isomorphic in $\mathrm{D}(Y)$ to a bounded complex of locally free modules of finite type. It follows, in Expose IV, that the canonical map from the naive $K^\bullet(Y)$ to $K(\mathrm{Parf}(Y))$ is an isomorphism. Thus a perfect complex $L_\bullet$ on Y has a class

$$\mathrm{Cl}(L_\bullet) \in K^\bullet(Y),$$

which is represented by

$$\sum_i (-1)^i \,\mathrm{Cl}(L'_i)$$

for any bounded locally free complex $L'_\bullet$ quasi-isomorphic to it.

Consequently, if f is proper of finite Tor-dimension between noetherian schemes and Y has an ample invertible module, one defines (7) by

$$f_*(\mathrm{Cl}(F)) = \mathrm{Cl}(\mathrm{R}f_*(F)) \tag{8}$$

for locally free F on X. If X also has an ample invertible module, this is exactly the map on K-groups induced by the exact functor of triangulated categories

$$\mathrm{R}f_* : \mathrm{Parf}(X) \to \mathrm{Parf}(Y).$$

1.5 The sophisticated group $K^\bullet(X)$ and the cotangent complex

For an arbitrary scheme, or locally ringed topos, the right invariant replacing the ring $K(X)$ of RRR is not the naive group defined only from locally free modules, but rather the group $K(\mathrm{Parf}(X))$ of perfect complexes. We continue to denote it by $K^\bullet(X)$, reserving $K^\bullet_{\mathrm{naive}}(X)$ for the group defined by locally free modules alone. This viewpoint is forced by the need to define direct images for proper morphisms of finite Tor-dimension: with perfect complexes, f_* is simply induced by $\mathrm{R}f_*$.

The new definition raises a homological-algebra problem: one must define a λ-ring structure on $K^\bullet(X)$, compatible with the ring structure coming from the derived tensor product. In other words one needs exterior-power operations in the derived category sending perfect complexes to perfect complexes. This question was not clarified in the seminar and is identified in the introduction as its most serious gap. Once it is solved, the construction of the filtration and of Chern classes proceeds as before.

To state the Riemann–Roch formula for a proper locally complete intersection morphism $f : X \to Y$, one must also define the virtual relative tangent class

$$T_f \in K^\bullet(X).$$

When f factors as in (4), by a closed immersion into a smooth Y-scheme, the formula (5) applies. It has a more derived form. The exterior differential gives a canonical morphism

$$N_{X/X'} \to \Omega^1_{X'/Y} \otimes_{\mathcal{O}_{X'}} \mathcal{O}_X. \tag{9}$$

By placing the source in degree 1 and the target in degree 0, one obtains a two-term complex $L_{X/Y}$. Then

$$T_f = \mathrm{Cl}(L_{X/Y}^{\vee}) \tag{10}$$

in $K^{\bullet}(X)$. For a locally complete intersection morphism, $L_{X/Y}$ is perfect, and Expose VIII proves that it is independent, up to canonical isomorphism in $\mathrm{D}(X)$, of the chosen factorization.

For an arbitrary morphism of schemes, and more generally for a morphism of commutatively ringed topoi, one can construct a canonical object $L_{X/Y}$ of $\mathrm{D}(X)$. Let $A = f^*\mathcal{O}_Y$ and let B be an A-algebra. Choose a map of sheaves of sets $T \to B$ generating B as an A-algebra, so that

$$A[T] \to B$$

is an epimorphism. If J is its kernel, consider the complex

$$J/J^2 \to \Omega^1_{A[T]/A} \otimes_{A[T]} B. \tag{11}$$

It is not hard to see that its isomorphism class in $\mathrm{D}(X)$ is independent of the choice of T and $T \to B$. This is the relative cotangent complex. In the locally complete intersection case it is perfect and gives the class T_f by (10). This gives another justification for using the sophisticated $K^{\bullet}(X)$ of perfect complexes instead of the naive group.

Subject to the exterior-power question above, all structural ingredients are available for giving a meaning to (2) for a proper locally complete intersection morphism of arbitrary schemes. This does not mean that the formula is proved in that generality, even for a proper smooth three-dimensional variety over an algebraically closed field of characteristic $p > 0$; see Expose XIV, no. 2.

1.6 Analytic variant

The form of Riemann–Roch just discussed also makes sense for a proper locally complete intersection morphism of complex analytic spaces, or of rigid analytic spaces in Tate's sense. The proof in RRR applies only when the morphism is projective. As in the algebraic case, a new idea seems necessary for the general case.

For nonsingular complex analytic spaces there is another version. If X is a complex analytic space and X^{top} denotes the same space with the sheaf of continuous complex-valued functions, there is a canonical morphism of ringed spaces

$$\epsilon : X^{\mathrm{top}} \to X$$

and hence a homomorphism

$$K^\bullet(X) \to K^\bullet(X^{\mathrm{top}}). \tag{12}$$

If X^{top} is compact, Expose II shows that every perfect complex on X^{top} is globally represented by a bounded complex of complex vector bundles. Thus $K^\bullet(X^{\mathrm{top}})$ is the usual topological K-group.

For a proper morphism $f : X \to Y$ of compact nonsingular complex analytic spaces, Atiyah–Hirzebruch attach a homomorphism

$$f_!^{\mathrm{top}} : K^\bullet(X^{\mathrm{top}}) \to K^\bullet(Y^{\mathrm{top}}).$$

Using also the analytic direct image $f_!$, defined by (8) and Grauert's finiteness theorem, one obtains a square

$$\begin{array}{ccc} K^\bullet(X) & \longrightarrow & K^\bullet(X^{\mathrm{top}}) \\ \downarrow f_! & & \downarrow f_!^{\mathrm{top}} \\ K^\bullet(Y) & \longrightarrow & K^\bullet(Y^{\mathrm{top}}). \end{array}$$

The Riemann–Roch assertion in this form is the commutativity of the square. Unlike the Chow-theoretic formula, this assertion does not discard torsion and takes place in a discrete group. It is known for nonsingular projective varieties. It is plausibly a variant of the Atiyah–Singer index theorem and should have a common generalization with it.

Once this square is known to commute, the differentiable Riemann–Roch theorem of Atiyah–Hirzebruch gives a cohomological version, namely the commutativity of

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{\text{ch}} & H^{2*}(X,\mathbb{Q}) \\ \downarrow f_! & & \downarrow f_* \\ K^\bullet(Y) & \xrightarrow{\text{ch}} & H^{2*}(Y,\mathbb{Q}), \end{array}$$

where the horizontal maps come from ordinary topological Chern classes and the right vertical arrow is the Gysin homomorphism in rational cohomology.

1.7 The covariant invariant $K_\bullet(X)$

So far the discussion has focused on the contravariant invariant $K^\bullet(X)$. The covariant invariant $K_\bullet(X)$ plays a complementary role. If $K^\bullet(X)$ is regarded as analogous to integral cohomology, then $K_\bullet(X)$ is analogous to integral homology, and its $K^\bullet(X)$-module structure corresponds to cap product. When X is regular, the natural map $K^\bullet(X) \to K_\bullet(X)$ being an isomorphism is analogous to Poincare duality.

For a noetherian scheme X, $K_\bullet(X)$ is given an increasing filtration by setting $\operatorname{Fil}_i K_\bullet(X)$ to be generated by classes $\operatorname{Cl}_\bullet(F)$ of coherent sheaves with $\dim\operatorname{supp} F \leq i$. Expose X proves compatibility with the filtration of $K^\bullet(X)$:

$$\operatorname{Fil}^i K^\bullet(X) \cdot \operatorname{Fil}_j K_\bullet(X) \subset \operatorname{Fil}_{j-i} K_\bullet(X). \tag{13}$$

The associated graded group $\operatorname{Gr}_\bullet(X)$ is thereby a module over $\operatorname{Gr}^\bullet(X)$. Its elements may be interpreted as classes of algebraic cycles, for an equivalence relation coarser than rational equivalence.

For a proper morphism $f : X \to Y$ of noetherian schemes, the direct image

$$f_* : K_\bullet(X) \to K_\bullet(Y)$$

is compatible with the filtrations and gives a graded map

$$f_* : \mathrm{Gr}_\bullet(X) \to \mathrm{Gr}_\bullet(Y),$$

satisfying a projection formula. In the proposed theory, a manageable intersection theory on nonregular noetherian schemes should be the combined study of the contravariant invariants $K^\bullet(X), \mathrm{Gr}^\bullet(X)$ and the covariant invariants $K_\bullet(X), \mathrm{Gr}_\bullet(X)$. They have different functorial characters and complement each other. For example, the covariant groups behave better under certain nonproper fibrations through homotopy exact sequences; one has canonical isomorphisms

$$K_\bullet(X[t]) \simeq K_\bullet(X), \qquad \mathrm{Gr}_\bullet(X[t]) \simeq \mathrm{Gr}_\bullet(X),$$

which generally fail for $K^\bullet$ when X is nonregular.

If X is proper over a field, the projection $\pi_X : X \to$ point defines the Euler–Poincare characteristic

$$\chi = \pi_{X*} : K_\bullet(X) \to K_\bullet(\text{point}) = \mathbb{Z},$$

and on $\mathrm{Gr}_0(X)$ it induces the degree of zero-cycles

$$\deg : \mathrm{Gr}_0(X) \to \mathbb{Z}.$$

Multiplication followed by degree gives pairings

$$\mathrm{Gr}^i(X) \times \mathrm{Gr}_i(X) \to \mathbb{Z},$$

explaining in what sense $\mathrm{Gr}^\bullet(X)$ and $\mathrm{Gr}_\bullet(X)$ are dual, as cohomology and homology are dual. The numerical relations in intersection theory over a field come from these maps and pairings. Expose X develops this point of view and also studies specialization.

1.8 The determinant functor

As in every intersection theory, one must locate the role of the Picard group. Under a mild restriction, automatically satisfied for instance when X is quasi-compact, Expose X proves a canonical isomorphism

$$c_1 : \mathrm{Pic}(X) \xrightarrow{\sim} \mathrm{Gr}^1(X),$$

which gives the exact role of the Picard group in the proposed theory. This motivates the more detailed study of the Picard group in Exposes XII and XIII, which are logically independent of the main text. Kleiman proves in Expose XIII finiteness theorems and results on numerical equivalence for the Picard group, using methods of Kleiman and Mumford that are substantially simpler than Grothendieck's original arguments. The one result not covered by those methods is EGA III, 7.7, whose proof and refinements, including representability theorems for the Picard functor, occupy Expose XII.

Finally, Expose XI was to study on an arbitrary locally ringed topos a determinant functor

$$\det : \mathrm{Parf}(X) \to \mathrm{Inv}(X),$$

from the category of perfect complexes, with only isomorphisms in the derived category as morphisms, to the category of invertible modules on X. It is essentially characterized by locality and by the formula

$$\det(L_\bullet) = \bigotimes_i \det(L_i)^{(-1)^i}$$

for a bounded complex of locally free modules. Although this expose is independent of the rest of the seminar except Expose I, it is an important part of the general "yoga" developed in Exposes I–XI.

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2 Appendix to Expose 0 – Classes of Sheaves and the Riemann–Roch Theorem

by A. Grothendieck

This appendix is the reproduction of the report cited in the introduction. Footnotes marked by special signs were added in October 1967.

Contents of the appendix

2.1 Chapter I. λ-rings: formal preliminaries

2.1.1 Definitions

A λ-ring is a commutative ring K with unit, equipped with maps

$$\lambda^i : K \to K \qquad (i \geq 0)$$

satisfying

$$\lambda^0(x) = 1, \qquad \lambda^1(x) = x, \qquad \lambda^n(x+y) = \sum_{i=0}^{n} \lambda^i(x)\lambda^{n-i}(y). \tag{14}$$

Equivalently, if

$$\lambda_t(x) = \sum_{n \geq 0} \lambda^n(x) t^n \in K[[t]],$$

then $x \mapsto \lambda_t(x)$ is a homomorphism from the additive group of K to the multiplicative group of formal power series of augmentation 1, lifting $x \mapsto 1 + xt$.

The ring $\mathbb{Z}$ has a unique λ-structure compatible with its ring structure and satisfying $\lambda_t(1) = 1 + t$; hence

$$\lambda_t(n) = (1+t)^n, \qquad \lambda^i(n) = \binom{n}{i}.$$

A λ-homomorphism is a ring homomorphism compatible with all the λ^i. An augmented λ-ring is a λ-ring equipped with a λ-homomorphism to $\mathbb{Z}$.

To define the special λ-rings, let k be a commutative ring with unit and put

$$\widetilde{K} = 1 + k[[t]]^+,$$

the group of formal series of augmentation 1. We put on this group the structure of a λ-ring for which addition is multiplication of formal series. The multiplication, denoted $f \circ g$, is characterized by universal integral polynomials in the coefficients and by the formula on linear factors

$$(1 + at) \circ (1 + bt) = 1 + abt. \tag{15}$$

The polynomials are obtained from elementary symmetric functions. Similarly, the operations $\lambda^i(f)$ are determined by universal polynomials and by the rule

$$\lambda^i(1 + at) = 1 \qquad (i > 1),$$

where 1 is the zero element of the additive group $1 + k[[t]]^+$. With these operations $\widetilde{K}$ is a λ-ring, with zero element 1 and unit $1+t$, and

$$(1+at)\circ f = f(at). \tag{16}$$

A λ-ring K is called *special* if the additive homomorphism

$$K \to 1 + K[[t]]^{+}, \qquad x \mapsto \lambda_t(x),$$

is a λ-homomorphism. Thus

$$\lambda_t(1) = 1+t, \tag{17}$$
$$\lambda_t(xy) = \lambda_t(x)\circ\lambda_t(y), \tag{18}$$
$$\lambda_t(\lambda^n(x)) = \lambda^n(\lambda_t(x)). \tag{19}$$

Equivalently, the operations satisfy universal polynomial identities

$$\begin{aligned} \lambda^i(1) &= 0 \quad (i>1),\\ \lambda^n(xy) &= P_n(\lambda^1 x,\dots,\lambda^n x;\lambda^1 y,\dots,\lambda^n y),\\ \lambda^m(\lambda^n x) &= P_{m,n}(\lambda^1 x,\dots,\lambda^{mn}x). \end{aligned}$$

The P_n and $P_{m,n}$ are precisely the universal polynomials furnished by symmetric-function computations.

2.1.2 Examples

Let G be a group and k a commutative ring. The group $K(G)$ is the quotient of the free group generated by finite type k-linear representations of G by relations

$$(\rho\oplus\rho') - (\rho) - (\rho').$$

Tensor product and exterior powers define a λ-ring structure. One can also pass to a stronger quotient by imposing the relation

$$(\rho) - (\rho') - (\rho'')$$

whenever ρ is an extension of ρ' by ρ'' which is split as an extension of k-modules. This gives the reduced group of representation classes, denoted here $K_\bullet(G)$, and the λ-operations descend to it.

If k is algebraically closed of characteristic zero, then $K(G)$, and hence also $K_\bullet(G)$, is a special λ-ring. In positive characteristic this need not remain true for $K(G)$, but $K_\bullet(G)$ is still special under the hypotheses used in the report. Variants include algebraic groups and rational representations, topological groups or Lie groups with continuous representations, and analogous constructions for quadratic forms or Lie-algebra representations.

Let X be an algebraic variety over k. One may take the quotient of the free group generated by vector bundles on X by the direct-sum relations. Exterior powers and tensor products again give a λ-ring. If X is complete, Krull–Remak–Schmidt identifies this group with the free group generated by indecomposable vector bundles, so that its knowledge is essentially the classification of vector bundles together with their elementary tensor operations. One may also impose extension relations and obtain the group $K(\mathcal{E}(X))$ attached to the additive category of vector bundles. Under the same hypotheses as above this gives a special λ-ring.

A particularly important computation is the following. Let k be algebraically closed, and let G be a connected affine algebraic group which is globally reductive, meaning that its radical is a torus. Only rational linear representations are considered. In characteristic zero, complete reducibility shows that

$$K(G) = K_\bullet(G)$$

is the free group on irreducible rational representations. If T is a maximal torus, then

$$K(T) = K_\bullet(T) = \mathbb{Z}[\widehat{T}],$$

where $\widehat{T}$ is the character group. The Weyl group W acts on $\widehat{T}$, hence on $K(T)$, and restriction gives a homomorphism

$$K_\bullet(G) \to K_\bullet(T)^W. \tag{20}$$

Theorem. *The homomorphism* (20) *is an isomorphism.*

It follows that $K_\bullet(G)$ depends only on the type of G, i.e. on the action of W on a free abelian group. It is a special λ-ring. If $\rho_1, \ldots, \rho_r$ are the fundamental representations of $G/\operatorname{rad} G$ and $\sigma_1, \ldots, \sigma_s$ a basis of rational homomorphisms from G to k^*, then

$$K_\bullet(G) \subset \mathbb{Z}[\rho_1, \ldots, \rho_r; \sigma_1, \sigma_1^{-1}, \ldots, \sigma_s, \sigma_s^{-1}]$$

inside the rational function field generated by these symbols. In particular $K_\bullet(G)$ has no zero divisors. For $G = \prod_i \mathrm{GL}(n_i, k)$ this description becomes completely explicit in terms of the exterior powers of the standard representations.

Remarks. In the representation examples the augmentation is the degree, i.e. the dimension of the representation. In the vector-bundle examples the augmentation is the rank. Chevalley supplied the positive characteristic form of the theorem just used.

2.1.3 The λ-ring defined by a graded ring

Let A be a graded commutative ring with unit, for simplicity concentrated in nonnegative degrees with $A^0 = \mathbb{Z}$. Write A_+ for the positive-degree part and consider

$$\mathbf{A} = \mathbb{Z} \times (1 + A_+),$$

with elements $[n, x]$ and addition

$$[n, x] + [n', x'] = [n + n', xx'].$$

The zero element is $[0, 1]$. We put on $\mathbf{A}$ the structure of an augmented λ-ring with augmentation $[n, x] \mapsto n$.

The product is characterized by a universal formula

$$[m, x]\,[n, y] = [mn, x^n y^m (x * y)], \tag{21}$$

where the components of $x * y \in 1 + A_+$ are integral universal polynomials in the homogeneous components of x and y. The operation $x * y$ is distributive with respect to multiplication in $1 + A_+$ and, on linear factors, is specified by

$$[1, 1 + x]\,[1, 1 + y] = [1, 1 + x + y]. \tag{22}$$

The polynomials are the usual polynomials expressing Chern classes of a tensor product in terms of the Chern classes and ranks of the factors.

The filtration of $\mathbf{A}$ is defined by $\mathbf{A}^0 = \mathbf{A}$ and by letting $\mathbf{A}^n$ be the set of $[0, x]$ with $x^i = 0$ for $1 \leq i < n$. Then

$$[0, x] \in \mathbf{A}^m,\ [0, y] \in \mathbf{A}^n \quad \Longrightarrow \quad [0, x][0, y] - [0, 1] \in \mathbf{A}^{m+n}.$$

The associated graded group is additively identified with A, but not directly as a ring. The map

$$\eta : A \to G(\mathbf{A}), \qquad a \in A^i \longmapsto \mathrm{cl}([1, 1 + a])$$

is a ring homomorphism.

The operations λ^i on $\mathbf{A}$ are determined by universal integral polynomials in the components of x, together with

$$\lambda^i[1, 1 + x] = [0, 1]quad(i > 1).$$

These polynomials are the symmetric-function formulas for the Chern classes of exterior powers of a vector bundle. Thus $\mathbf{A}$ is a special augmented λ-ring and the filtration ideals $\mathbf{A}^i$ are stable under all λ^j.

For $x = [n, 1 + x^1 + \cdots + x^n]$ with $x^i = 0$ for $i > n$, one has $\lambda^i(x) = 0$ for $i > n$, $\lambda^n(x) = [1, 1 + x^n]$, and

$$\lambda_t(x) = \sum_{i=0}^{n} [1, 1 + x^i] t^i \lambda_{-t}(\bar{x}^{n-i}), \tag{23}$$

where the bar denotes the principal involution of the graded ring, changing signs according to degree. Formula (23) remains valid without the finite truncation assumption as soon as the augmentation is n.

For an arbitrary λ-ring K define

$$\gamma^n(x) = (-1)^n \sum_{i=0}^{n} (-1)^i \binom{n-1+i}{n-i} \lambda^i(x) = \lambda^n(x + n - 1). \tag{24}$$

Then $\gamma^0(x) = 1$, $\gamma^1(x) = x$, and

$$\gamma_t(x) = \sum_{n \geq 0} \gamma^n(x) t^n = \lambda_{t/(1-t)}(x).$$

Conversely $\lambda_s(x) = \gamma_{s/(1+s)}(x)$. Therefore the γ^i define another λ-structure. For the ring $\mathbf{A}$ one obtains

$$\gamma^n\big([0, 1 + x] - \varepsilon([0, 1 + x])\big) = [0, 1 + (n-1)! x^n + \cdots], \tag{25}$$

so the degree n component can be recovered, up to the factor $(n - 1)!$, by reducing modulo $\mathbf{A}^{n+1}$.

The Chern homomorphism. For every graded ring A as above, define an augmented ring homomorphism

$$\mathrm{ch} : \mathbf{A} \to A \otimes \mathbb{Q}. \tag{26}$$

It is characterized by additivity, by sending 1 to 1, by universal homogeneous rational polynomials in positive degrees, and by

$$\mathrm{ch}([1, 1 + x^1]) = \exp(x^1 t) = \sum_{n \geq 0} \frac{(x^1)^n}{n!} t^n. \tag{27}$$

Let η be the additive endomorphism of $A \otimes \mathbb{Q}$ sending a homogeneous element x^i of degree i to $(-1)^{i-1}(i-1)! x^i$. Then

$$\mathrm{ch}\big([n, 1 + \sum_{i=1}^{N} x^i]\big) = n + \eta^{-1} \log\big(1 + \sum_{i=1}^{N} \eta(x^i)\big). \tag{28}$$

If A has only finitely many homogeneous components, this gives an isomorphism

$$\mathbf{A} \otimes \mathbb{Q} \simeq A \otimes \mathbb{Q}.$$

A formal power series $f \in \mathbb{Q}[[t]]$ defines by universal formulas an additive homomorphism $f_* : \mathbf{A} \otimes \mathbb{Q} \to \mathbf{A} \otimes \mathbb{Q}$. For

$$f(t) = \frac{t}{1 - e^{-t}}$$

one writes $f_* = \mathrm{Todd}$ and extends Todd to all $x \in \mathbf{A}$. The following identity is central.

Proposition. *Let*

$$N = [q, 1 + N^1 + \cdots + N^q]$$

with $N^i = 0$ *for* $i > q$. *If*

$$\gamma_t(N-q) = \sum_{i\geq 0} \gamma^i(N-q)t^i, \qquad \lambda_t(N) = \sum_{i=0}^{q} \lambda^i(N)t^i,$$

then

$$\mathrm{ch}(\lambda_{-t}(N)) = (-1)^q \,\mathrm{Todd}(N)\ \mathrm{ch}(\gamma_t(N-q)). \tag{29}$$

In particular,

$$\mathrm{ch}(\lambda_{-1}(N)) = (-1)^q \,\mathrm{Todd}(N). \tag{30}$$

2.1.4 The operations $\rho^p(N, x)$

Let **A** now denote the ring of formal power series with integral coefficients in two infinite families of variables, written $\lambda^i N$ and $\lambda^j x$ for $i, j \geq 1$, with $\lambda^0 N = N$ and $\lambda^0 x = x$. Equip it with the unique completed special λ-ring structure for which the variables represent the exterior powers of N and x and the operations are continuous. Form the invertible series

$$\lambda_{-t}(N) = \sum_{i\geq 0} (-1)^i \lambda^i(N)t^i.$$

The element $\rho^p(N, x)$ is defined by

$$\rho^p(N,x)\lambda_{-t}(N) = \lambda_t\big(\rho^p(N,x)\lambda_{-1}(N)\big). \tag{31}$$

Theorem. *Let $\mathfrak{I}$ be the closed ideal generated by the $\lambda^i N$ with $i > q$. After reducing modulo $\mathfrak{I}$, and identifying the quotient with the ring of formal series in the remaining variables, the element $\rho^p(N, x)$ is a polynomial, homogeneous of weight p in the variables attached to x.*

The proof reduces first to $x = 1$ and then to the representation ring of $G = \mathrm{GL}(N)$, where N is a vector space of dimension q over an algebraically closed field of characteristic zero. Since $K(G)$ identifies with a subring generated by the exterior powers of N and the inverse of the top exterior power, one obtains a polynomial identity by comparing in the common quotient field. The general case follows by checking the same identity in the representation ring of $\mathrm{GL}(N) \times \mathrm{GL}(F)$ and using that this ring has no zero divisors.

More explicitly, if N and F are finite-dimensional vector spaces, let n_0 be the kernel of

$$(a_1, \dots, a_q) \longmapsto a_1 + \cdots + a_q.$$

The symmetric group acts on suitable exterior powers, and taking the alternating part gives a graded module for $\mathrm{GL}(N) \times \mathrm{GL}(F)$. Then

$$\rho^p(N, F) = \sum_i (-1)^i \, \mathrm{Cl}\big(\Lambda^i n_0^p \otimes \Lambda^p F\big). \tag{32}$$

This equality characterizes the universal polynomial $\rho^p(N, x)$ when $\dim F \geq p$. Grothendieck notes that the same assertion should hold over a non-algebraically-closed field of characteristic zero, and that the alternating-part construction is naturally a characteristic-zero operation.

For any special λ-ring K and any element N such that $\lambda^i N = 0$ for $i \gg 0$, the operations $x \mapsto \rho^p(N, x)$ are defined by universal integral polynomials. They satisfy $\rho^1(N, x) = x$ and (31). It is useful to introduce a new special λ-ring K_N, obtained by adjoining a unit to the additive group of K with multiplication determined by the ρ-operations and with

$$\lambda_t(x) = \sum_{p \geq 0} \rho^p(N, x) t^p, \qquad \rho^0(N, x) = 1.$$

This construction turns K_N into a special λ-ring.

Guided by the preceding formulas, put

$$\gamma^n(N, x) = \sum_{i=0}^{n} \binom{n}{i} \rho^i(N, x). \tag{33}$$

Equivalently, it is $\rho^n(N, x_1)$, where x_1 denotes x regarded as an element of K_N.

Proposition. *Let A be a graded commutative ring with unit, let*

$$N = [q, 1 + N^1 + \cdots + N^q]$$

and let

$$x = [n, 1 + \sum_{i \geq 1} x^i]$$

be an arbitrary element of $\mathbf{A}$. Then for every $j \geq 0$ one has

$$\gamma^j(N, x) \in \mathbf{A}^j,$$

and its degree j component has the form

$$\big(\gamma^j(N, x)\big)^j = C_{q,j}(N^1, \ldots, N^q; x^1, \ldots, x^j)\, x^j, \tag{34}$$

where $C_{q,j}$ is a universal polynomial with integer coefficients characterized by

$$C_{q,j}(N^1, \ldots, N^q; t, t^2, \ldots, t^j) = \binom{n + q + j - 1}{j} t^j. \tag{35}$$

The proof reduces to a polynomial ring and uses

$$\gamma^p(N, x)\lambda_{-1}(N) = \gamma^p(N\lambda_{-1}(N)),$$

together with the filtration estimates supplied by the preceding formulas.

2.2 Chapter II. Coherent algebraic sheaves and Chern classes

2.2.1 Chow theory

For simplicity assume in this paragraph that the ground field k is algebraically closed. Most, and perhaps all, of what follows remains valid without this restriction. All algebraic spaces and regular maps are taken over k.

An algebraic space is called quasi-projective if it is isomorphic to a locally closed subset of a projective space. Let X be a connected nonsingular quasi-projective variety of dimension n. The Chow ring $A(X)$ is the graded ring of cycles on X modulo rational equivalence, with $A^i(X)$ consisting of classes of cycles of dimension $n - i$. Its ring structure is based on Chow's moving theorem:

Theorem (Chow)**.** *Every cycle on X is rationally equivalent to a nonsingular cycle. Given classes in $A^i(X)$ and $A^j(X)$, one can choose nonsingular representatives Y^i and Z^j which meet transversely everywhere.*

A cycle is called nonsingular when its irreducible components are nonsingular varieties. Two cycles meet transversely when every component of one meets every component of the other transversely, i.e. at intersection points the relevant tangent spaces are in general position.

If $f : X \to Y$ is a morphism between nonsingular quasi-projective varieties, the preceding theorem defines a graded ring homomorphism

$$f^* : A(Y) \to A(X). \tag{36}$$

Thus $A(X)$ is contravariant. If f is proper, it also defines an additive direct image

$$f_*: A(X) \to A(Y), \tag{37}$$

which preserves dimensions of cycles, hence shifts codimension by $\dim Y - \dim X$. The classical projection formula is

$$f_*(x\, f^*y) = f_*(x)y. \tag{38}$$

Chow theory also attaches Chern classes to vector bundles. For a vector bundle E on X, one has

$$c^i(E) \in A^i(X), \qquad c^0(E) = 1,$$

and

$$c_t(E) = \sum_{i\geq 0} c^i(E)t^i.$$

The defining properties are the Whitney product formula for extensions

$$c(E) = c(E')c(E'') \tag{39}$$

whenever E is an extension of E' by E''; compatibility with the λ-ring structure,

$$c(E \otimes F) = c(E) \circ c(F), \qquad c(\Lambda^i E) = \lambda^i(c(E)); \tag{40}$$

the normalization

$$c^1(L(D)) = \mathrm{Cl}(D) \tag{41}$$

for the line bundle associated with a divisor D; and functoriality

$$c^i(f^*E) = f^*c^i(E). \tag{42}$$

Equivalently, if $K(Y) \to K(X)$ is the λ-homomorphism induced by inverse image of vector bundles, then the diagram

$$\begin{array}{ccc} K(Y) & \longrightarrow & K(X) \\ \downarrow c_Y & & \downarrow c_X \\ A(Y) & \longrightarrow & A(X) \end{array}$$

commutes.

Theorem. *There exists a theory of Chern classes satisfying* (39)*,* (40)*,* (41)*, and* (42)*.*

We shall also use the following proposition.

Proposition. *Let Y be a closed subset of X and let $U = X \setminus Y$. If $x \in A(X)$ restricts to zero on U, then x has a representative which is a cycle supported on Y.*

Corollary. *Let $p = \dim X - \dim Y$. If $i < p$, then $A^i(X) \to A^i(U)$ is bijective. Every element of the kernel of $A^p(X) \to A^p(U)$ is a linear combination of the components of dimension $n - p$ of Y; in particular it is proportional to $\mathrm{Cl}(Y)$ when Y is irreducible.*

2.2.2 Definition of Chern classes of coherent algebraic sheaves

Let $\mathcal{C}$ be an abelian category and $\mathcal{C}_1$ a subclass of its objects. The group $K(\mathcal{C}_1)$ is the quotient of the free abelian group generated by isomorphism classes of objects of $\mathcal{C}_1$ by relations

$$(E) - (E') - (E'')$$

whenever E is an extension of E' by E'' and all three objects lie in $\mathcal{C}_1$. We write $\gamma_{\mathcal{C}_1}(E)$ for the class of E, omitting the subscript when no confusion is possible. This group has the expected universal property for additive functions on $\mathcal{C}_1$.

If $F : \mathcal{C} \to \mathcal{C}'$ is an additive functor with derived functors R^qF, and if $R^qF(A)$ belongs to a suitable subclass $\mathcal{C}'_1$ and vanishes for $q \gg 0$ for all $A \in \mathcal{C}_1$, then F defines a homomorphism

$$K(\mathcal{C}_1) \to K(\mathcal{C}'_1), \qquad \gamma_{\mathcal{C}_1}(E) \mapsto \sum_{i \geq 0} (-1)^i \gamma_{\mathcal{C}'_1}(R^iF(E)). \tag{43}$$

The same applies to cohomological functors with only finitely many nonzero values. If F is exact, this reduces to

$$F(\gamma_{\mathcal{C}_1}(E)) = \gamma_{\mathcal{C}'_1}(F(E)).$$

It also gives the canonical homomorphism $K(\mathcal{C}'_1) \to K(\mathcal{C}_1)$ when $\mathcal{C}'_1 \subset \mathcal{C}_1$.

Theorem. *Let $\mathcal{C}$ be an abelian category and let $\mathcal{C}'_1 \subset \mathcal{C}_1$ be two subclasses whose isomorphism classes form sets. Suppose:*

(i) In every exact sequence $0 \to A' \to A \to A'' \to 0$, if A and A'' lie in $\mathcal{C}_1$ (respectively in $\mathcal{C}'_1$), then A' also lies in $\mathcal{C}_1$ (respectively in $\mathcal{C}'_1$).

(ii) If $u : A \to B$ and $v : C \to B$ with $A, B, C \in \mathcal{C}_1$ and u surjective, then there exist $L \in \mathcal{C}'_1$ and maps $u' : L \to C$, $v' : L \to A$ such that u' is surjective and $vu' = uv'$.

(iii) For every $A \in \mathcal{C}_1$ there is an integer $d(A)$ such that, for every left resolution L of A by objects of $\mathcal{C}'_1$, the $d(A)$-cycles $Z_{d(A)}(L)$ lie in $\mathcal{C}'_1$.

Then the canonical homomorphism $K(\mathcal{C}'_1) \to K(\mathcal{C}_1)$ is an isomorphism.

Principle of proof. For $A \in \mathcal{C}_1$, choose a finite resolution L of A by objects of $\mathcal{C}'_1$ and put

$$f(A) = \sum_i (-1)^i \gamma_{\mathcal{C}'_1}(L^i).$$

The conditions allow one to compare any two such resolutions by a third resolution mapping surjectively to both. The difference of the corresponding

Euler characteristics is the Euler characteristic of an acyclic complex in $\mathcal{C}_1'$, hence zero. Thus $f(A)$ is independent of the resolution. One checks similarly that f is additive, hence gives a homomorphism $K(\mathcal{C}_1) \to K(\mathcal{C}_1')$ inverse to the canonical map.

For checking condition (ii), it is enough to know that $\mathcal{C}_1$ is contained in a subclass $\mathcal{C}_2$ satisfying: every object of $\mathcal{C}_2$ is a quotient of an object of $\mathcal{C}_1'$; and kernels of morphisms between objects of $\mathcal{C}_2$ again lie in $\mathcal{C}_2$.

Corollary. *Let X be a quasi-projective algebraic space. Let $\mathcal{F}(X)$ be the category of coherent algebraic sheaves on X, let $\xi(X)$ be the category of locally free coherent sheaves, and let $\mathcal{F}_0(X)$ be the coherent sheaves whose stalks have finite cohomological dimension. Then the canonical map*

$$K(\xi(X)) \to K(\mathcal{F}_0(X)) \tag{44}$$

is an isomorphism.

The conditions of the theorem are verified as follows. Conditions (i) and (iii) come from the definition of cohomological dimension and from the fact that a finite projective module over a noetherian local ring is free. For (ii) it suffices to use the following theorem of Serre.

Proposition (Serre)**.** *Every coherent algebraic sheaf on a quasi-projective algebraic space is a quotient of a locally free coherent sheaf.*

For a closed subspace of projective space this is the usual theorem that $F(n)$ is generated by global sections for $n \gg 0$. The general quasi-projective case is reduced to it by extending coherent sheaves from an open subset to the ambient algebraic space. If F is coherent on an open $U \subset X$, one extends by Zorn's lemma to a maximal open, reduces to X affine, and constructs a quasi-coherent extension whose restriction to U is generated by finitely many sections; these sections generate a coherent subsheaf restricting to F.

If now X is quasi-projective and nonsingular, the syzygy theorem gives $\mathcal{F}_0(X) = \mathcal{F}(X)$. Hence

$$K(\xi(X)) \simeq K(\mathcal{F}(X)). \tag{45}$$

Equivalently, for any abelian group A, additive functions on vector bundles on X are in bijection with additive functions on coherent algebraic sheaves on X. If

$$0 \to \mathcal{O}_X(E_m) \to \cdots \to \mathcal{O}_X(E_0) \to F \to 0$$

is a finite locally free resolution of a coherent sheaf F, then the corresponding value is

$$g(F) = \sum_i (-1)^i f(E_i). \tag{46}$$

In particular the total Chern class function on vector bundles extends additively to coherent sheaves. Its components

$$F \longmapsto c^i(F) \in A^i(X)$$

are called the Chern classes of the coherent sheaf F. They are computed from a finite locally free resolution by (46), interpreted multiplicatively for total Chern classes. The degree-zero component is the rank of F.

2.2.3 Functorial generalities on $K(X)$

For an algebraic space X write

$$K(X) = K(\mathcal{F}(X)).$$

If X is nonsingular and quasi-projective, we identify this group with $K(\xi(X))$ by (45); hence $K(X)$ inherits a λ-ring structure. It is important, however, to describe the operations directly in terms of coherent sheaves rather than by passing to vector bundles.

For a vector bundle E or a coherent sheaf F on X, write $\gamma(E)$ or $\gamma(F)$ for its class in $K(X)$. If Y is an irreducible subvariety of X, put

$$\gamma(Y) = \gamma(\mathcal{O}_Y),$$

where $\mathcal{O}_Y$ is regarded as a coherent sheaf on X; extend linearly to cycles. There is a filtration on $K(X)$ by codimension of support: $K(X)^i$ is generated by classes of coherent sheaves F for which $\dim \operatorname{supp} F \leq \dim X - i$. The associated graded group compares with the Chow group by sending a cycle to the class of its structure sheaf.

For nonsingular quasi-projective X and Y and a morphism $f : X \to Y$, inverse image of vector bundles gives a map

$$f^! : K(Y) \to K(X),$$

which is compatible with filtrations. On associated gradeds it matches the usual pullback of Chow classes:

$$\begin{array}{ccc} A(Y) & \xrightarrow{f^*} & A(X) \\ \downarrow & & \downarrow \\ G(K(Y)) & \xrightarrow{G(f^!)} & G(K(X)). \end{array}$$

The proof reduces, via the graph of f, to projections and immersions.

Proposition (Cellular decomposition)**.** *Let P be an algebraic space with an increasing family of closed subsets*

$$P_0 \subset P_1 \subset \cdots \subset P_n = P$$

such that the connected components of $P^i - P^{i-1}$ are affine spaces. Then for *every algebraic space X, the natural map*

$$K(X) \otimes K(P) \to K(X \times P)$$

coming from tensor products of sheaves is surjective. Moreover the classes of the strata generate $K(P)$.

The proof is by induction on n, using the exact sequence for an open complement and a closed subset. Grothendieck notes that he does not know in this generality whether the map is always injective.

2.2.4 Riemann–Roch theorem in the report

Let $f : Y \to X$ be a proper morphism from a nonsingular quasi-projective variety Y to another such variety X, and let $y \in K(Y)$. The Riemann–Roch formula is

$$f_*\big(\mathrm{ch}(y)\,\mathrm{Todd}(Y)\big) = \mathrm{ch}(f_! y)\,\mathrm{Todd}(X) \quad \text{in } A(X) \otimes \mathbb{Q}. \tag{47}$$

The proof begins with three formal reductions. If $Z \xrightarrow{g} Y \xrightarrow{f} X$ are proper morphisms, then validity for f and g implies validity for fg; under suitable surjectivity or injectivity hypotheses one can also infer one of the two from the other two. Since a proper morphism is factored as an immersion followed by a projection $X \times \mathbb{P}^r \to X$, one reduces to the cases of a projection and an immersion. The projection case is a standard computation, using the cellular decomposition of projective space.

For a closed immersion $i : Y \hookrightarrow X$, the formula becomes

$$\mathrm{ch}(i_! y) = i_*\big(\mathrm{ch}(y)\,\mathrm{Todd}(E)^{-1}\big), \tag{48}$$

where E is the normal bundle of Y in X. If $y = i^* x$ for some $x \in K(X)$, the formula follows from

$$\operatorname{ch}(\lambda_{-1}(E^*)) = c^p(E)\,\operatorname{Todd}(E)^{-1}, \tag{49}$$

where $p = \operatorname{codim}_X Y$. In particular, it holds for hypersurfaces in the appropriate form.

The general immersion case is reduced by blowing up X along Y. Let X' be the blow-up, Y' the inverse image of Y, and use the notation customary for the exceptional divisor and the normal bundle. One has identities of the form

$$\begin{aligned}
i'_!(y) &= j_!\big(y\lambda_{-1}(F^*)\big),\\
\lambda_{-1}(F^*) &\equiv \xi^{p-1}(E) \pmod{(1-\mathcal{L})},\\
f_!(1) &= 1, \qquad f_! f^! = 1,\\
g^* i_!(a) &= j_!(a\, c^p(E)) \pmod{\operatorname{Ker} f_!}.
\end{aligned}$$

The first follows from the Tor spectral sequence, the second from projective-space cohomology and Kunneth formulas, and the last from the compatibility of top Chern classes with the exceptional divisor. These formulas imply Riemann–Roch in the case where a certain filtration condition is satisfied; in particular when $\dim Y < \dim X - 1$.

Finally, one reduces to that inequality by replacing the immersion with a related one after product with a projective space and a suitable deformation. The report ends by noting that a remaining equality in the blow-up computation should follow directly from the same formula used above, and that at worst the argument is correct after passing to the associated graded, losing only possible torsion.

3 Expose I – Generalities on Finiteness Conditions in Derived Categories

by L. Illusie

0. Introduction

The purpose of Exposes I–IV is to develop, with the degree of generality required in this seminar, the formalism of finiteness conditions in the derived categories of ringed topoi.

As was said in Expose 0, the need to define, on arbitrary schemes, "Grothendieck groups" with good variance properties is what forces one to generalize and loosen the finiteness notions used up to now. A classical notion such as that of a coherent sheaf on a ringed space $(X, \mathcal{O}_X)$ loses much of its usefulness as soon as $\mathcal{O}_X$ itself is no longer coherent. One might try to replace it by finite presentation, but finite presentation has the defect that the kernel of an epimorphism of finitely presented modules need not be finitely presented.

If X is a ringed space whose local rings are noetherian, a coherent $\mathcal{O}_X$-module F locally admits a finite left resolution by finite free modules; in other words, locally there is an exact sequence

$$0 \longrightarrow L_n \longrightarrow L_{n-1} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow F \longrightarrow 0,$$

where the L_i are finite free. When $\mathcal{O}_X$ is no longer assumed coherent, we shall say that a $\mathcal{O}_X$-module F is *pseudo-coherent* if it satisfies the analogous condition in all finite stages, i.e. if it is of finite n-presentation for every n. Pseudo-coherence then has, with respect to short exact sequences, the same stability property as coherence: if two terms of a short exact sequence of

$\mathcal{O}_X$-modules are pseudo-coherent, then so is the third.

This notion is extended without difficulty to complexes of sheaves. A complex F of $\mathcal{O}_X$-modules is called *pseudo-coherent* if, locally, it can be approximated in all degrees by complexes whose terms are finite free in the relevant range. The precise formulation is given below in terms of n-pseudo-coherence.

Pseudo-coherent complexes have excellent stability properties, described in Sections 1 and 2 of this expose. First, pseudo-coherence is a local property. Secondly, if

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

is a morphism of ringed spaces, the derived inverse image Lf^* of a pseudo-coherent complex on Y is pseudo-coherent on X, provided that the complex has finite Tor-dimension relative to f, i.e. relative to the sheaf of rings $f^{-1}(\mathcal{O}_Y)$.

A coherent complex over a noetherian local ringed space also admits locally a finite left resolution by finite free modules. It is therefore natural, for an arbitrary sheaf of local rings $\mathcal{O}_X$, to single out the modules having this property. Such modules are called *perfect*. More generally, a complex F of $\mathcal{O}_X$-modules is called *perfect* if, locally, there is a quasi-isomorphism

$$L \longrightarrow F,$$

where L is a bounded complex whose components are finite free modules. Section 3 proves that the full subcategory $D(X)_{\mathrm{parf}}$ of $D(X)$ formed by perfect complexes is, like the pseudo-coherent part, a triangulated subcategory, stable under inverse image and under the derived tensor product.

Global resolutions of perfect complexes are treated in Expose II. In particular, global resolutions exist in the following cases: X is the Zariski topos of an affine scheme; more generally, X is the Zariski topos of a quasi-compact scheme carrying an ample invertible sheaf, or even an ample family of invertible sheaves in the sense of Kleiman; and also when X is the topos of sheaves of sets on a compact topological space and $\mathcal{O}_X$ is the sheaf of continuous complex-valued functions. As an illustration, and to show the flexibility of the notions introduced, the appendix gives a purely sheaf-theoretic definition of the index of a family of elliptic operators.

Expose III studies the stability of finiteness conditions under derived direct image. To obtain useful results one is led to introduce the notion of a complex *pseudo-coherent relative* to a morphism $f : X \to Y$: this means pseudo-coherent relative to f and of finite Tor-dimension relative to f, again relative to the sheaf of rings $f^{-1}(\mathcal{O}_Y)$.

The central result of Expose III is the *finiteness theorem*. In a slightly more precise form, it says that if $f : X \to Y$ is a proper morphism of S-schemes locally of finite type, then the functor Rf_* carries complexes pseudo-coherent relative to S to complexes pseudo-coherent relative to S. In this generality the theorem remains a conjecture in the seminar, but it is proved in the two following important cases:

a) S is locally noetherian;

b) f is projective.

The extension to the general case seems to be of the same order of difficulty as the corresponding theorem in analytic geometry, namely Grauert's finiteness theorem.

Combining the finiteness theorem with an essentially formal identity called the *projection formula*, one obtains usable criteria for the stability of relative perfection under direct image. As a corollary one recovers Grauert's continuity and semicontinuity theorems, in the form of EGA III, 7.6.

Expose IV translates the results of Exposes I–III into the language of

Grothendieck groups. For a ringed topos $(X, \mathcal{O}_X)$ one denotes by $K^*(X)$, respectively by $K_*(X)$, the Grothendieck group of the category of perfect complexes of finite Tor-dimension, respectively of pseudo-coherent complexes with bounded cohomology. These groups carry the functorialities needed in intersection theory.

1. Preliminary definitions

1.1. Fibred categories with additive, abelian, or triangulated fibres

Definition 1.1.1. Let S be a category, and let $\mathbf{C}$ be an S-fibred category, in the sense of SGA 1, VI, 6.1, whose fibres are additive categories, respectively abelian categories, respectively triangulated categories. We shall say that $\mathbf{C}$ is an additive, respectively abelian, respectively triangulated, S-category if, for every arrow $f : X \to Y$ of S, the inverse-image functor

$$f^* : \mathbf{C}_Y \longrightarrow \mathbf{C}_X$$

which is determined up to unique isomorphism, is additive, respectively exact, respectively triangulated.

Definition 1.1.2. Let $\mathbf{C}$ be a triangulated S-category, and let $\mathbf{C}'$ be a full sub-S-category of $\mathbf{C}$. We say that $\mathbf{C}'$ is a triangulated sub-S-category of $\mathbf{C}$ if, for every object X of S, the fibre $\mathbf{C}'_X$ is a triangulated subcategory of $\mathbf{C}_X$.

Definition 1.1.3. Let $\mathbf{C}$ be an additive S-category. One defines an additive, respectively triangulated, S-category

$$\mathbf{C}(\mathbf{C}), \qquad \mathbf{K}(\mathbf{C}),$$

whose fibre at $X \in S$ is the category of complexes in $\mathbf{C}_X$, respectively the homotopy category of complexes in $\mathbf{C}_X$. If $\mathbf{C}$ is a flat abelian S-category, one also defines a triangulated S-category $\mathbf{D}(\mathbf{C})$ whose fibre at X is the derived category $\mathbf{D}(\mathbf{C}_X)$. For an arrow $f : X \to Y$ of S, the inverse-image functor on derived categories

$$f^* : \mathbf{D}(\mathbf{C}_Y) \longrightarrow \mathbf{D}(\mathbf{C}_X)$$

is obtained from the functor on homotopy categories by passing to the quotient.

1.4. Quasi-isomorphisms and finiteness conditions

Lemma (1.4.1)**.** *Let S be a scheme, let $n \in \mathbb{Z}$, and let $u : E \to F$ be a morphism of perfect complexes on S. Suppose that $H^i(u) = 0$ for $i > n$ and that $H^n(u)$ is of finite presentation. Then there exists a commutative diagram of perfect complexes*

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u\downarrow & & \downarrow u' \\ F & \xrightarrow{=} & F, \end{array}$$

such that $H^i(u') = 0$ for $i \geq n$ and v is an isomorphism in degrees $< n$.

Lemma (1.4.2)**.** *Let $\mathbf{C}$ be an abelian category, let $u : E \to F$ be a morphism of complexes of $\mathbf{C}$, or a morphism in $\mathbf{D}(\mathbf{C})$, and let G be the cone of u. For a fixed integer n, the following conditions are equivalent:*

(i) $H^i(G) = 0$ *for* $i \geq n$*;*

(ii) $H^i(u)$ *is an isomorphism for* $i > n$ *and an epimorphism for* $i = n$.

The proof is immediate from the long exact cohomology sequence.

Definition 1.4.3. A morphism u satisfying the equivalent conditions of Lemma 1.4.2 is called an *n-quasi-isomorphism*, respectively an *n-isomorphism*, according as it is considered at the level of complexes, respectively in the derived category.

From Lemma 1.4.1 one obtains the following local improvement criterion.

Corollary (1.4.4)**.** *Let S, $\mathbf{C}$, and $\mathbf{C}_0$ be as in the preceding discussion. Let $u : E \to F$ be an $(n+1)$-quasi-isomorphism of complexes in $\mathbf{C}_S$, with $n \in \mathbb{Z}$, and let G be its cone. Suppose that $H^n(G)$ is of finite $\mathbf{C}_0$-type. Then the collection of objects X over S on which one can modify u as in 1.4.1 so as to obtain an n-quasi-isomorphism is a refinement of S.*

In less formal terms, the finite-type condition on $H^n(G)$ permits one, locally, to gain one degree: one changes u in degree n by adding to E^n an object of $\mathbf{C}_0$, so that the resulting morphism becomes an n-quasi-isomorphism.

Corollary (1.4.5)**.** *Let $\mathbf{C}$ be an abelian category, let $n \in \mathbb{Z}$, and let $u : E \to F$ be an n-quasi-isomorphism of complexes of $\mathbf{C}$. There is a commutative diagram in the category of complexes*

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u \downarrow & & \downarrow u' \\ F & \xrightarrow{=} & F, \end{array}$$

such that u' is an n-isomorphism and v is an $(n+1)$-quasi-isomorphism.

1.5. Finiteness conditions in derived categories

Definition 1.5.0. We shall say that $\mathbf{C}_0$ is *locally stable under kernels of epimorphisms* if the following condition holds. For every object X of S and every epimorphism $u : E \to F$ in $\mathbf{C}_X$ with E and F objects of $\mathbf{C}_{0,X}$, the kernel of u is locally in $\mathbf{C}_{0,X}$; that is, the family of arrows $f : U \to X$ such that

$$f^*(\operatorname{Ker} u) \in \mathbf{C}_{0,U}$$

is a refinement of X.

By decomposing a long exact sequence into short exact sequences, this condition is equivalent to the apparently stronger condition that, for every exact sequence

$$0 \longrightarrow E^0 \longrightarrow E^1 \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0$$

in $\mathbf{C}_X$, if E^i lies in $\mathbf{C}_{0,X}$ for $1 \leq i \leq n$, then E^0 is locally in $\mathbf{C}_{0,X}$.

We shall say that $\mathbf{C}_0$ is *locally quasi-stable under kernels of epimorphisms* if, for every exact sequence of the displayed form, E^0 is of finite $\mathbf{C}_0$-type.

Definition 1.5.0 bis. We shall say that $\mathbf{C}_0$ is *locally quasi-liftable* in $\mathbf{C}$ if the following weaker condition holds. For every object X of S and every epimorphism $u : E \to F$ in $\mathbf{C}_X$, with F in $\mathbf{C}_{0,X}$ and E in $\mathbf{C}_X$, there exists locally an epimorphism $v : E' \to E$ with E' in $\mathbf{C}_{0,X}$ such that uv is an epimorphism in $\mathbf{C}_{0,X}$.

Theorem (1.5.1)**.** *Assume that $\mathbf{C}_0$ is locally stable under kernels of epimorphisms, respectively locally quasi-stable under kernels of epimorphisms, respectively locally quasi-liftable in $\mathbf{C}$. Then the full subcategory $\mathbf{D}_{\mathbf{C}_0}(\mathbf{C})$ of $\mathbf{D}(\mathbf{C})$ formed by the complexes whose cohomology is locally in $\mathbf{C}_0$, respectively of finite $\mathbf{C}_0$-type, respectively locally quasi-liftable, is a triangulated subcategory of $\mathbf{D}(\mathbf{C})$.*

SGA 6 – Expose I

Generalities on Finiteness Conditions in Derived Categories

Sections 2–8

Corollary 1.4.5, proof continued

In the situation of Corollary 1.4.5, u' is a quasi-isomorphism and v is a monomorphism inducing the identity in degrees $\geq n$, that is, such that

$$E'^i = E^i, \qquad v^i = \mathrm{Id}(E^i) \quad (i \geq n).$$

Proof. Applying (1.4.4) in the case where $\underline{S}$ is the punctual site and $\underline{C}_0 = \underline{C}$, one obtains a commutative triangle in $C(\underline{C})$

$$\begin{array}{ccc} E & \xrightarrow{v_{n-1}} & E'_{n-1} \\ u\downarrow & & \\ F & \underset{u'_{n-1}}{\longleftarrow} & \end{array}$$

where u'_{n-1} is an $(n-1)$-quasi-isomorphism and v_{n-1} is a monomorphism inducing the identity in degrees $\geq n$. Iterating this procedure, one obtains, for $i \leq n-1$, a sequence of commutative triangles in $C(\underline{C})$

$$\begin{array}{ccc} E'_i & \xrightarrow{v_{i-1}} & E'_{i-1} \\ u'_i\downarrow & & \\ F & \underset{u'_{i-1}}{\longleftarrow} & \end{array}$$

where u'_i is an i-quasi-isomorphism and v_{i-1} is a monomorphism inducing the identity in degrees $\geq i$. The corollary follows by passing to the limit. More precisely, writing d'_i for the differential of E'_i, define E' by

$$E'^i = E^i, d'^i = d^i \quad \text{for } i \geq n,$$

and

$$E'^i = (E'_i)^i, \qquad d'^i = (d'_i)^i \quad \text{for } i < n.$$

Then define v and u' by

$$v^j = (v_i)^j, \qquad u'^j = (u'_i)^j \quad \text{for } i \leq j.$$

It is then immediate to check that E', v, and u' so defined have the required properties.

2. Pseudo-coherent complexes

2.0

In this section $\underline{S}$ denotes a site, S an object of $\underline{S}$, $\underline{C}$ a flat abelian $\underline{S}$-category in the sense of (1.1), and $\underline{C}_0$ a strictly full additive sub-$\underline{S}$-category of $\underline{C}$. We assume that $\underline{C}_0$ is locally quasi-stable under kernels of epimorphisms (1.2), and locally quasi-liftable in $\underline{C}$ (1.2).

Definition (2.1)**.** *Let $S \in \operatorname{ob} \underline{S}$, let E be a complex of $\underline{C}_S$, and let $n \in \mathbb{Z}$. We shall say that E is* strictly $\underline{C}_0$-n-pseudo-coherent *(respectively $\underline{C}_0$-pseudo-coherent, respectively $\underline{C}_0$-perfect) if $E^i = 0$ for i sufficiently large and $E^i \in \operatorname{ob} \underline{C}_{0S}$ for $i \geq n$ (respectively for every i, respectively for every i and $E^i = 0$ for almost all i).*

When no confusion is possible, we shall sometimes say simply "strictly n-pseudo-coherent" instead of "strictly $\underline{C}_0$-n-pseudo-coherent", and similarly for the other variants.

Lemma (2.1.1)**.** *Let $E \in \operatorname{ob} C(\underline{C}_S)$ and let $n \in \mathbb{Z}$. If E is strictly n-pseudo-coherent and if $H^i(E) = 0$ for $i > n$, then $H^n(E)$ is of finite type.*

Proof. The sequence

$$0 \longrightarrow B^n(E) \longrightarrow Z^n(E) \longrightarrow H^n(E) \longrightarrow 0$$

is exact, and $E^i \in \operatorname{ob} \underline{C}_{0S}$ for $i \geq n$. Since $\underline{C}_{0S}$ is locally quasi-stable under kernels of epimorphisms, $Z^n(E)$ is of finite type; so is its quotient $H^n(E)$.

Proposition (2.2)**.** *Let $F \in \operatorname{ob} D(\underline{C}_S)$ and let $n \in \mathbb{Z}$.*

a) *If F is strictly n-pseudo-coherent, relatively to $\underline{C}_0$, and if $u : E \to F$ is a quasi-isomorphism, then the family of objects X over S for which there exists a quasi-isomorphism*

$$v : F' \longrightarrow E_X$$

with F' strictly n-pseudo-coherent is a refinement of S.

b) *The following conditions are equivalent:*

(i) the family of X over S for which there exists a quasi-isomorphism $u : E \to F_X$ in $C(\underline{C}_S)$, with E strictly n-pseudo-coherent, is a refinement of S;

(ii) the same condition as (i), but with u an isomorphism in $D(\underline{C}_S)$;

(iii) the family of X over S for which there exists an n-quasi-isomorphism (1.4.3) $u : E \to F_X$ in $C(\underline{C}_S)$, with E strictly perfect, is a refinement;

(iv) the same condition as (iii), but with u an n-isomorphism in $D(\underline{C}_S)$.

Proof. For a), choose, for an integer i large enough, an i-quasi-isomorphism $v_i : E_i \to E$ with E_i strictly n-pseudo-coherent; one may initially take $E_i = 0$. Suppose $i > n$. We show that, after refining S, there is an $(i-1)$-quasi-isomorphism $v_{i-1} : E'_{i-1} \to E$, with E'_{i-1} strictly n-pseudo-coherent.

By (1.4.4), it is enough to show that $H^i(M^\bullet)$ is of $\underline{C}_0$-finite type, where $M^\bullet$ is the cone of v_i. If $M'^\bullet$ denotes the cone of $u \circ v_i$, the fact that u is a quasi-isomorphism and (1.4.2) imply

$$H^j(M^\bullet) \xrightarrow{\sim} H^j(M'^\bullet) \quad \text{for all } j,$$

and $H^j(M'^\bullet) = 0$ for $j > i$. Moreover $M'^\bullet$ is strictly $(i-1)$-pseudo-coherent. Hence, by (2.1.1), $H^i(M^\bullet) \simeq H^i(M'^\bullet)$ is of finite type. This proves the assertion. Iterating, and after refining S, one obtains an n-quasi-isomorphism $v_0 : E_0 \to E$, where E_0 is strictly n-pseudo-coherent. By (1.4.5) one can then extend v_0 to a quasi-isomorphism $v : F' \to E$ without changing anything in degrees $> n$; in particular F' is strictly n-pseudo-coherent. This proves a).

For b), the equivalence of (i) and (ii) follows immediately from a) and from the definition of isomorphisms in $D(\underline{C}_S)$. If $E \in \operatorname{ob} D(\underline{C}_S)$ is strictly n-pseudo-coherent, there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

where E' is strictly perfect and $E''^i = 0$ for $i > n$; hence $E' \to E$ is an n-quasi-isomorphism, which proves (i)⇒(iii). The implication (iii)⇒(iv) is immediate.

It remains to prove (iv)⇒(i). Let $u : E \to F$ be an n-isomorphism in $D(\underline{C}_S)$, with E strictly perfect. It is represented by a diagram in $C(\underline{C}_S)$

$$\begin{tikzcd} E \arrow[dr] & F \\ & E' \arrow[u, "\wr"'] \end{tikzcd}$$

where s is a quasi-isomorphism and u' is an n-quasi-isomorphism. By a), after refining S there exists a quasi-isomorphism $t : E'' \to E'$ with E'' strictly n-pseudo-coherent. Applying (1.4.5) to $u' \circ t : E'' \to F$, one extends it to a quasi-isomorphism $E''' \to F$ with E''' strictly n-pseudo-coherent. This proves (iv)⇒(i), and completes the proof of (2.2).

Definition (2.3)**.** *Under the hypotheses of (2.2), an object $F \in \operatorname{ob} D(\underline{C}_S)$ is called n-$\underline{C}_0$*-pseudo-coherent *if it satisfies the equivalent conditions b) of (2.2). It is called $\underline{C}_0$*-pseudo-coherent *if it is n-$\underline{C}_0$-pseudo-coherent for every $n \in \mathbb{Z}$.*

When there is no risk of confusion, we shall simply say n-pseudo-coherent, pseudo-coherent, instead of n-$\underline{C}_0$-pseudo-coherent, $\underline{C}_0$-pseudo-coherent.

Remark 2.3.1. Let $F \in \operatorname{ob} D(\underline{C}_S)$ be a pseudo-coherent complex. The family of X over S for which there exists a quasi-isomorphism $E \to F_X$ with E strictly pseudo-coherent (2.1) is not, in general, a refinement of S. Later, however, we shall see important cases in which this is true, namely in (2.7.a) and in Expose II.

The proof of (2.2 a)) gives the following result.

Proposition (2.4)**.** *Let $n, i \in \mathbb{Z}$, and let $u : E \to F$ be an i-quasi-isomorphism in $C(\underline{C}_S)$, with F n-pseudo-coherent and E strictly n-pseudo-coherent. Then, after refining S, there exists a commutative triangle in $C(\underline{C}_S)$*

$$\begin{tikzcd} E \arrow[r, "v"] \arrow[dr] & E' \arrow[d, "\wr"] \\ & F \end{tikzcd}$$

where E' is strictly n-pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degrees $> i$.

Proposition (2.5)**.** a) *If $F \in \operatorname{ob} D(\underline{C}_S)$ is n-pseudo-coherent, respectively pseudo-coherent, then $F[m]$ is $(n-m)$-pseudo-coherent, respectively pseudo-coherent.*

b) *Let*

$$\begin{array}{ccc} & & G \\ & \nearrow & \downarrow \\ E & \longrightarrow & F \end{array}$$

be a distinguished triangle in $D(\underline{C}_S)$. *If* E *and* G *are* n*-pseudo-coherent, respectively pseudo-coherent, then* F *is* n*-pseudo-coherent, respectively pseudo-coherent.*

Proof. The assertion a) is immediate from the definitions. For b), it is enough to prove the non-parenthetical assertion. By rotation of the triangle, it is enough to show that if E is $(n+1)$-pseudo-coherent and F is n-pseudo-coherent, then G is n-pseudo-coherent. We may localize and suppose that E, respectively F, is strictly $(n+1)$-pseudo-coherent, respectively strictly n-pseudo-coherent. The morphism $E \to F$ in the derived category is represented by a pair of morphisms in $C(\underline{C}_S)$,

$$\begin{array}{ccc} & & E' \\ & \swarrow & \downarrow \\ E & & F, \end{array}$$

where one arrow is a quasi-isomorphism. By (2.2 a)), after localizing further, there is a quasi-isomorphism $t : E'' \to E'$ with E'' strictly $(n+1)$-pseudo-coherent. We are therefore reduced to the case where the morphism is represented by an actual morphism in $C(\underline{C}_S)$. The assertion then follows from the definitions and from the formula for the cone,

$$C(u)^\bullet = E[1] \oplus F^\bullet.$$

Scholium 2.6. By rotating triangles, pseudo-coherence properties imposed on two of the three objects in a triangle imply the corresponding property for the third. The following table summarizes the cases; the entries are pseudo-coherence degrees, and each row expresses an implication in a distinguished triangle $E \to F \to G \to E[1]$:

E	F	G
$n+1$	n	n
n	n	$n-1$.

We write $D(\underline{C})_{n\text{-coh}}$, respectively $D(\underline{C})_{\text{coh}}$, for the strictly full fibred subcategory of $D(\underline{C})$ whose fibre over $X \in \operatorname{ob} \underline{S}$ consists of the objects of $D(\underline{C}_X)$ that are n-pseudo-coherent, respectively pseudo-coherent, relative to $\underline{C}_0$. By (2.5), $D(\underline{C})_{\text{coh}}$ is a triangulated sub-$\underline{S}$-category of $D(\underline{C})$.

Proposition (2.7)**.** *Suppose that* $\underline{S}$ *is the punctual site, so that* $\underline{C} = C$ *and* $\underline{C}_0 = C_0$.

a) *If* $F \in \operatorname{ob} D(C)$ *is pseudo-coherent, there exists a quasi-isomorphism* $E \to F$ *with* E *strictly pseudo-coherent.*

b) *Let* $\mathcal{A}^{\text{ac}}(C_0)$ *be the full triangulated subcategory of* $K(C_0)$ *formed by the acyclic complexes. The canonical functor*

$$K(C_0)/\mathcal{A}^{\text{ac}}(C_0) \longrightarrow D(C)_{\text{coh}}$$

is fully faithful and has essential image $D(C)_{\text{coh}}$.

Proof. For a), let $F \in \operatorname{ob} D(C)_{\text{coh}}$. Suppose that an i-quasi-isomorphism $u_i : E_i \to F$ has been constructed in $C(C)$, with E_i strictly pseudo-coherent; this is possible for i large enough with $E_i = 0$. Let G_i be the cone of u_i. By (1.4.2) and (2.1.1), $H^i(G_i) \in \operatorname{ob} C_0$ is of finite C_0-type. Hence, by (1.4.4), there exists a commutative triangle in $C(C)$

$$\begin{array}{ccc} E_i & \xrightarrow{v_i} & E'_i \\ u_i \downarrow & & \swarrow u'_i \\ F & & \end{array}$$

where E'_i is strictly pseudo-coherent, u'_i is an $(i-1)$-quasi-isomorphism, and v_i is a monomorphism inducing the identity in degrees $> i$. Passing to the limit, as in the proof of (1.4.5), gives a quasi-isomorphism $u : E \to F$ with E strictly pseudo-coherent.

For b), one applies Verdier's criterion, chapter I, §2, 4.2 b)(i), together with a).

The proof of a) gives the following more precise result, generalizing (1.4.5).

Proposition (2.7.1)**.** *Under the hypotheses of (2.7), let $u : E \to F$ be an n-quasi-isomorphism in $C(C)$, with F pseudo-coherent and E strictly pseudo-coherent. There exists a commutative triangle in $C(C)$*

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ & \searrow & \swarrow \wr \\ & & F \end{array}$$

where E' is strictly pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degrees $> n$.

Definition (2.8)**.** *Let $n \in \mathbb{N}$ and $F \in \operatorname{ob} \underline{C}_S$. We say that F is of n-$\underline{C}_0$-*finite presentation *if the family of objects X over S on which there exists an exact sequence*

$$E^{-n} \longrightarrow E^{-n+1} \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F_X \longrightarrow 0,$$

*with $E^i \in \operatorname{ob} \underline{C}_{0X}$ for $-n \leq i \leq 0$, is a refinement of S. We say that F is of ∞-$\underline{C}_0$-*finite presentation *if F is of n-$\underline{C}_0$-finite presentation for every $n \in \mathbb{N}$.*

When no ambiguity is possible, we shall simply say n-finite presentation. We shall say "finite presentation" for "1-finite presentation", and "finite type" for "0-finite presentation"; the latter notion coincides with the one introduced in (1.2).

Proposition (2.9)**.** *Let $n \in \mathbb{N}$ and $F \in \operatorname{ob} \underline{C}_S$. The following conditions are equivalent:*

(i) F is of n-finite presentation, respectively of ∞-finite presentation;

(ii) F is $(-n)$-pseudo-coherent, respectively pseudo-coherent.

When $\underline{S}$ is the punctual site, F is pseudo-coherent if and only if there exists an exact sequence

$$\cdots \longrightarrow E^{-n} \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F \longrightarrow 0$$

with $E^i \in \operatorname{ob} C_0$ for every i.

Proof. For the equivalence it is enough to treat the non-parenthetical case. The implication (i)⇒(ii) is immediate from the definitions. Conversely, (ii)⇒(i) follows at once from (2.4): the morphism $0 \to F$ is a quasi-isomorphism, and one obtains, after localizing, a quasi-isomorphism $E \to F$ with E strictly

n-pseudo-coherent and $E^i = 0$ for $i \leq -1$, that is, an n-finite presentation of F. The second assertion is proved in the same way, using (2.7.1).

Proposition (2.10)**.** a) *Let $m, n \in \mathbb{Z}$ with $n \leq m$. If $F \in \operatorname{ob} D^-(\underline{C}_S)$ is n-pseudo-coherent and acyclic in degrees $\geq m$, then locally there is a quasi-isomorphism $E \to F$ with E strictly n-pseudo-coherent and $E^i = 0$ for $i > m$. If $\underline{S}$ is punctual, and F is pseudo-coherent and acyclic in degrees $\geq m$, then there is a quasi-isomorphism $E \to F$ with E strictly pseudo-coherent and $E^i = 0$ for $i > m$.*

b) *Let $F \in \operatorname{ob} D^-(\underline{C}_S)$ be acyclic in degrees $> n$. Then F is n-pseudo-coherent if and only if $H^n(F)$ is of finite type.*

Proof. For a), since F is acyclic in degrees $> m$, the morphism $0 \to F$ is an m-quasi-isomorphism. The assertion follows from (2.4) and (2.7.1).

For b), there is a distinguished triangle in $D(\underline{C}_S)$

$$\begin{array}{ccc} & & F' \\ & \nearrow^{+1} & \downarrow \\ F & \longrightarrow & H^n(F)[-n]. \end{array}$$

To see this, first truncate F by replacing F^n by $Z^n(F)$ and by putting $F^i = 0$ for $i > n$. The long exact cohomology sequence gives $H^i(F') = 0$ for $i \geq n-1$, hence F' is $(n-1)$-pseudo-coherent. The conclusion follows from (2.5) and (2.6).

Proposition (2.11)**.** *Let $n \in \mathbb{Z}$.*

a) *Let $F \in \operatorname{ob} D(\underline{C}_S)$. If, for every $i \geq n$, respectively for every i, the component F^i is of $(i-n)$-finite presentation, respectively of ∞-finite presentation, then F is n-pseudo-coherent, respectively pseudo-coherent.*

b) *Let $F \in \operatorname{ob} D^-(\underline{C}_S)$. If, for every $i \geq n$, respectively for every i, the cohomology object $H^i(F)$ is of $(i-n)$-finite presentation, respectively of ∞-finite presentation, then F is pseudo-coherent.*

Proof. It is enough to treat the non-parenthetical assertions. For a), reason by induction on an integer N such that $F^i = 0$ for $i > n + N$. If $N = 0$, the assertion follows immediately from (2.10 b). If $N \geq 1$, one has an exact sequence

$$0 \longrightarrow F''[-n-N] \longrightarrow F \longrightarrow F' \longrightarrow 0.$$

By induction, F' is n-pseudo-coherent; on the other hand $F''[-n-N]$ is n-pseudo-coherent by the hypothesis, (2.9), and (2.5 a). Hence F is n-pseudo-coherent by (2.5 b).

For b), reason by induction on N such that $H^i(F) = 0$ for $i > n + N$. The case $N = 0$ is (2.10 b). If $N \geq 1$, there is a distinguished triangle

$$\begin{array}{ccc} & & F' \\ & \nearrow & \downarrow \\ F & \longrightarrow & H^{n+N}(F)[-n-N]. \end{array}$$

The long exact sequence gives

$$H^i(F') = 0 \quad \text{for } i \geq n+1,$$

and $H^i(F') \xrightarrow{\sim} H^i(F)$ for $i \le n$. By the induction hypothesis, applied with $n+1$ in place of n, F' is $(n+1)$-pseudo-coherent. Also $H^{n+N}(F)[-n-N]$ is n-pseudo-coherent by (2.9) and (2.5 a). The conclusion follows from (2.6).

Remark 2.11.1. If F is pseudo-coherent, the components F^i, and even the cohomology objects $H^i(F)$, need not in general be pseudo-coherent, nor even of finite type. In the next section, however, in the sorites on coherence, we shall see important examples where pseudo-coherence of F implies pseudo-coherence of the $H^i(F)$.

Proposition (2.12)**.** *Let $F', F'' \in \operatorname{ob} D(\underline{C}_S)$ and put $F = F' \oplus F''$. Then F is n-pseudo-coherent, respectively pseudo-coherent, if and only if F' and F'' are n-pseudo-coherent, respectively pseudo-coherent.*

Proof. It is enough to prove the non-parenthetical assertion. Sufficiency is obvious. For necessity, suppose by induction that F' and F'' have already been shown to be $(n+1)$-pseudo-coherent. After localizing, suppose even that F' and F'' are strictly $(n+1)$-pseudo-coherent and that $F = F' \oplus F''$ in $C(\underline{C}_S)$. Let F'_+, respectively F'_-, be the complex obtained from F' by replacing F'^i by 0 for $i \ge n$, respectively for $i \le n$; define F''_+ and F''_- similarly. There is an exact sequence

$$0 \longrightarrow F'_+ \oplus F''_+ \longrightarrow F \longrightarrow F'_- \oplus F''_- \longrightarrow 0.$$

Since $F'_+ \oplus F''_+$ is strictly perfect and F is n-pseudo-coherent, (2.6) implies that $F'_- \oplus F''_-$ is n-pseudo-coherent. By (2.10 b), this means that $H^n(F'_- \oplus F''_-)$ is of finite type. But

$$H^n(F'_- \oplus F''_-) = H^n(F'_-) \oplus H^n(F''_-),$$

so $H^n(F'_-)$ and $H^n(F''_-)$ are of finite type. Again by (2.10 b), F'_- and F''_- are n-pseudo-coherent. Applying (2.5) to

$$0 \longrightarrow F'_+ \longrightarrow F' \longrightarrow F'_- \longrightarrow 0$$

and similarly for F'', one concludes that F' and F'' are n-pseudo-coherent.

Proposition (2.13)**.** *Let $(\underline{S}', \underline{C}', \underline{C}'_0)$ be a triple satisfying the same hypotheses (2.0) as $(\underline{S}, \underline{C}, \underline{C}_0)$. Let $f : \underline{S}' \to \underline{S}$ be a morphism of sites, corresponding to an underlying functor $\psi : \underline{S} \to \underline{S}'$, and let*

$$u : D^*(\underline{C}) \longrightarrow D^*(\underline{C}') \qquad (* = -, +, b)$$

be a Cartesian S-functor above f. Assume that u is exact on each fibre and that there exist integers a, b satisfying the following conditions for every $S \in \operatorname{ob} \underline{S}$:

(i) if $E \in \operatorname{ob} \underline{C}_{0S}$, then $u(E)$ is a-$\underline{C}'_0$-pseudo-coherent, respectively $\underline{C}'_0$-pseudo-coherent;

(ii) if $F \in \operatorname{ob} D^*(\underline{C}_S)$ is acyclic in degrees > 0, then $u(F)$ is acyclic in degrees $\le b$.

Then, if $F \in \operatorname{ob} D^*(\underline{C}_S)$ is m-pseudo-coherent relative to $\underline{C}_0$ and acyclic in degrees $> n$, the object $u(F) \in \operatorname{ob} D^*(\underline{C}'_{S'})$, where $S' = f^{-1}(S)$, is

$$\sup(m+a, n+b)\text{-}\textit{pseudo-coherent}$$

relative to $\underline{C}'_0$, respectively $(m+b)$-pseudo-coherent, and is acyclic in degrees $> n+b$. In particular, in the parenthetical case, if F is pseudo-coherent relative to $\underline{C}_0$, then $u(F)$ is pseudo-coherent relative to $\underline{C}'_0$.

Proof. It is enough to treat the non-parenthetical case. First suppose that F is strictly perfect and that $F^i = 0$ for $i > n$. We prove that $u(F)$ is $(n+a)$-$\underline{C}'_0$-pseudo-coherent by induction on the number N of indices i for which $F^i \neq 0$. If $N = 1$, then $F = F^p[-p]$ for some $p \leq n$, and $u(F)$ is $(p+a)$-$\underline{C}'_0$-pseudo-coherent, hence a fortiori $(n+a)$-pseudo-coherent. If $N > 1$, write F in a short exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

where F' and F'' are strictly perfect of length $< N$ and vanish in degrees $> n$. Applying (2.5 b) to the corresponding triangle gives the claim by induction.

Now let F be m-$\underline{C}_0$-pseudo-coherent and acyclic in degrees $> n$. After localizing and using (2.10 a), we may suppose that F is strictly m-pseudo-coherent and that $F^i = 0$ for $i > n$. There is an exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

with F'' zero in degrees $> m$, and F' strictly perfect and zero in degrees $> n$. In the corresponding distinguished triangle for $u(F)$, the object $u(F'')$ is acyclic in degrees $> m+b$, hence $(m+b)$-pseudo-coherent, while $u(F')$ is $(n+a)$-pseudo-coherent by the preceding paragraph. The conclusion follows again from (2.5 b).

Corollary (2.14)**.** *Let $\underline{C}_1$ be a strictly full additive sub-$\underline{S}$-category of $\underline{C}$ such that $\underline{C}_0 \subset \underline{C}_1 \subset \underline{C}$. Assume that every object of $\underline{C}_1$ is $\underline{C}_0$-pseudo-coherent. Then:*

(i) $\underline{C}_1$ is locally quasi-liftable in $\underline{C}$ and locally quasi-stable under kernels of epimorphisms;

(ii) for $S \in \operatorname{ob} \underline{S}$ and $F \in \operatorname{ob} D(\underline{C}_S)$, F is n-pseudo-coherent, respectively pseudo-coherent, relative to $\underline{C}_1$ if and only if it is so relative to $\underline{C}_0$.

Proof. For (i), let $E \to F$ be an epimorphism in $\underline{C}_S$ with $F \in \operatorname{ob} \underline{C}_{1S}$. Since F is $\underline{C}_0$-pseudo-coherent, hence in particular of $\underline{C}_0$-finite type, after localizing there is an epimorphism $E' \to F$ with $E' \in \operatorname{ob} \underline{C}_{0S}$. Using quasi-liftability of $\underline{C}_0$ in $\underline{C}$, one may, after localizing again, suppose that this epimorphism factors through E; this proves quasi-liftability of $\underline{C}_1$. Similarly, if $E \in \operatorname{ob} \underline{C}_{1S}$ admits a finite right resolution by objects of $\underline{C}_0$, then E is $\underline{C}_0$-pseudo-coherent by (2.11 a), hence of $\underline{C}_0$-finite type.

For (ii), it is enough to prove necessity. One may apply either (2.11 a) or (2.13), with $\underline{S}' = \underline{S}$, $\underline{C}' = \underline{C}$, $f = \mathrm{Id}$, $u = \mathrm{Id}$, and with $\underline{C}_0$ and $\underline{C}'_0$ replaced by $\underline{C}_1$ and $\underline{C}_0$ respectively.

Scholium 2.14.1. There is therefore a largest strictly full additive sub-$\underline{S}$-category of $\underline{C}$ giving the same notions of n-pseudo-coherence and pseudo-coherence as $\underline{C}_0$, namely the subcategory formed by the $\underline{C}_0$-pseudo-coherent objects. This category is locally stable under kernels of epimorphisms.

Example 2.15. Let $(\underline{S}, A)$ be a ringed topos, and let $\underline{C}, \underline{C}_0$ be the $\underline{S}$-categories considered in (1.3.d). The fibred category $D(\underline{C})$ will often be denoted $D(\underline{S}, A)$, or $D(\underline{S})$, or even $D(S)$ if there is no ambiguity about A. If S is the final object of $\underline{S}$, an object of $D(S)$ will be called n-pseudo-coherent, respectively pseudo-coherent, if it is n-$\underline{C}_0$-pseudo-coherent, respectively $\underline{C}_0$-pseudo-coherent. The same convention applies to the local notions. By (2.14) the same notion is obtained if $\underline{C}_0$ is replaced by the $\underline{S}$-category whose fibre over U is the category of A_U-modules locally direct summands of finite free modules. A direct summand of a finite free module is not, in general, locally free of finite type; nevertheless one has the following theorem.

Theorem (2.15.1)**.** *Assume that $\underline{S}$ satisfies one of the following two conditions:*

(i) $\underline{S}$ has enough points, and for every point s of $\underline{S}$, the finitely generated projective left A_s-modules are free. This last condition holds, for example, if $A_s/\operatorname{rad}(A_s)$ is a field, or if $\operatorname{rad}(A_s)$ is nilpotent.

(ii) $(\underline{S}, A)$ is a locally ringed topos in the sense of Giraud: A is commutative and, for every $X \in \operatorname{ob}\underline{S}$ and every $f \in \Gamma(X, A)$, one has $X = X_f \amalg X_{1-f}$, where X_f, respectively X_{1-f}, denotes the largest subobject of X on which f, respectively $1-f$, is invertible.

Then every direct summand of a finite free left A-module is locally free of finite type.

Proof. Assume first (i). If E and F are A-modules and E is of finite presentation, the canonical map

$$\operatorname{Hom}(E, F)_s \longrightarrow \operatorname{Hom}(E_s, F_s)$$

is an isomorphism for every point s of S. It follows that if E and F are both of finite presentation and $E_s \simeq F_s$ at a point s, then there exists an s-pointed object U of $\underline{S}$ such that $E|U \simeq F|U$. The conclusion is immediate.

Assume (ii). Let $n \in \mathbb{N}$ and let D^n be the functor on locally ringed topoi defined by

$$T \longmapsto \Gamma(T, \mathcal{O}_T)^n.$$

It is known that the functor $\underline{\operatorname{End}}_{\mathcal{O}_T}(D^n)$ is represented by $X_n = \operatorname{Spec}\mathbb{Z}[T_{ij}]$ with its endomorphism u of $\mathcal{O}^n_{X_n}$ defined by the matrix (T_{ij}). Let $p : A^n \to A^n$ be an idempotent. We must show that $p(A^n)$ is locally free of finite type. The preceding representability gives a morphism $s : \underline{S} \to X_n$ of locally ringed topoi such that $s^*(u) = p$. Since the corresponding matrix is idempotent, s factors through the closed subscheme $M \subset X_n$ of idempotent matrices. Hence $p = s'^* i^*(u)$, and the claim follows because, by the case already proved, the image of $i^*(u)$ is locally free of finite type.

Here an s-pointed object means one through which the point s factors, i.e. an object U with s factoring through $S_{/U}$.

Corollary (2.16)**.** *Let $f : (\underline{S}', A') \to (\underline{S}, A)$ be a morphism of ringed topoi, defined by a functor $\psi : \underline{S} \to \underline{S}'$ and a morphism of sheaves of rings $\psi^*(A) \to A'$. Let $D(\underline{S}', A')$ denote the derived category of A'-A-bimodules, left over A' and right over A, and let*

$$\otimes_A : D(\underline{S}', A') \times D(\underline{S}, A) \longrightarrow D(\underline{S}', A')$$

be the derived functor of $(E', E) \mapsto E' \otimes_{\psi^* A} \psi^*(E)$.

Let $X \in \operatorname{ob}\underline{S}$, $X' = \psi(X)$, $E' \in \operatorname{ob} D(\underline{S}', A')$, and $E \in \operatorname{ob} D(\underline{S}, A)$. If E' is m'-pseudo-coherent and acyclic in degrees $> n'$ as an object of $D(\underline{S}', A')$, and if E is m-pseudo-coherent and acyclic in degrees $> n$, then $E' \otimes_A E$ is $\sup(m+n', m'+n)$-pseudo-coherent and acyclic in degrees $> n+n'$.

Proof. Replacing $\underline{S}$ by $\underline{S}_{/X}$, one may suppose $X = S$. Apply (2.13) to the functor

$$u = E' \otimes_A - : D(\underline{S}, A) \longrightarrow D(\underline{S}', A').$$

If $U \in \operatorname{ob} \underline{S}$, $U' = \psi(U)$, and E is a complex of left A-modules which is acyclic in degrees $> n$, then $E' \otimes_A E$ is acyclic in degrees $> n + n'$. Moreover, for $E = (A_U)^\sharp$, one has $u(E) = (E'_{U'})^\sharp$, which is m'-pseudo-coherent by hypothesis and (2.12). Thus the hypotheses of (2.13) are satisfied with $a = m'$ and $b = n'$.

Corollary (2.16.1)**.** *Under the hypotheses of (2.16):*

a) *The functor $Lf^* : D^-(\underline{S}, A) \to D^-(\underline{S}', A')$, which is the functor $A' \otimes_A -$, induces a functor*

$$Lf^* : D(\underline{S}, A)_{\mathrm{coh}} \longrightarrow D(\underline{S}', A')_{\mathrm{coh}}.$$

b) *If A is commutative, the functor $\otimes_A : D^-(\underline{S}) \times_{\underline{S}} D^-(\underline{S}) \to D^-(\underline{S})$ induces a functor*

$$\otimes_A : D(\underline{S})_{\mathrm{coh}} \times_{\underline{S}} D(\underline{S})_{\mathrm{coh}} \longrightarrow D(\underline{S})_{\mathrm{coh}}.$$

Exercise 2.16.2. Let $\underline{S}$ be a topos and let $A \to A'$ be a morphism of rings of $\underline{S}$. Assume either:

(i) $A \to A'$ is surjective with nilpotent kernel;

(ii) the functor $A' \otimes_A -$ on left A-modules is exact and faithful.

Show that $E \in \operatorname{ob} D(\underline{S}, A)$ is n-pseudo-coherent, respectively pseudo-coherent, if and only if $A' \otimes_A E \in \operatorname{ob} D(\underline{S}, A')$ is n-pseudo-coherent, respectively pseudo-coherent.

Indication. First show that a left A-module L is of finite type if and only if $A' \otimes_A L$ is of finite type, and then reduce to this case by truncation, using (2.5) and (2.10 b).

3. Relation with the classical notion of coherence

3.0

The hypotheses and notation are those of (2.0), except that $\underline{C}_0$ is no longer assumed locally quasi-stable under kernels of epimorphisms.

Definition (3.1)**.** *Let $F \in \operatorname{ob} \underline{C}_S$. We say that F is* precoherent *relative to $\underline{C}_0$ if, for every object U over S and every morphism $u : E \to F|U$ with $E \in \operatorname{ob} \underline{C}_{0U}$, the kernel of u is of $\underline{C}_0$-finite type (1.2). We say that F is* coherent *relative to $\underline{C}_0$ if F is precoherent and of finite type.*

The notion of precoherence is introduced here only as a technical intermediary; the notion to retain is coherence.

Remark 3.2. It is clear from the definitions that if F is precoherent, respectively of finite type, then every subobject, respectively every quotient, of F is precoherent, respectively of finite type.

Lemma (3.3)**.** a) *Let*

$$0 \longrightarrow F \longrightarrow G \longrightarrow H$$

be an exact sequence in $\underline{C}_S$*. If* G *is of finite type and* H *is precoherent, then* F *is of finite type.*

b) *Let*

$$F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be exact in $\underline{C}_S$*. If* F *is of finite type and* G *is precoherent, then* H *is precoherent.*

c) *Let*

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be exact. If F *and* H *are precoherent, respectively of finite type, then so is* G*.*

Proof. For a), after localizing, choose an epimorphism $G' \to G$ with $G' \in \operatorname{ob} \underline{C}_{0S}$. This gives a commutative diagram with exact rows

$$\begin{array}{ccccc} 0 & \longrightarrow & F' & \longrightarrow & G' \\ & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F & \longrightarrow & G \longrightarrow H. \end{array}$$

Since H is precoherent, F' is of finite type; since $F' \to F$ is an epimorphism, F is of finite type.

For b), let U be over S and let $H' \to H|U$ be a morphism with $H' \in \operatorname{ob} \underline{C}_{0U}$. After changing notation, suppose $U = S$. By quasi-liftability of $\underline{C}_0$ in $\underline{C}$, and after localizing, there is a commutative square

$$\begin{array}{ccc} G' & \longrightarrow & H' \\ \downarrow & & \downarrow \\ G & \longrightarrow & H, \end{array}$$

where the horizontal arrows are epimorphisms and $G' \in \operatorname{ob} \underline{C}_{0S}$. Since $\operatorname{Ker}(f)$ is a quotient of $\operatorname{Ker}(fp)$, it is enough to show that $\operatorname{Ker}(fp)$ is of finite type; one may therefore suppose $H' = G'$. The maps $F \to G$ and $H' \to H$ define $g : F \oplus H' \to G$. If $K = \operatorname{Ker}(G \to H)$, the snake lemma applied to the natural diagram with exact rows shows that $\operatorname{Ker}(g) \to \operatorname{Ker}(f)$ is an epimorphism. But $F \oplus H'$ is of finite type and G is precoherent, so $\operatorname{Ker}(g)$ is of finite type by a). This proves b).

For c), first prove the assertion for precoherence. Up to notation, one must show that if $\varphi : G' \to G$ is a morphism with $G' \in \operatorname{ob} \underline{C}_{0S}$, then $\operatorname{Ker}(\varphi)$ is of finite type. Let H' be the image of $G' \to H$. The snake lemma applied to

$$\begin{array}{ccccc} 0 & \longrightarrow & F' & \longrightarrow & G' \longrightarrow H' \longrightarrow 0 \\ & & \downarrow & & \downarrow \varphi \quad \downarrow h \\ 0 & \longrightarrow & F & \longrightarrow & G \longrightarrow H \longrightarrow 0 \end{array}$$

with h a monomorphism, identifies $\operatorname{Ker}(f)$ with $\operatorname{Ker}(\varphi)$. Since F' is of finite type by a), another application of a) gives the required finite type property. The finite-type assertion follows by choosing, after localizing, epimorphisms $F' \twoheadrightarrow F$ and $H' \twoheadrightarrow H$ with $F', H' \in \operatorname{ob} \underline{C}_{0S}$, and then by lifting further so that the map to H factors through G; the resulting epimorphism $F' \oplus H' \to G$ proves that G is of finite type.

Corollary (3.4)**.** *The strictly full subcategory of* $\underline{C}_S$ *formed by the coherent objects is stable under finite projective and finite inductive limits. In particular, the kernel, cokernel, and image of a morphism between* *two coherent objects are coherent. If*

$$E^0 \longrightarrow E^1 \longrightarrow E^2 \longrightarrow E^3 \longrightarrow E^4$$

is exact and if E^0, E^1, E^3, E^4 are coherent, then E^2 is coherent.

Corollary (3.5)**.** *Assume that the objects of $\underline{C}_0$ are $\underline{C}_0$-coherent. This implies in particular that $\underline{C}_0$ is locally quasi-stable under kernels of epimorphisms, so that the hypotheses of (2.0) hold.*

a) *For $F \in \operatorname{ob} \underline{C}_S$, the following conditions are equivalent:*

(i) F is of finite presentation;

(ii) F is coherent;

(iii) F is of ∞-finite presentation (2.8).

b) *For $F \in \operatorname{ob} D(\underline{C}_S)$ and $n \in \mathbb{Z}$, one has*

$$F \textit{ is } n\textit{-pseudo-coherent} \Longleftrightarrow H^i(F) \textit{ is coherent for } i \geq n+1,$$

and $H^n(F)$ is of finite type.

In particular, F is pseudo-coherent if and only if $H^n(F)$ is coherent for every n.

Proof. The assertion a), and the implication $\Leftarrow$ of b), follow directly from (3.4). The implication $\Rightarrow$ in b) follows immediately from a) and (2.11 b).

Examples 3.6. Let $(\underline{S}, A)$ be a ringed topos, and let $\underline{C}_0 \subset \underline{C}$ be the $\underline{S}$-categories of (1.3.d), cf. also (2.15). The objects of $\underline{C}_0$ are coherent if and only if A is coherent, i.e. precoherent. This holds, for example, in the following cases:

- A is constant with value a left noetherian ring;
- $\underline{S}$ is the topos of sheaves of sets for the Zariski topology on a locally noetherian scheme S, and $A = \mathcal{O}_S$;
- $\underline{S}$ is the topos of sheaves of sets on a complex analytic space S, and $A = \mathcal{O}_S$.

By contrast, the sheaf of continuous complex-valued functions on a topological space is in general not coherent.

If $(\underline{S}, A)$ is defined by a ringed space, the notion of coherence considered here coincides with Serre's usual notion. In accordance with the newer conventions, the word "scheme" is used for what older texts called a "prescheme", and "separated scheme" for what older texts called a "scheme".

Proposition (3.7)**.** a) *Let $f : (\underline{S}', A') \to (\underline{S}, A)$ be a morphism of ringed topoi. Assume that f is flat on the right, i.e. that $f^* = A' \otimes_A -$ is exact. For $F \in \operatorname{ob} D(\underline{S}, A)$ and $G \in \operatorname{ob} D^-(\underline{S}, A)$ there is a canonical functorial morphism*

$$f^* \mathrm{R}\underline{\mathrm{Hom}}(F, G) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(f^*F, f^*G).$$

It is an isomorphism when F is pseudo-coherent.

b) *Assume A commutative and coherent, and let $F \in \operatorname{ob} D(\underline{S})$ and $G \in \operatorname{ob} D^-(\underline{S})$. If $H^i(F)$ and $H^i(G)$ are coherent for every i, then so is $\underline{\mathrm{Ext}}^i(F, G)$.*

Proof. For a), to define the morphism one may suppose G has injective components, and then take the composite

$$f^*\underline{\mathrm{Hom}}(F, G) \longrightarrow \underline{\mathrm{Hom}}(f^*F, f^*G) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(f^*F, f^*G).$$

We prove that it is an isomorphism for F pseudo-coherent. Let $T = f^*\mathrm{R}\underline{\mathrm{Hom}}(-, G)$ and $T' = \mathrm{R}\underline{\mathrm{Hom}}(f^*(-), f^*G)$, and choose $a \in \mathbb{Z}$ such that $H^i(G) = 0$ for $i < a$. The assertion is clear for F strictly perfect. Also, if F is acyclic in degrees $> n$, then TF and $T'F$ are acyclic in degrees $< a - n$. Since the question is local, we may suppose that F is strictly n-pseudo-coherent, so that it fits into a distinguished triangle

$$\begin{tikzcd} & F' \arrow[d] \\ F'' \arrow[ur] \arrow[r] & F, \end{tikzcd}$$

with F' strictly perfect and F'' acyclic in degrees $> n-1$. The induced morphism of distinguished triangles

$$\begin{tikzcd} TF'' \arrow[r] \arrow[d] & TF \arrow[r] \arrow[d] & TF' \arrow[r] \arrow[d] & {} \\ T'F'' \arrow[r] & T'F \arrow[r] & T'F' \arrow[r] & {} \end{tikzcd}$$

has an isomorphism on the right, while the left terms are acyclic in degrees $< a - n$. Therefore $TF \to T'F$ is an isomorphism on cohomology in degrees $< a - n$. Since n can be taken arbitrarily negative, the morphism is an isomorphism.

For b), the proof is left to the reader; see, for example, Hartshorne, III, 15.3.5.

Remark 3.7.1. A different compatibility of $\mathrm{R}\underline{\mathrm{Hom}}$ with Lf^* will be given below in (7.1.2).

Corollary (3.7.2)**.** *Let $(\underline{S}, A)$ be a ringed topos, let s be a point of $\underline{S}$, and let $E, F \in \operatorname{ob} D(\underline{S}, A)_{\mathrm{coh}}$. If $E_s \simeq F_s$, then there exists an s-pointed object U of $\underline{S}$ such that $E|U \simeq F|U$.*

Proof. Let $u : E \to F$ and $v : F \to E$ be inverse isomorphisms on the fibre at s. By (2.2 a) and the compatibility of Hom with fibres for coherent objects, one has

$$\mathrm{Hom}(E, F)_s \simeq \mathrm{Hom}(E_s, F_s), \qquad \mathrm{Hom}(F, E)_s \simeq \mathrm{Hom}(F_s, E_s).$$

Thus, on a suitable neighbourhood U of s, there exist sections u and v of the corresponding Hom sheaves. The endomorphisms vu of $E|U$ and uv of $F|U$ induce the identity at s; after passing to a further neighbourhood V of s, they are the identity. Hence $E|V \simeq F|V$.

4. Perfect complexes

4.0

In this section $\underline{S}$ denotes a site, S an object of $\underline{S}$, $\underline{C}$ a flat abelian $\underline{S}$-category, and $\underline{C}_0$ a strictly full additive sub-$\underline{S}$-category of $\underline{C}$. Unless otherwise specified, $\underline{C}_0$ is assumed locally liftable in $\underline{C}$ and locally stable under kernels of epimorphisms.

Lemma (4.1)**.** *Let $u : E \to F$ be a morphism in $C(\underline{C}_S)$. If E is strictly $\underline{C}_0$-perfect and F is acyclic, then u is locally homotopic to zero.*

This lemma was communicated to the editor by D. Ferrand.

Proof. We must construct locally a homotopy k with $u = dk + kd$. Since E is strictly perfect, there exist $a \leq b$ with $E^i = 0$ for $i \notin [a, b]$; take $k^i = 0$ outside this range. Construct $k^i : E^i \to F^{i-1}$ by descending induction on i. Suppose k^i has been constructed so that

$$u^j = d^{j-1}k^j + k^{j+1}d^j \quad (j > i \geq a).$$

Then $u^i - d^{i-1}k^i$ factors through $Z^i(F)$. Since F is acyclic, $Z^i(F) = B^i(F)$. Because $\underline{C}_0$ is locally liftable in $\underline{C}$, after localizing there exists $k^{i-1} : E^{i-1} \to F^{i-2}$ such that

$$u^i = d^{i-1}k^i + k^{i+1}d^i.$$

The induction gives the required homotopy.

Corollary (4.2)**.** *Let*

$$\begin{tikzcd} P \arrow[r, "p"] \arrow[dr] & F \\ & E' \arrow[u, "\wr"'] \end{tikzcd}$$

be a diagram in $C(\underline{C}_S)$, where P is strictly $\underline{C}_0$-perfect and the vertical arrow is a quasi-isomorphism. The family of X over S for which there exists a morphism $u : P_X \to E_X$ in $C(\underline{C}_X)$ making the evident square commute in $K(\underline{C}_X)$ is a refinement of S.

Proof. Let G be the cone of the quasi-isomorphism $E' \to F$. The exact sequence

$$\cdots \to \mathrm{Hom}_{K(\underline{C}_S)}(P, E') \to \mathrm{Hom}_{K(\underline{C}_S)}(P, F) \to \mathrm{Hom}_{K(\underline{C}_S)}(P, G) \to \cdots$$

shows that the obstruction to lifting p lies in $\mathrm{Hom}_K(P, G)$. Since G is acyclic, this obstruction is locally zero by (4.1). Thus, locally, the required lift exists.

Corollary (4.3)**.** *Let $f : E \to F$ be a quasi-isomorphism in $C(\underline{C}_S)$ with F strictly perfect. The family of X over S for which there exists a quasi-isomorphism $g : F_X \to E_X$ such that $f_X g$ is homotopic to the identity is a refinement of S.*

Proof. This is the special case of (4.2) obtained by taking $P = F$ and $p = \mathrm{Id}_F$.

Scholium 4.4. Let $F \in \operatorname{ob} K(\underline{C}_S)$. Using, for instance, a cleavage of $\underline{C}$, one can define a category (Qis/F) fibred over $\underline{S}_{/S}$ whose fibre at X over S is the opposite of the category of quasi-isomorphisms with target F_X. Then (4.3) says that, if F is strictly perfect, the morphism Id_F is locally cofinal in (Qis/F).

Notation 4.5. Let $\underline{M}$ be an $\underline{S}$-fibred category. For $X \in \operatorname{ob} \underline{S}$ and $E, F \in \operatorname{ob} \underline{M}_X$, write $\underline{\mathrm{Hom}}_X(E, F)$ for the sheaf on $\underline{S}_{/X}$ associated with the presheaf

$$U \longmapsto \mathrm{Hom}_{\underline{M}_U}(E|U, F|U).$$

Even when $\underline{C}$ is a stack, the corresponding presheaf for $\underline{M} = K(\underline{C})$ or $D(\underline{C})$ is not in general a sheaf.

Corollary (4.6)**.** *Let $E, F \in \operatorname{ob} C(\underline{C}_S)$, with E strictly perfect. The canonical map*

$$\underline{\mathrm{Hom}}_{K(\underline{C}_S)}(E, F) \longrightarrow \underline{\mathrm{Hom}}_{D(\underline{C}_S)}(E, F)$$

is an isomorphism.

Proof. This follows immediately from (4.4) and from the formula

$$\mathrm{Hom}_{D(\underline{C}_X)}(E_X, F_X) = \varinjlim_{X' \to X \in (\mathrm{Qis}/E_X)^\circ} \mathrm{Hom}_{K(\underline{C}_{X'})}(E_{X'}, F_{X'})$$

for X over S.

Definition (4.7)**.** *Let $F \in \operatorname{ob} D(\underline{C}_S)$ and let $a, b \in \mathbb{Z}$ with $a \le b$. We say that F has $\underline{C}_0$-*perfect amplitude contained in $[a, b]$*, and write*

$$\underline{C}_0\text{-}\mathit{parf}\text{-}\mathit{amp}(F) \subset [a, b],$$

if the family of X over S for which there exists a quasi-isomorphism $F' \to F$ in $C(\underline{C}_X)$ with F' strictly $\underline{C}_0$-perfect and $F'^i = 0$ for $i \notin [a, b]$, is a refinement of S. By (4.3), this is equivalent to the condition obtained by replacing "quasi-isomorphism in $C(\underline{C}_X)$" by "isomorphism in $D(\underline{C}_X)$". We say that F has finite $\underline{C}_0$-perfect amplitude if this holds for some a, b, and that F is $\underline{C}_0$-perfect if it is locally of finite $\underline{C}_0$-perfect amplitude.

When no confusion is possible, the subscript $\underline{C}_0$ will be omitted.

Example 4.8. Let $(\underline{S}, A)$ be a ringed topos. Let $\underline{C}$ be the $\underline{S}$-category whose fibre over U is the category of left A_U-modules, and let $\underline{C}_0$ be the sub-$\underline{S}$-category of left A_U-modules which are direct summands of finite free modules. By (1.3.1), the pair $(\underline{C}, \underline{C}_0)$ satisfies the hypotheses of (4.0). An object F of $D(\underline{S}, A)$ is said to have perfect amplitude contained in $[a, b]$, to have finite perfect amplitude, or to be perfect, if this holds relative to $\underline{C}_0$. When direct summands of finite free modules are locally free of finite type, to say that F is perfect means that F is locally isomorphic in the derived category to a bounded complex whose components are finite free modules.

Notation 4.9. We write

$$D(\underline{C})_{\underline{C}_0\text{-parf}} \quad \text{or} \quad \mathrm{Parf}(\underline{C}, \underline{C}_0)$$

for the strictly full sub-$\underline{S}$-category of $D(\underline{C})$ formed by the $\underline{C}_0$-perfect objects. In the situation of (4.8) we also write $D(\underline{S})_{A\text{-parf}}$, $\mathrm{Parf}(\underline{S}, A)$, or simply $\mathrm{Parf}(S)$ when A is clear.

Proposition (4.10)**.** *The category* $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ *is a triangulated sub-*$\underline{S}$*-category of* $D(\underline{C})$*. The canonical functor*

$$K^b(\underline{C}_0) \longrightarrow \mathrm{Parf}(\underline{C}, \underline{C}_0)$$

and similarly the induced functor on categories of distinguished triangles, induce $\underline{S}$*-equivalences on the associated stacks. Explicitly:*

(i) for every $X \in \mathrm{ob}\,\underline{S}$ *and* $E, F \in \mathrm{ob}\,K^b(\underline{C}_{0X})$*, the canonical map of Hom sets to* $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ *is an isomorphism, and the same holds for triangles;*

(ii) every object of $\mathrm{Parf}(\underline{C}, \underline{C}_0)$*, and similarly every triangle in it, is locally isomorphic in* $D(\underline{C})$ *to an object of* $K^b(\underline{C}_0)$*, respectively to a triangle in* $K^b(\underline{C}_0)$*.*

Proof. The non-triangular form of (i) is a special case of (4.6), and the non-triangular form of (ii) is true by definition. The category $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ is clearly stable under translation. If

$$\begin{array}{ccc} & & G \\ & \nearrow^{+1} & \downarrow \\ E & \xrightarrow{u} & F \end{array}$$

is a distinguished triangle with E and F perfect, then locally E and F may be assumed to lie in $K^b(\underline{C}_0)$. After further localizing, the morphism u may be represented by a morphism of $K^b(\underline{C}_0)$; then G is isomorphic to the cone of u, hence is perfect. This proves that $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ is triangulated and also proves the triangular version of (ii).

For the triangular version of (i), let $E = (E_1 \to E_2 \to E_3 \to E_1)$ and $F = (F_1 \to F_2 \to F_3 \to F_1)$ be distinguished triangles in $K(\underline{C}_{0X})$. The injectivity of the map of Hom sets follows from the non-triangular assertion. If $(\alpha_i) : E \to F$ is a morphism of triangles in $D(\underline{C})$, then locally each α_i is represented by a morphism in $K^b(\underline{C}_0)$. The differences measuring commutativity of the squares vanish in $D(\underline{C})$ and hence, locally, in $K(\underline{C}_0)$ by the non-triangular assertion. Thus the morphism of triangles is locally represented in $K(\underline{C}_0)$. This proves surjectivity.

Remark 4.10.1. In (4.10) one may replace $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ by $D^*(\underline{C})_{\underline{C}_0\text{-parf}} = D^*(\underline{C}) \cap \mathrm{Parf}(\underline{C}, \underline{C}_0)$ for $* = -$ or b.

Complement 4.11. The triangulated structure of $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ can be sharpened in terms of perfect amplitude. If E has perfect amplitude in $[a, b]$, then $E[n]$ has perfect amplitude in $[a - n, b - n]$. If

$$\begin{array}{ccc} & & G \\ & \nearrow & \downarrow \\ E & \longrightarrow & F \end{array}$$

is a distinguished triangle, the following table, analogous to (2.6), records the possible implications; the entries are perfect amplitudes:

E	F	G
$[a, b+1]$	$[a, b]$	$[a, b]$
$[a, b]$	$[a, b]$	$[a-1, b-1]$.

Corollary (4.12)**.** *Assume that $\underline{S}$ is the punctual site. Then $F \in \operatorname{ob} D(\underline{C})$ has perfect amplitude relative to $\underline{C}_0$ contained in $[a, b]$ if and only if there exists a quasi-isomorphism $F' \to F$ with F' strictly perfect and $F'^i = 0$ for $i \notin [a, b]$. The canonical functors*

$$K^b(\underline{C}_0) \longrightarrow \operatorname{Parf}(\underline{C}, \underline{C}_0), \qquad \operatorname{Tr}(K^b(\underline{C}_0)) \longrightarrow \operatorname{Tr}(\operatorname{Parf}(\underline{C}, \underline{C}_0))$$

are equivalences of categories.

Remark 4.12.1. When $\underline{S}$ is punctual, the objects of $\underline{C}_0$ are projective. If $\underline{C}_0$ contains all projective objects of $\underline{C}$, one says "projective amplitude" instead of "$\underline{C}_0$-perfect amplitude".

Remark 4.12.2. Let C be an abelian category, viewed as fibred over the punctual site, and let C_0 be a strictly full additive subcategory. Suppose that C_0 is stable under kernels of epimorphisms and quasi-liftable in C. An acyclic object of $K(C_0)$ is in general not zero, so the functor $K^b(C_0) \to D(C)$ cannot be faithful. However, if $K^{\mathrm{ac}}(C_0)$ denotes the thick subcategory of $K^b(C_0)$ formed by the complexes which are acyclic in C, the canonical functor

$$K^b(C_0)/K^{\mathrm{ac}}(C_0) \longrightarrow D(C)$$

is fully faithful. Indeed, one already knows from (2.7) that the functor to $D^b(C)$ is fully faithful; applying Verdier's criterion shows that the quotient functor has the required full faithfulness.

Lemma (4.13)**.** *Let $F \in \operatorname{ob} D(\underline{C}_S)$ and let $a, b, c \in \mathbb{Z}$ with $a \leq b$ and $a \leq c$. If F has perfect amplitude contained in $[a, c]$ and if $H^i(F) = 0$ for $i \geq b$, then F has perfect amplitude contained in $[a, d]$, where $d = \inf(b, c)$.*

Proof. We may suppose $b \leq c$ and that F is strictly perfect with $F^i = 0$ outside $[a, c]$. Then F is isomorphic to its truncation F' defined by $F'^i = F^i$ for $i \leq b$, $F'^i = 0$ for $i > b$, and $F'^b = Z^b(F^\bullet)$. The latter object has a finite right resolution by objects of $\underline{C}_0$, hence is locally an object of $\underline{C}_0$. This gives the claim.

Corollary (4.14)**.** *Let $F \in \operatorname{ob} \underline{C}_S$ and let $n \in \mathbb{N}$. The following conditions are equivalent:*

(i) *$\underline{C}_0$-parf-amp$(F) \subset [-n, 0]$;*

(ii) *locally, F admits a left resolution of length n by objects of $\underline{C}_{0S}$; that is, the family of X over S for which there is an exact sequence*

$$0 \longrightarrow L^{-n} \longrightarrow L^{-n+1} \longrightarrow \cdots \longrightarrow L^0 \longrightarrow F_X \longrightarrow 0$$

with all $L^i \in \operatorname{ob} \underline{C}_{0X}$ is a refinement of S.

Proof. This is an immediate consequence of (4.7) and (4.13).

Definition (4.14.1)**.** *We say that F has* perfect dimension $\leq n$, *and write* $\operatorname{parf.dim}(F) \leq n$, *if F satisfies the equivalent conditions of (4.14). The perfect dimension of F is the smallest such n, if it exists; otherwise it is infinite.*

When S is the punctual site and $\underline{C}_0$ is the full subcategory of projective objects of $\underline{C}$, perfect dimension is just the usual projective dimension.

Proposition (4.15)**.** *Let $a, b, n \in \mathbb{Z}$ with $a \leq b$ and $n \geq 0$, and let $F \in \operatorname{ob} D(\underline{C}_S)$ be such that $F^i = 0$, respectively $H^i(F) = 0$, for $i \notin [a, b]$. If, for every $i \in [a, b]$, the object F^i, respectively $H^i(F)$, has perfect dimension $\leq n + (i - a)$, then F has perfect amplitude contained in $[a - n, b]$. Consequently, if all F^i, respectively all $H^i(F)$, are perfect, then F is perfect.*

Proof. Proceed by induction on b, with a and n fixed, using truncations and (4.11), as in the proof of (2.11).

Lemma (4.16)**.** *Let $a, b \in \mathbb{Z}$ with $a \leq b$, and let $F \in \operatorname{ob} D(\underline{C}_S)$ be such that $F^i = 0$ for $i \notin [a, b]$. Assume that $\underline{C}_0$-parf-amp$(F) \subset [a, b]$ and that $F^i \in \operatorname{ob} \underline{C}_S$ for $i > a$. Then F^a is locally an object of $\underline{C}_{0S}$.*

Proof. Truncating F gives an exact sequence

$$0 \longrightarrow F' \longrightarrow F^\bullet \longrightarrow F^a[-a] \longrightarrow 0$$

with $\underline{C}_0$-parf-amp$(F') \subset [a+1, b+1]$. From (4.11) and (4.13) it follows that $\underline{C}_0$-parf-dim$(F') = 0$, i.e. F' is locally an object of $\underline{C}_{0S}$, and the assertion follows.

Proposition (4.17)**.** *Let $F', F'' \in \operatorname{ob} D(\underline{C}_S)$ and let $a, b \in \mathbb{Z}$, $a \leq b$. Consider:*

(i) $\underline{C}_0$-parf-amp$(F') \subset [a, b]$ and $\underline{C}_0$-parf-amp$(F'') \subset [a, b]$;

(ii) $\underline{C}_0$-parf-amp$(F' \oplus F'') \subset [a, b]$.

Then (ii) implies (i). If $\underline{C}_0$ is locally stable under direct summands, then the two conditions are equivalent.

Proof. The implication (ii)$\Rightarrow$(i) is a special case of (4.11). Conversely, put $F = F' \oplus F''$. Since F is pseudo-coherent, F' and F'' are pseudo-coherent by (2.12). Also $H^i(F) = 0$ for $i \notin [a, b]$ implies the same for F' and F''. By (2.10 a), after localizing one may suppose $F'^i, F''^i \in \operatorname{ob} \underline{C}_S$ for $i \geq a + 1$ and zero for $i > b$, and after truncating also zero for $i < a$. Since F has perfect amplitude in $[a, b]$, (4.16) shows that F^a is locally in $\underline{C}_{0S}$. But $F^a = F'^a \oplus F''^a$, hence F'^a and F''^a are locally in $\underline{C}_{0S}$ if $\underline{C}_0$ is locally stable under direct summands.

Proposition (4.18)**.** *Let $(\underline{S}', \underline{C}', \underline{C}'_0)$ satisfy the same hypotheses (4.0) as $(\underline{S}, \underline{C}, \underline{C}_0)$. Let $f : \underline{S}' \to \underline{S}$ be a morphism of sites, defined by $\psi : \underline{S} \to \underline{S}'$, and let*

$$u : D^*(\underline{C}) \longrightarrow D^*(\underline{C}') \qquad (* = -, +, b)$$

be an S-functor above f. If u sends the objects of $\underline{C}_0$ to $\underline{C}'_0$-perfect complexes, then u defines a functor

$$D^*(\underline{C})_{\underline{C}_0\text{-parf}} \longrightarrow D^*(\underline{C}')_{\underline{C}'_0\text{-parf}}.$$

Proof. Let $F \in \operatorname{ob} D^*(\underline{C})_{\underline{C}_0\text{-parf}}$. The question is local, so one may suppose F strictly perfect. The assertion follows by devissage, using the fact that $D^*(\underline{C}')_{\underline{C}'_0\text{-parf}}$ is a triangulated subcategory of $D^*(\underline{C}')$.

Remark 4.18.1. If there exist $m, n \in \mathbb{Z}$, $m \leq n$, such that $F \in \operatorname{ob} \underline{C}_{0S}$ implies

$$\underline{C}'_0\text{-parf-amp}(u(F)) \subset [m, n],$$

then

$$\underline{C}_0\text{-parf-amp}(F) \subset [a,b] \Longrightarrow \underline{C}'_0\text{-parf-amp}(u(F)) \subset [a+m, b+n].$$

The proof is an exercise, using (4.11).

Corollary (4.19)**.** *Under the hypotheses of (2.16), if $E' \in \operatorname{ob} D(\underline{S}', A')$ is perfect as an object of $D(\underline{S}', A')$ and if $E \in \operatorname{ob} D(\underline{S}, A)$ is perfect, then $E' \otimes_A E$ is perfect.*

Proof. One may suppose that $E' \in \operatorname{ob} D(\underline{S}', A')$, where S' is the final object of $\underline{S}'$, and apply (4.18) to $u = E' \otimes_A - : D(\underline{S}, A) \to D(\underline{S}', A')$.

Corollary (4.19.1)**.** *In particular, if $f : (\underline{S}', A') \to (\underline{S}, A)$ is a morphism of ringed topoi, the derived inverse image Lf^* sends perfect complexes to perfect complexes. If A is commutative, the derived tensor product of two perfect complexes is perfect.*

5. Finite Tor-dimension and perfection

5.0

The reader will have wondered what is missing from a pseudo-coherent complex in order for it to be perfect. Is it enough, for example, that it have bounded cohomology? The answer is no. Let $\underline{C}$ be the category of modules over a commutative noetherian ring A, and let $\underline{C}_0$ be the full subcategory of finite projective A-modules. Any finite A-module, viewed as a complex concentrated in degree zero, is pseudo-coherent and has bounded cohomology, but it is perfect only if it has finite projective dimension, equivalently finite Tor-dimension. This example suggests that the difference between pseudo-coherence and perfection is a cohomological-dimension condition. We now examine this in the category of modules on a ringed topos. The preliminary facts on Tor-dimension are recalled briefly; see Hartshorne, III, 4, and SGA 4, XVII, 4.9.

Proposition (5.1)**.** *Let $(\underline{S}, A)$ be a ringed topos, let $X \in \operatorname{ob} \underline{S}$, let $F \in \operatorname{ob} D^-(X)$, and let $m, n \in \mathbb{Z}$ with $m \leq n$. The following conditions are equivalent:*

(i) F is isomorphic in $D^-(X)$ to a complex F' such that each F'^i is flat and $F'^i = 0$ for $i \notin [m,n]$;

(ii) for every right A_X-module E, one has

$$H^i(E \otimes_{A_X} F) = 0 \quad \textit{for } i \notin [m,n].$$

Proof. The implication (i)⇒(ii) is trivial. The converse follows, by truncation, from the following lemma.

Lemma (5.1.1)**.** *Let $F \in \operatorname{ob} D^-(X)$. Assume that F satisfies condition (ii) of (5.1), that $F^i = 0$ for $i \notin [m,n]$, and that F^i is flat for $i \neq m$. Then F^m is flat.*

Proof. There is a distinguished triangle

$$\begin{array}{ccc} & & F^m[-m] \\ & \nearrow & \downarrow \\ F' & \longrightarrow & F, \end{array}$$

where F' has flat components and is zero outside $[m+1,n]$. Apply $E \otimes_{A_X} -$ for an arbitrary right A_X-module E, and write the associated exact cohomology sequence.

Definition (5.2)**.** *An object F of $D^-(X)$ satisfying the equivalent conditions of (5.1) is said to have* flat amplitude *contained in $[m,n]$, and we write*

$$\text{tor-amp}(F) \subset [m,n].$$

We say that F has finite flat amplitude, or finite Tor-dimension, if this holds for some $m \leq n$.

5.3. For $X \in \operatorname{ob} \underline{S}$, denote by $D^-(X)_{\text{tordf}}$ the full subcategory of $D^-(X)$ formed by the objects of finite Tor-dimension. By criterion (ii) of (5.1), it is a triangulated subcategory of $D^-(X)$. As X varies, the categories $D^-(X)_{\text{tordf}}$ form a triangulated sub-$\underline{S}$-category of $D(\underline{S})$, denoted $D(\underline{S})_{\text{tordf}}$. The tensor product induces an exact $\underline{S}$-functor

$$\otimes_A : D(\underline{S})_{\text{tordf}} \times_{\underline{S}} D(\underline{S}) \longrightarrow D(\underline{S}).$$

For a left A-module F, to have flat amplitude contained in $[-n,0]$ is equivalent to having Tor-dimension $\leq n$ in the ordinary sense, i.e. to admitting a left resolution of length n by flat A-modules.

5.4. Let $X \in \operatorname{ob} \underline{S}$ and $m \leq n$. An object $F \in \operatorname{ob} D^-(X)$ has flat amplitude contained in $[m,n]$ if and only if there is a covering family $(X_i \to X)$ such that $F|_{X_i}$ has flat amplitude contained in $[m,n]$ for all i. In particular, if $\underline{S}_{/X}$ is of finite type in the sense of SGA 4, VI, 1.7, and if F is locally of finite Tor-dimension, then F is of finite Tor-dimension.

Proposition (5.5)**.** *Under the hypotheses of (2.16), if E' has flat amplitude contained in $[m',n']$ as an object of $D(\underline{S}',A')$ and E has flat amplitude contained in $[m,n]$, then $E' \otimes_A E$ has flat amplitude contained in the corresponding interval obtained by adding the bounds.*

Proof. This is immediate from the form (5.1)(i) of finite Tor-dimension and from the fact that if E is a flat left A-module, then ψ^*E is a flat left ψ^*A-module.

Corollary (5.5.1)**.** a) *The functor $Lf^* : D(\underline{S},A) \to D(\underline{S}',A')$ induces*

$$Lf^* : D(\underline{S},A)_{\text{tordf}} \longrightarrow D(\underline{S}',A')_{\text{tordf}}.$$

b) *If A is commutative, the tensor product induces*

$$\otimes_A : D(\underline{S})_{\text{tordf}} \times_{\underline{S}} D(\underline{S}) \longrightarrow D(\underline{S})_{\text{tordf}}.$$

Remark 5.5.2. Let $E \in \operatorname{ob} D^-(\underline{S},A)$, with S the final object of $\underline{S}$, and let $m \leq n$. If E has flat amplitude contained in $[m,n]$, then so does every fibre E_s at a point s of $\underline{S}$. The converse holds if $\underline{S}$ has enough points, using criterion (5.1)(ii) and the fact that tensor product commutes with passage to fibres.

Exercises 5.6. a) Assume A commutative and $(\underline{S}',A')$ faithfully flat over $(\underline{S},A)$. Then $E \in \operatorname{ob} D^-(\underline{S},A)$ has flat amplitude contained in $[m,n]$ if and only if f^*E does.

b) For $F \in \operatorname{ob} D^+(\underline{S},A)$, show the equivalence of:

(i) F is isomorphic in $D^+(\underline{S}, A)$ to a complex F' whose components are injective and which is zero outside $[m, n]$;

(ii) for every left A-module E, one has $\underline{\mathrm{Ext}}^i(E, F) = 0$ for $i \notin [m, n]$.

Such an F is said to have injective amplitude contained in $[m, n]$. The full subcategory of $D^+(\underline{S}, A)$ formed by the objects of finite injective amplitude is triangulated and is denoted $D^+(\underline{S}, A)_{\mathrm{injdf}}$.

c) If A is commutative, show that

$$\mathrm{R}\underline{\mathrm{Hom}} : D(\underline{S}, A)^{\circ}_{\mathrm{tordf}} \times D(\underline{S}, A)_{\mathrm{injdf}} \longrightarrow D(\underline{S}, A)_{\mathrm{parf}}$$

is defined.

We can now answer the question raised in (5.0) and state the main result of this section.

Theorem (5.7)**.** *Let $(\underline{S}, A)$ be a ringed topos, let $E \in \operatorname{ob} D(\underline{S}, A)$, and let $m, n \in \mathbb{Z}$ with $m \leq n$. The following conditions are equivalent:*

(i) $\operatorname{parf.amp}(E) \subset [m, n]$*;*

(ii) E is $(m-1)$-pseudo-coherent and $\operatorname{tor-amp}(E) \subset [m, n]$.

Since "perfect" plainly implies "pseudo-coherent", one obtains:

Corollary (5.7.1)**.** *An object E is perfect if and only if it is pseudo-coherent and locally of finite Tor-dimension.*

For the proof we need a few lemmas.

Lemma (5.7.2)**.** *Let B be a ring of $\underline{S}$. If E is a left A-module, G a right B-module, and F an A-B-bimodule, there is a canonical functorial homomorphism*

$$G \otimes_B \underline{\mathrm{Hom}}_A(E, F) \longrightarrow \underline{\mathrm{Hom}}_A(E, G \otimes_B F).$$

It is an isomorphism when E is a direct summand of a finite free A-module.

Proof. The morphism is the evident evaluation map. The assertion is clear for $E = A$ and is stable under finite sums and direct summands.

Lemma (5.7.3)**.** *Let $E \in \operatorname{ob} D^-(\underline{S}, A)$ have finite Tor-dimension, and suppose E is $(m-1)$-pseudo-coherent. Then locally, after replacing E by a quasi-isomorphic complex, one may arrange that $E^i = 0$ for $i > n$ and that the terms in the relevant range are finite locally free, provided the Tor-amplitude is contained in $[m, n]$.*

Proof. This is obtained by applying the approximation process of Section 2 and then using (5.1.1) to force flatness in the bottom degree. The pseudo-coherent hypothesis supplies finite-type generators; the Tor-amplitude hypothesis eliminates the last obstruction.

Proof of Theorem 5.7. If E has perfect amplitude contained in $[m, n]$, then it is locally represented by a complex of finite locally free modules in degrees $[m, n]$. It is therefore $(m-1)$-pseudo-coherent, and its Tor-amplitude is contained in $[m, n]$ by (5.1).

Conversely, assume (ii). By pseudo-coherence, locally E is represented, up to the required order, by a bounded-above complex of finite locally free modules. Since the Tor-amplitude is contained in $[m, n]$, the truncation argument of (5.1.1) and the lifting results of Section 2 show that this representation may be chosen with no non-zero terms outside $[m, n]$. Thus E has perfect amplitude contained in $[m, n]$.

Proposition (5.8)**.** *Let $E \in \operatorname{ob} D(\underline{S}, A)$ be perfect. Then perfect amplitude and Tor-amplitude agree: if* $\operatorname{parf.amp}(E) \subset [a, b]$*, then* $\operatorname{tor-amp}(E) \subset [a, b]$*, and conversely finite Tor-amplitude together with pseudo-coherence gives the corresponding perfect amplitude by (5.7).*

Corollary (5.8.4)**.** *Perfect complexes are locally of finite Tor-dimension. They are stable under Lf^* for morphisms of ringed topoi and, when the ring is commutative, under derived tensor product.*

Corollary (5.9)**.** *Let $(\underline{S}, A)$ be a ringed topos. Suppose that every coherent A-module has locally finite projective dimension bounded in the required sense. Then every pseudo-coherent object of $D(\underline{S}, A)$ with bounded cohomology is perfect.*

Examples 5.10. Corollary (5.9) applies in particular when $(\underline{S}, A)$ is defined by a ringed space whose local rings are regular, for example a regular scheme, or a smooth complex analytic space over $\mathbb{C}$.

6. Rank of a perfect complex

6.0

Let $(X, \mathcal{O}_X)$ be a ringed space in local rings. To each locally free sheaf of finite type L on an open set U one attaches a locally constant integer-valued function $\operatorname{rank}_U(L)$, called the rank of L. The family of such functions is characterized by compatibility with restriction, additivity in short exact sequences of finite locally free modules, and the condition $\operatorname{rank}(\mathcal{O}_X) = 1$. We now extend the notion of rank to perfect complexes.

Definition (6.1)**.** *Let C be an additive category, respectively a triangulated category, and let F be an abelian group. A function $f : \operatorname{ob}(C) \to F$ is called* additive *if, whenever $L \simeq L' \oplus L''$, respectively whenever there is a distinguished triangle*

$$L' \longrightarrow L \longrightarrow L'' \longrightarrow L'[1],$$

one has

$$f(L) = f(L') + f(L'').$$

The additive functions from C to F form an abelian group, denoted $\operatorname{Add}(C, F)$.

6.2. If C is triangulated, any additive function on C is additive on the underlying additive category. In particular $f(0) = 0$, $f(L) = f(M)$ when $L \simeq M$, and

$$f(L[n]) = (-1)^n f(L)$$

for all L and all $n \in \mathbb{Z}$.

6.3. The functor $\mathrm{Add}(C, -)$ is representable by an abelian group denoted $k(C)$, endowed with a universal additive function $\mathrm{ob}(C) \to k(C)$. If C is additive, $k(C)$ is the group completion of the monoid of isomorphism classes of objects. If C is triangulated, $k(C)$ is the quotient of the free abelian group on $\mathrm{ob}(C)$ by the relations $L = L' + L''$ for each distinguished triangle $L' \to L \to L'' \to L'[1]$.

We do not use the usual notation $K^0(C)$ here, because for an additive category C the notation $K^0(C)$ already denotes the homotopy category of complexes in C in the conventions being used. The group $k(C)$ is functorial in C for additive, respectively exact, functors, and isomorphic functors induce the same map on $k(C)$.

Lemma (6.4)**.** *Let C be an additive category. The canonical functor $C \to K^b(C)$ induces a functorial isomorphism*

$$k(C) \xrightarrow{\sim} k(K^b(C)).$$

Proof. This follows immediately from the formula

$$f(L) = \sum_i (-1)^i f(L^i) \tag{6.4.1}$$

for $L \in \mathrm{ob}\, K(C)$ and for any additive function f on $K(C)$.

Definition (6.5)**.** *Let $\underline{S}$ be a category, $\underline{C}$ an additive, respectively triangulated, $\underline{S}$-category, and F an abelian presheaf on $\underline{S}$. A family of functions*

$$f_X : \mathrm{ob}(\underline{C}_X) \longrightarrow F(X), \qquad X \in \mathrm{ob}\, \underline{S},$$

*is called an $\underline{S}$-*additive function *from $\underline{C}$ to F if:*

(i) each f_X is additive in the sense of (6.1);

(ii) for every arrow $u : X \to Y$ of $\underline{S}$, the diagram

$$\begin{array}{ccc} \mathrm{ob}\, \underline{C}_Y & \xrightarrow{f_Y} & F(Y) \\ \downarrow & & \downarrow F(u) \\ \mathrm{ob}\, \underline{C}_X & \xrightarrow{f_X} & F(X) \end{array}$$

commutes.

The $\underline{S}$-additive functions from $\underline{C}$ to F form an abelian group denoted $\mathrm{Add}_{\underline{S}}(\underline{C}, F)$.

Remarks 6.6. Let $\underline{C}^{\mathrm{iso}}$ be the fibred $\underline{S}$-category whose fibre over X has the same objects as $\underline{C}_X$ and whose morphisms are only the isomorphisms, and let $\underline{F}$ be the fibred category with discrete fibre $F(X)$ over X. An $\underline{S}$-additive function is a Cartesian $\underline{S}$-functor $\underline{C}^{\mathrm{iso}} \to \underline{F}$ whose restrictions to the fibres are additive.

If $\underline{C}$ is triangulated, additivity can be expressed by the exact sequence

$$0 \longrightarrow \mathrm{Add}_{\underline{S}}(\underline{C}, F) \longrightarrow \mathrm{Cart}_{\underline{S}}(\underline{C}^{\mathrm{iso}}, \underline{F}) \xrightarrow{\Delta} \mathrm{Cart}_{\underline{S}}(\mathrm{Tr}(\underline{C})^{\mathrm{iso}}, \underline{F}), \tag{6.6.1}$$

where $\Delta(f)(T) = f(L) - f(L'') - f(L')$ for a distinguished triangle $T = (L' \to L \to L'' \to L'[1])$.

For a morphism $u : X \to Y$ in $\underline{S}$, inverse-image functors, all isomorphic to one another, define a unique homomorphism $k(\underline{C}_Y) \to k(\underline{C}_X)$. Thus $X \mapsto k(\underline{C}_X)$ is an abelian presheaf on $\underline{S}$, denoted $k(\underline{C})$, and

$$\mathrm{Add}_{\underline{S}}(\underline{C}, F) = \mathrm{Hom}(k(\underline{C}), F).$$

Proposition (6.7)**.** *Let $\underline{S}$ be a site, $\underline{C}$ a flat abelian $\underline{S}$-category, and $\underline{C}_0$ a strictly full additive sub-$\underline{S}$-category satisfying the hypotheses of (4.0). The morphism of presheaves*

$$k(\underline{C}_0) \longrightarrow k(\mathrm{Parf}(\underline{C}, \underline{C}_0))$$

induced by the inclusion $\underline{C}_0 \subset \mathrm{Parf}(\underline{C}, \underline{C}_0)$ becomes an isomorphism after sheafification. Equivalently, for every abelian sheaf $\underline{F}$ on $\underline{S}$, the restriction homomorphism

$$\mathrm{Add}_{\underline{S}}(\mathrm{Parf}(\underline{C}, \underline{C}_0), \underline{F}) \longrightarrow \mathrm{Add}_{\underline{S}}(\underline{C}_0, \underline{F})$$

is an isomorphism.

Proof. The restriction factors as

$$\mathrm{Add}_{\underline{S}}(\mathrm{Parf}(\underline{C}, \underline{C}_0), \underline{F}) \xrightarrow{a} \mathrm{Add}_{\underline{S}}(K^b(\underline{C}_0), \underline{F}) \xrightarrow{b} \mathrm{Add}_{\underline{S}}(\underline{C}_0, \underline{F}).$$

By (6.4), b is an isomorphism. It remains to see that a is an isomorphism. Consider the commutative diagram whose rows are the exact sequences (6.6.1) and whose vertical arrows are restrictions from perfect complexes to bounded complexes in $\underline{C}_0$. Since $\underline{F}$ is a sheaf, $\underline{\underline{F}}$ is a stack; by (4.10), the two right vertical arrows are isomorphisms. Hence so is a.

Remark 6.7.1. In (6.7) one may replace $\mathrm{Parf}(\underline{C}, \underline{C}_0)$ by $D^*(\underline{C})_{\underline{C}_0\text{-parf}}$ for $* = -$ or b.

6.8. Let $(\underline{S}, A)$ be a ringed topos, and let $\underline{C}$ and $\underline{C}_0$ be the $\underline{S}$-categories considered in (4.8). There are homomorphisms of abelian presheaves

$$k(\underline{C}_0) \xrightarrow{\alpha} k(D^-(A\text{-parf})) \xrightarrow{\beta} k(\mathrm{Parf}(A\text{-parf})),$$

which induce isomorphisms after sheafification by (6.7). The homomorphism α is compatible with inverse image: for a morphism of ringed topoi $u : (\underline{S}', A') \to (\underline{S}, A)$, the square obtained from Lu^* commutes. If A is commutative, the derived tensor product gives ring-presheaf structures and α is a homomorphism of ring presheaves.

6.9. Assume now that $(\underline{S}, A)$ is locally ringed. Then, by (2.15.1), $\underline{C}_{0X}$ is the category of locally free A_X-modules of finite type. Let $\underline{C}_1$ be the $\underline{S}$-category whose fibre over X is the category of finite free A_X-modules. An object of $D(\underline{S})$ is perfect if and only if it is perfect relative to $\underline{C}_1$. The homomorphisms

$$k(\underline{C}_1) \longrightarrow k(\underline{C}_0) \longrightarrow k(D^-(\underline{S})_{\mathrm{parf}}) \longrightarrow k(\mathrm{Parf}(\underline{S}))$$

induce isomorphisms after sheafification. We shall show that these associated sheaves are canonically isomorphic to the constant sheaf $\mathbb{Z}_{\underline{S}}$.

Lemma (6.10)**.** *Let $(\underline{S}, A)$ be a locally ringed topos, let U be an object different from the empty object, and let $p, q \in \mathbb{N}$. If $A_U^p \simeq A_U^q$, then $p = q$.*

Proof. By the same kind of argument as in the proof of (2.15.1), one reduces to the locally ringed topos defined by a prescheme. There the assertion is trivial, by evaluating at a point of U.

Definition (6.11)**.** *The* rank*, denoted* rg*, is the $\underline{S}$-additive function from $\underline{C}_1$ to $\mathbb{Z}_{\underline{S}}$ defined by*

$$\mathrm{rg}(L) = p \quad \textit{if } L \simeq A_X^p, \quad X \neq \emptyset,$$

and by $\mathrm{rg}(L) = 0$ for objects over the empty object.

By (6.7), the rank function extends uniquely to an $\underline{S}$-additive function from $\mathrm{Parf}(\underline{S})$ to $\mathbb{Z}_{\underline{S}}$, still denoted rg.

Proposition (6.12)**.** a) *For $L, M \in \mathrm{ob}\, D^-(X)_{\mathrm{parf}}$, one has*

$$\mathrm{rg}(L \,\mathbf{L}\!\otimes M) = \mathrm{rg}(L)\,\mathrm{rg}(M).$$

b) *If $u : (\underline{S}', A') \to (\underline{S}, A)$ is a morphism of locally ringed topoi and $M \in \mathrm{ob}\, D^-(\underline{S})_{\mathrm{parf}}$, then*

$$\mathrm{rg}(Lu^* M) = u^*(\mathrm{rg}(M)).$$

Equivalently, rank defines a homomorphism of ring presheaves

$$k(D^-(\underline{S})_{\mathrm{parf}}) \longrightarrow \mathbb{Z}_{\underline{S}}$$

compatible with inverse images.

Proof. It is enough to check these statements after passing to the associated sheaves. By (6.9), one may replace $k(D^-(\underline{S})_{\mathrm{parf}})$ by $k(\underline{C}_1)$, where the assertions reduce to finite free modules and are immediate.

6.14. Let $(\underline{S}, A)$ be locally ringed, let $X \in \mathrm{ob}\, \underline{S}$, and let $n \in H^0(X, \mathbb{Z}_{\underline{S}})$. Define an object A_X^n of $\mathrm{Parf}(X)$ as follows. If n is constant with value an integer $n \geq 0$, set $A_X^n = A_X^{\oplus n}$; if $n < 0$, set $A_X^n = A_X^{\oplus |n|}[1]$. In general, decompose X into disjoint pieces on which n is constant and glue the corresponding objects. This gives a homomorphism of ring presheaves

$$i : \mathbb{Z}_{\underline{S}} \longrightarrow k(\mathrm{Parf}(\underline{S})), \qquad i_X(n) = \mathrm{cl}(A_X^n).$$

Proposition (6.14)**.** *Under the hypotheses above,*

$$\mathrm{rg} \circ i = \mathrm{Id}_{\mathbb{Z}_{\underline{S}}}.$$

Thus $k(\mathrm{Parf}(\underline{S}))$, viewed as a presheaf of $\mathbb{Z}_{\underline{S}}$-algebras via i, is augmented by rank. Moreover,

$$\mathrm{rg} : k(\mathrm{Parf}(\underline{S})) \longrightarrow \mathbb{Z}_{\underline{S}}$$

becomes an isomorphism after sheafification.

Proof. The first assertion follows from the relations $\mathrm{cl}(L[1]) = -\mathrm{cl}(L)$ and $\mathrm{cl}(L^n) = n\,\mathrm{cl}(L)$. For the second, i factors through $k(\underline{C}_0)$, and since $\mathrm{rg}\circ i$ is the identity, it is enough, by (6.9), to prove that $i\circ\mathrm{rg} : k(\underline{C}_0) \to k(\underline{C}_0)$ induces an isomorphism after sheafification. This says that for $L \in \mathrm{ob}\,\underline{C}_0$, the object $i\circ\mathrm{rg}(L)$ is locally isomorphic to L, which is immediate.

Corollary (6.15)**.** *Under the hypotheses of (6.13), for every abelian sheaf $\underline{F}$ on $\underline{S}$, the group of $\underline{S}$-additive functions from* $\mathrm{Parf}(\underline{S})$ *to $\underline{F}$ is canonically identified with $H^0(\underline{S}, \underline{F})$, where $\underline{S}$ denotes the final object of $\underline{S}$. For $\underline{F} = \mathbb{Z}_{\underline{S}}$, the rank function corresponds to the unit section of $\mathbb{Z}_{\underline{S}}$.*

7. Duality of perfect complexes

7.0

In this section $(\underline{S}, A)$ denotes a ringed topos in commutative rings, and S is the final object of $\underline{S}$. The aim is to generalize to perfect complexes the usual duality notions for locally free sheaves of finite type.

Proposition (7.1)**.** *Let $E \in \mathrm{ob}\,D(\underline{S})_{\mathrm{parf}}$ and $F \in \mathrm{ob}\,D^+(\underline{S})$.*

a) *If F has locally bounded cohomology, so does* $\mathrm{R}\underline{\mathrm{Hom}}(E, F)$*. If F is pseudo-coherent, respectively locally of finite Tor-dimension, respectively perfect, then the same is true of* $\mathrm{R}\underline{\mathrm{Hom}}(E, F)$*.*

7.1.1. Assume moreover that E has finite Tor-dimension, as holds by (5.8.4 b). Let a, b, m, n be integers such that

$$\text{tor-amp}(E) \subset [a, b]$$

(equivalently, by (5.8), $\text{parf-amp}(E) \subset [a, b]$), and $H^i(F) = 0$ for $i \notin [m, n]$, respectively $\text{parf-amp}(F) \subset [m, n]$. Then

$$H^i(\mathrm{R}\underline{\mathrm{Hom}}(E, F)) = 0 \quad \text{for } i \notin [m-b, n-a],$$

i.e. $\text{tor-amp}(\mathrm{R}\underline{\mathrm{Hom}}(E, F)) \subset [m-b, n-a]$.

Proof. The assertion a) is local; we may suppose E strictly perfect and then argue by devissage on E, reducing to $E = A$. For b), one may suppose E strictly perfect with $E^i = 0$ for $i \notin [c, d]$, and choose a representative of F in the indicated degrees, or with flat components in the indicated interval. Then the assertion follows from the next lemma.

Lemma (7.1.1)**.** *Let $E \in \mathrm{ob}\,D(\underline{S})$ and $F \in \mathrm{ob}\,D^+(\underline{S})$. If E is strictly pseudo-coherent, with components direct summands of finite free modules, the canonical map in $D^+(\underline{S})$*

$$\mathrm{Hom}^\bullet(E, F) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(E, F)$$

is an isomorphism.

Proof. The spectral sequence

$$E_2^{pq} = H^p(\mathrm{Ext}^q(E^\bullet, F^\bullet)) \Rightarrow \mathrm{Ext}^{p+q}(E, F)$$

degenerates because $\mathrm{Ext}^q(E^i, F^j) = 0$ for $q \neq 0$. This gives the required isomorphisms between the cohomology of $\mathrm{Hom}^\bullet(E, F)$ and of $\mathrm{R}\underline{\mathrm{Hom}}(E, F)$.

Proposition (7.1.2)**.** *Let $f : (\underline{S}', A') \to (\underline{S}, A)$ be a morphism of ringed topoi. Let $E \in \operatorname{ob} D^-(\underline{S})$ and $F \in \operatorname{ob} D^+(\underline{S})$ be such that $\mathrm{R}\underline{\mathrm{Hom}}(E, F) \in \operatorname{ob} D^+(\underline{S})$ and $Lf^*(F) \in \operatorname{ob} D^+(\underline{S}')$. There is a canonical functorial morphism*

$$Lf^*\mathrm{R}\underline{\mathrm{Hom}}(E, F) \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(Lf^*E, Lf^*F).$$

It is an isomorphism if E is perfect.

Proof. By the adjunction between Lf^* and Rf_*, it is enough to define a map

$$\mathrm{R}\underline{\mathrm{Hom}}(E, F) \longrightarrow Rf_*\mathrm{R}\underline{\mathrm{Hom}}(Lf^*E, Lf^*F).$$

Use the adjunction morphism $F \to Rf_*Lf^*F$ and the duality isomorphism

$$Rf_*\mathrm{R}\underline{\mathrm{Hom}}(Lf^*E, Lf^*F) \simeq \mathrm{R}\underline{\mathrm{Hom}}(E, Rf_*Lf^*F).$$

If E is perfect, the assertion is local and follows by devissage from the trivial case $E = A$.

Proposition (7.2)**.** *For $E \in \operatorname{ob} D(\underline{S})$ and $F \in \operatorname{ob} D^+(\underline{S})$ there is a canonical functorial morphism*

$$E \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(\mathrm{R}\underline{\mathrm{Hom}}(E, F), F).$$

It is an isomorphism if $E \in \operatorname{ob} D(\underline{S})_{\mathrm{parf}}$ and $F = A$.

Proof. Resolve F injectively and use the canonical morphism in the homotopy category

$$E \longrightarrow \mathrm{Hom}^\bullet(\mathrm{Hom}^\bullet(E, F), F).$$

For E perfect and $F = A$, the assertion is local and follows by devissage from $E = A$.

7.3. For $E \in \operatorname{ob} D(\underline{S})$ put

$$E^\vee = \mathrm{R}\underline{\mathrm{Hom}}(E, A),$$

and call this the *dual* of E. By (7.1) and (7.2), if E is perfect, respectively perfect and of finite Tor-dimension, then $E^\vee$ has the same property, and the canonical morphism $E \to E^{\vee\vee}$ is an isomorphism. In other words, $\mathrm{R}\underline{\mathrm{Hom}}(-, A)$ is an anti-equivalence of $D(\underline{S})_{\mathrm{parf}}$, respectively of the finite Tor-dimension part, with itself.

Proposition (Cartan isomorphism 7.4)**.** *For $E \in \operatorname{ob} D(\underline{S})$, $F \in \operatorname{ob} D^-(\underline{S})$, and $G \in \operatorname{ob} D^+(\underline{S})$, there are canonical functorial isomorphisms*

$$\mathrm{R}\underline{\mathrm{Hom}}(E \overset{\mathbf{L}}{\otimes} F, G) \simeq \mathrm{R}\underline{\mathrm{Hom}}(E, \mathrm{R}\underline{\mathrm{Hom}}(F, G)),$$

and the corresponding non-underlined Hom-isomorphisms obtained by taking global sections.

Proof. This is standard: resolve G injectively and F by flat modules.

Lemma (7.5)**.** *For $E \in \operatorname{ob} D(\underline{S})$ and $F \in \operatorname{ob} D^+(\underline{S})$, there is a canonical functorial homomorphism*

$$E^\vee \overset{\mathbf{L}}{\otimes} \mathrm{R}\underline{\mathrm{Hom}}(E, F) \longrightarrow F.$$

Proposition (7.6)**.** *Let $E \in \operatorname{ob} D(\underline{S})$ and let $F, G \in \operatorname{ob} D^+(\underline{S})$. Under either of the following hypotheses:*

(i) G has finite Tor-dimension;

(ii) E is perfect, F has finite Tor-dimension, and $G \in \operatorname{ob} D(\underline{S})$,

there is a canonical functorial homomorphism

$$\mathrm{R}\underline{\mathrm{Hom}}(E, F) \overset{\mathbf{L}}{\otimes} G \longrightarrow \mathrm{R}\underline{\mathrm{Hom}}(E, F \overset{\mathbf{L}}{\otimes} G).$$

It is an isomorphism when E is perfect.

Proof. Under (i), define the map by resolving F injectively and G flatly. Under (ii), by (7.4) it is equivalent to define a map

$$E \overset{\mathbf{L}}{\otimes} \mathrm{R}\underline{\mathrm{Hom}}(E, F) \overset{\mathbf{L}}{\otimes} G \longrightarrow F \overset{\mathbf{L}}{\otimes} G,$$

which is obtained by tensoring the map of (7.5) with G. When E is perfect, reduce locally by devissage to $E = A$.

Corollary (7.7)**.** *For $E \in \operatorname{ob} D(\underline{S})_{\mathrm{parf}}$ and $F \in \operatorname{ob} D^+(\underline{S})$, there is a functorial isomorphism*

$$E^\vee \overset{\mathbf{L}}{\otimes} F \xrightarrow{\sim} \mathrm{R}\underline{\mathrm{Hom}}(E, F).$$

Proof. Apply (7.6) with F and G replaced by A and F, and use (7.1) to know that $E^\vee$ is perfect.

Corollary (7.8)**.** *For $E, F \in \operatorname{ob} D(\underline{S})_{\mathrm{parf}}$, there is a natural morphism*

$$\mathrm{R}\underline{\mathrm{Hom}}(E, F) \overset{\mathbf{L}}{\otimes} \mathrm{R}\underline{\mathrm{Hom}}(F, E) \longrightarrow A$$

which is a perfect duality: modulo (7.4), it identifies each of the two complexes with the dual of the other.

Proof. The natural pairings $E^\vee \overset{\mathbf{L}}{\otimes} E \to A$ and $F^\vee \overset{\mathbf{L}}{\otimes} F \to A$ give, by tensor product, a pairing

$$E^\vee \overset{\mathbf{L}}{\otimes} F \overset{\mathbf{L}}{\otimes} F^\vee \overset{\mathbf{L}}{\otimes} E \longrightarrow A.$$

Using (7.7) this is the asserted pairing on $\mathrm{R}\underline{\mathrm{Hom}}$ complexes. Perfectness of the duality follows from the canonical isomorphisms

$$\mathrm{R}\underline{\mathrm{Hom}}(F^\vee \overset{\mathbf{L}}{\otimes} E, A) \simeq \mathrm{R}\underline{\mathrm{Hom}}(F^\vee, E^\vee) \simeq F^{\vee\vee} \overset{\mathbf{L}}{\otimes} E^\vee \simeq F \overset{\mathbf{L}}{\otimes} E^\vee,$$

with the usual compatibility check; alternatively, reduce locally to $E = F = A$.

Exercise 7.9. For $E, F \in \operatorname{ob} D(\underline{S})_{\mathrm{parf}}$, show that there is a canonical isomorphism

$$(E \overset{\mathbf{L}}{\otimes} F)^\vee \simeq E^\vee \overset{\mathbf{L}}{\otimes} F^\vee.$$

8. Traces and cup-products

8.0

In this section we generalize to perfect complexes the trace of an endomorphism of a finitely generated projective module. Applications are given in SGA 5, Expose III. The hypotheses and notation are those of (7.0).

8.1. Let $E \in \operatorname{ob} D(\underline{S})_{\mathrm{parf}}$. There is a natural morphism

$$\mathrm{tr} : \mathrm{R}\underline{\mathrm{Hom}}(E, E) \longrightarrow A,$$

obtained by composing the inverse of the isomorphism (7.7),

$$\mathrm{R}\underline{\mathrm{Hom}}(E, E) \simeq E^\vee \overset{\mathbf{L}}{\otimes} E,$$

with the canonical map $E^\vee \overset{\mathbf{L}}{\otimes} E \to A$ of (7.5). For $U \in \operatorname{ob} \underline{S}$, applying $H^0(U, -)$ gives an $H^0(U, A)$-module homomorphism

$$\mathrm{Tr}_{E|U} : \mathrm{Hom}_{D(\underline{S})}(E|U, E|U) \longrightarrow H^0(U, A),$$

called the trace homomorphism. The finite Tor-dimension hypothesis on E is in fact unnecessary for defining this trace; see (8.5).

8.1.1. If $E = A$, then $\mathrm{Tr}_{A|U} : \mathrm{Hom}(A|U, A|U) \to H^0(U, A)$ is the usual isomorphism. If E and F are direct summands of finite free modules, and if an endomorphism of $(E \oplus F)|U$ is represented by a block matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then

$$\mathrm{Tr}_{E\oplus F|U}(u) = \mathrm{Tr}_{E|U}(a) + \mathrm{Tr}_{F|U}(d).$$

8.1.2. Let E be a strictly perfect complex, let $U \in \operatorname{ob} \underline{S}$, and let $u : E|U \to E|U$ be a morphism of complexes with components $u^i : E^i|U \to E^i|U$. Then

$$\mathrm{Tr}_{E|U}(u) = \sum_i (-1)^i \, \mathrm{Tr}_{E^i|U}(u^i).$$

Proof. The question is local. One may suppose that u is represented by a section of $H^0(E^\vee \otimes E)$, and even by a sum of elementary tensors $\sum_i x_i \otimes x_i'$, where x_i and x_i' are sections of E^i and of $E^{\vee i}$. Writing explicitly the map $E^\vee \otimes E \to A$, one obtains

$$\mathrm{Tr}_{E|U}(u) = \sum_i (-1)^i \langle x_i, x_i' \rangle = \sum_i (-1)^i \, \mathrm{Tr}_{E^i|U}(u^i).$$

8.2. Let $E, F \in \operatorname{ob} D(\underline{S})_{\mathrm{parf}}$. By (7.8) there is a canonical homomorphism

$$\cup : \mathrm{R}\underline{\mathrm{Hom}}(E,F) \overset{\mathbf{L}}{\otimes} \mathrm{R}\underline{\mathrm{Hom}}(F,E) \longrightarrow A.$$

For $U \in \operatorname{ob} \underline{S}$, and for

$$u \in H^0(U, \mathrm{R}\underline{\mathrm{Hom}}(E,F)), \qquad v \in H^0(U, \mathrm{R}\underline{\mathrm{Hom}}(F,E)),$$

the image under $H^0(\cup)$ defines an $H^0(U,A)$-bilinear pairing

$$\mathrm{Hom}_{D(\underline{S})}(E|U, F|U) \otimes_{H^0(U,A)} \mathrm{Hom}_{D(\underline{S})}(F|U, E|U) \longrightarrow H^0(U,A),$$

called the cup-product and denoted $\langle u, v \rangle$.

Proposition (8.3)**.** *In the situation of (8.2), for $u : E|U \to F|U$ and $v : F|U \to E|U$ one has*

$$\mathrm{Tr}_{E|U}(vu) = \mathrm{Tr}_{F|U}(uv) = \langle u, v \rangle.$$

In particular, for $E = F$,

$$\mathrm{Tr}_{E|U}(u) = \langle u, \mathrm{Id}_E \rangle = \langle \mathrm{Id}_E, u \rangle.$$

Proof. It is enough, for notation, to take $U = S$. The maps u and v may be viewed as

$$u : A \to E^\vee \overset{\mathbf{L}}{\otimes} F, \qquad v : A \to F^\vee \overset{\mathbf{L}}{\otimes} E.$$

The composites giving vu and uv are obtained by inserting the corresponding evaluation morphisms. The result follows from the commutativity of the standard diagrams comparing the two possible ways of evaluating

$$E^\vee \overset{\mathbf{L}}{\otimes} F \overset{\mathbf{L}}{\otimes} F^\vee \overset{\mathbf{L}}{\otimes} E$$

and the analogous expression with E and F interchanged. This verification is purely formal.

Proposition (8.4)**.** *Let E, F, U be as in (8.2). Let $s : E|U \to F|U$ and let $s' : E|U \to F|U$ be an isomorphism. Then*

$$\mathrm{Tr}_{F|U}(s's^{-1}) = \mathrm{Tr}_{E|U}(s^{-1}s').$$

Proof. This is immediate from (8.3).

8.5. Let $\mathcal{E}(\underline{C}_0)$ be the $\underline{S}$-fibred category whose fibre over U has objects the pairs (L, u) with $L \in \operatorname{ob} \underline{C}_0$ and $u \in \mathrm{End}_{\underline{C}_0}(L)$, and whose morphisms $(L,u) \to (M,v)$ are the morphisms $f : L \to M$ satisfying $fu = vf$. By (8.4), the trace may be viewed as an $\underline{S}$-functor from $\mathcal{E}(\underline{C}_0)^{\mathrm{iso}}$ to A, where the superscript means that only isomorphisms are kept and A is regarded as the fibred category with discrete fibres. Since every perfect complex is locally of finite Tor-dimension, the forgetful $\underline{S}$-functor

$$\mathcal{E}(\underline{C}_0)^{\mathrm{iso}} \longrightarrow \mathcal{E}(D(\underline{S})_{\mathrm{parf}})^{\mathrm{iso}}$$

induces an isomorphism on the associated stacks. Hence the trace homomorphism extends to perfect complexes without assuming finite Tor-dimension.

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SGA 6 – Expose II

Existence of Global Resolutions

L. Illusie

1. General globalization criteria

1.0

In this section we fix a site $\underline{S}$, a flat abelian $\underline{S}$-category $\underline{C}$ in the sense of I, 1.1, and a strictly full additive sub-$\underline{S}$-category $\underline{C}_0$. We suppose that $\underline{C}_0$ is locally quasi-liftable in $\underline{C}$ and locally quasi-stable under kernels of epimorphisms.

Proposition (1.1)**.** *Let $S \in \operatorname{ob} \underline{S}$, and assume:*

(i) $\underline{C}_{0S}$ is quasi-liftable in $\underline{C}_S$;

(ii) every object of $\underline{C}_S$ of $\underline{C}$-finite type, respectively of finite $\underline{C}_0$-presentation, is of $\underline{C}_{0S}$-finite type.

Then:

a) $\underline{C}_{0S}$ is quasi-stable under kernels of epimorphisms.

b) If $E \in \operatorname{ob} \mathrm{D}(\underline{C}_S)$ is n-$\underline{C}_0$-pseudo-coherent, then E is n-$\underline{C}_{0S}$-pseudo-coherent, respectively $(n+1)$-$\underline{C}_{0S}$-pseudo-coherent. Equivalently,

$$\mathrm{D}(\underline{C}_S)_{n-\underline{C}_0-\mathrm{coh}} \subset \mathrm{D}(\underline{C}_S)_{n-\underline{C}_{0S}-\mathrm{coh}}$$

with the corresponding $(n+1)$-variant.

c) An object $E \in \mathrm{D}(\underline{C}_S)$ is pseudo-coherent relative to $\underline{C}_0$ if and only if it is pseudo-coherent relative to $\underline{C}_{0S}$; thus

$$\mathrm{D}(\underline{C}_S)_{\underline{C}_0-\mathrm{coh}} = \mathrm{D}(\underline{C}_S)_{\underline{C}_{0S}-\mathrm{coh}}.$$

Moreover the canonical functor

$$\mathrm{K}(\underline{C}_{0S})/\mathrm{K}_{\underline{C}_S}^{\underline{C}_{0S}-\emptyset} \longrightarrow \mathrm{D}(\underline{C}_S)$$

is fully faithful and has essential image $\mathrm{D}(\underline{C}_S)_{\underline{C}_{0S}-\mathrm{coh}}$.

If $\underline{C}_{0S}$ is liftable in $\underline{C}_S$, then $\mathrm{K}_{\underline{C}_S}^{\underline{C}_{0S}-\emptyset}$ is reduced to zero and

$$\mathrm{K}(\underline{C}_{0S}) \longrightarrow \mathrm{D}(\underline{C}_S)_{\underline{C}_{0S}-\mathrm{coh}}$$

is fully faithful with the same essential image.

Proof. For a), let

$$0 \longrightarrow L^0 \longrightarrow L^1 \longrightarrow L^2 \longrightarrow \cdots$$

be an exact sequence in $\underline{C}_{0S}$ with all L^i in $\underline{C}_{0S}$. One has to prove that L is of $\underline{C}_{0S}$-finite type. By I, 2.2 it is $\underline{C}_S$-pseudo-coherent, hence by I, 2.9 of finite $\underline{C}_S$-presentation, and the assertion follows from (ii).

For b), one argues by descending induction on i for $i \geq n$ (respectively $i \geq n+1$). After replacing E by a strictly pseudo-coherent representative and taking the usual cone, one reduces to a complex with $E^i = 0$ above the critical degree. By I, 2.10, the relevant cohomology object is of $\underline{C}_0$-finite type, respectively of finite $\underline{C}_0$-presentation; by I, 1.4 it is then of $\underline{C}_{0S}$-finite type. Applying I, 2.10 again gives the required pseudo-coherence. Part c) follows from b) and I, 2.7.

Proposition (1.2)**.** *Assume that $\underline{C}_0$ is locally liftable in $\underline{C}$ and locally stable under kernels of epimorphisms, and keep the hypotheses of 1.1. Let $\underline{C}_{0,\mathrm{loc}}$ be the strictly full subcategory of $\underline{C}$ consisting of objects locally lying in $\underline{C}_0$. Then:*

- *a) $\underline{C}_{0,\mathrm{loc}}$ is stable under kernels of epimorphisms and quasi-liftable in $\underline{C}$.*
- *b) For $E \in \operatorname{ob} \mathrm{D}(\underline{C}_*)$ and an interval $[m,n] \subset \mathbb{Z}$, E has $\underline{C}_0$-perfect amplitude contained in $[m,n]$ if and only if it is isomorphic in $\mathrm{D}(\underline{C}_*)$ to a complex F with $F^i = 0$ outside $[m,n]$, with $F^i \in \underline{C}_{0,\mathrm{loc}}$ for $i > m$, and with $F^m \in \underline{C}_{0,\mathrm{loc}}$.*
- *c) The canonical functor*
$$\mathrm{K}^b(\underline{C}_{0,\mathrm{loc}})/\mathrm{K}^{b,\emptyset}(\underline{C}_{0,\mathrm{loc}}) \longrightarrow \mathrm{D}(\underline{C}_*)$$
is fully faithful and has essential image the full subcategory of complexes of finite $\underline{C}_0$-perfect amplitude.
- *d) If $\underline{C}_0$ is liftable, one may replace in b) "is isomorphic to F in $\mathrm{D}(\underline{C}_*)$" by "there exists a quasi-isomorphism $F \to E$", and $\mathrm{K}^{b,\emptyset}(\underline{C}_{0,\mathrm{loc}})$ is zero.*

Proof. Stability of $\underline{C}_{0,\mathrm{loc}}$ under kernels of epimorphisms is clear. Since objects of $\underline{C}_{0S}$ are $\underline{C}_S$-pseudo-coherent, hence $\underline{C}$-pseudo-coherent by 1.1 c), quasi-liftability follows from I, 2.14. For b), only necessity has to be proved. From 1.1 c) one already knows that E is $\underline{C}_{0,\mathrm{loc}}$-pseudo-coherent. Since E is acyclic above n, one may, after replacing it by an isomorphic complex, suppose it strictly $(m+1)$-$\underline{C}_{0,\mathrm{loc}}$-pseudo-coherent and zero above n. Since E has perfect amplitude in $[m,n]$, I, 4.16 implies that E^m is locally an object of $\underline{C}_0$, hence of $\underline{C}_{0,\mathrm{loc}}$. Part c) follows from a), b), and I, 4.12.2; part d) is immediate from I, 4.6.

2. Application to certain categories of modules

2.0. Notation

Let $(\underline{S}, \underline{A})$ be a ringed topos, where $\underline{A}$ is a sheaf of rings, not necessarily commutative. We write $\underline{C}$ for the $\underline{S}$-category of left $\underline{A}$-modules and $\underline{C}_0$ for the sub-$\underline{S}$-category of finite free left $\underline{A}$-modules. Then $\underline{C}$ is flat and $\underline{C}_0$ is locally liftable and locally stable under kernels.

2.1. Lifting lemmas

Lemma (2.1.1)**.** *Let A be a ring and let C be the category of left A-modules. Let C_0 be the full additive subcategory of finite free modules. Then:*

- *a) C_0 is liftable in C.*
- *b) C_0 is stable under kernels of epimorphisms if and only if every finite left A-module is finitely presented.*

The proof is the elementary lifting argument for free finite modules; b) is the usual kernel criterion for finite presentation.

Lemma (2.1.2)**.** *Let $(\underline{S}, \underline{A})$ be a ringed topos, $\underline{C}$ the $\underline{S}$-category of left $\underline{A}$-modules, and $\underline{C}_0$ the subcategory of finite free modules. Let $S \in \operatorname{ob} \underline{S}$, and let $\underline{C}'_S$ be a strictly full additive subcategory of $\underline{C}_S$ containing $\underline{C}_{0S}$ and stable under kernels, cokernels, and extensions. Suppose that*

(i) $\operatorname{Ext}^1_{\underline{C}_S}(G, F) = 0$ *for all* $F \in \underline{C}'_S$ *and all* $G \in \underline{C}_{0S}$*;*

(ii) every extension of an object $F \in \underline{C}'_S$ *by an object* $G \in \underline{C}_S$ *such that* $\operatorname{Hom}(G, \underline{A})$ *lies in* $\underline{C}_{0S}$ *and* G *is locally a direct factor of a finite free module is split.*

Then $\underline{C}'_S$ *is liftable in* $\underline{C}_S$*.*

Lemma (2.1.3)**.** *Let A and B be abelian categories and let $\theta : A \to B$ be an exact functor.*

a) For $M \in A$, the following conditions are equivalent:

(i) for every $L \in A$, the canonical maps

$$\operatorname{Ext}^n_A(L, M) \longrightarrow \operatorname{Ext}^n_B(\theta L, \theta M)$$

are isomorphisms for all n;

(ii) for every $L \in A$, the canonical map

$$\operatorname{Hom}_A(L, M) \longrightarrow \operatorname{Hom}_B(\theta L, \theta M)$$

is an isomorphism, and for every $n \geq 1$ and every $x \in \operatorname{Ext}^n_A(L, M)$ there is an epimorphism $L' \twoheadrightarrow L$ which kills x.

b) The following are equivalent:

(i) for all $L, M \in A$ and all n, the canonical map on Ext^n is an isomorphism;

(ii) the canonical functor

$$\mathrm{D}^*(A) \longrightarrow \mathrm{D}^*(B)$$

is fully faithful and has essential image the full subcategory of complexes F such that every $H^i(F)$ lies in the essential image of θ.

Proof. For a), the implication (i) implies (ii) because a class in $\operatorname{Ext}^n_A(L, M)$ is represented, in Yoneda's interpretation, by an exact sequence

$$0 \to M \to E_n \to \cdots \to E_1 \to L \to 0,$$

and the epimorphism $E_1 \twoheadrightarrow L$ kills the class. Conversely, set

$$F^n(L) = \operatorname{Ext}^n_A(L, M), \qquad G^n(L) = \operatorname{Ext}^n_B(\theta L, \theta M).$$

Effaceability gives both $F^n(L)$ and $G^n(L)$, for $n \geq 1$, as filtered colimits of the cokernels associated with short exact sequences

$$0 \to L'' \to L' \to L \to 0.$$

Since $F^0 \to G^0$ is an isomorphism, induction on n proves that $F^n \to G^n$ is an isomorphism for all n. Part b) follows by the standard devissage in the derived category. The dual formulation is obtained by reversing arrows; if the categories have enough injectives, the same conditions are equivalent to the corresponding assertion for $\mathrm{R}\underline{\mathrm{Hom}}$ on $\mathrm{D}^- \times \mathrm{D}^+$.

2.2. Noetherian schemes and divisorial schemes

Definition (2.2.1)**.** *For a scheme S, let $\mathrm{Qcoh}(S)$ and $\mathrm{Coh}(S)$ denote the categories of quasi-coherent and coherent $\mathcal{O}_S$-modules. For $E \in \mathrm{D}(\mathrm{Qcoh}(S))$, pseudo-coherence, perfect amplitude, and perfection are understood relative to the fibred category on the Zariski site of S given by $U \mapsto \mathrm{Loclib}(U)$. We write $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{pcoh}}$ and $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$ for the corresponding full triangulated subcategories.*

Proposition (2.2.2)**.** *Let S be a noetherian scheme. The canonical functor*

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful and has essential image $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{pcoh}}$.

Proof. Apply 1.1 on the Zariski site of S, with $\underline{C}(U) = \mathrm{Qcoh}(U)$ and $\underline{C}_0(U) = \mathrm{Coh}(U)$. A coherent object is pseudo-coherent, so pseudo-coherence in $\mathrm{D}(\mathrm{Qcoh}(U))$ is the same as pseudo-coherence relative to $\underline{C}_0$. The two hypotheses of 1.1 are the ordinary facts that a quotient of a coherent module is coherent and that every quasi-coherent module on a noetherian scheme is the filtered direct limit of its coherent submodules.

Corollary (2.2.2.1)**.** *For S noetherian, respectively noetherian of finite dimension, the canonical functor*

$$\mathrm{D}^-(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^-(S) \quad \textit{respectively} \quad \mathrm{D}^b(\mathrm{Coh}(S)) \longrightarrow \mathrm{D}^b(S)$$

is fully faithful and has essential image $\mathrm{D}^-(S)_{\mathrm{pcoh}}$, respectively $\mathrm{D}^b(S)_{\mathrm{pcoh}}$.

Proposition (2.2.3)**.** *Let S be quasi-compact and quasi-separated, and let $(\mathcal{L}_i)_{i\in I}$ be a family of invertible sheaves. For an $\mathcal{O}_S$-module F and $(i,n) \in I \times \mathbb{Z}$, let $F_{(i,n)}$ be the submodule of $F \otimes \mathcal{L}_i^{\otimes n}$ generated by its global sections. The following conditions are equivalent:*

(i) the opens S_f, where $f \in \Gamma(S, \mathcal{L}_i^{\otimes n})$, $i \in I$, $n \in \mathbb{N}$, form a basis for the topology of S;

(ii) those among the S_f which are affine cover S;

(iii) every quasi-coherent finite type $\mathcal{O}_S$-module F admits an epimorphism

$$\bigoplus_{(i,n)} \mathcal{L}_i^{\otimes -n} \longrightarrow F;$$

(iv) for every finite type quasi-coherent F and finite type quasi-coherent submodule $F' \subset F$, there are $(i,n) \in I \times \mathbb{N}^+$ and $u : \mathcal{L}_i^{\otimes -n} \to F$ with $\mathrm{Im}(u) \not\subset F'$.

Proof. This is a paraphrase of EGA II, 4.5.2 and 4.5.5. The implication from the existence of affine basic opens to the epimorphism is obtained by extending finitely many local generators after multiplying them by sufficiently high powers of the chosen sections. Conversely, applying the generator condition to ideals defining complements of open neighbourhoods gives the required basic opens.

Definition (2.2.4)**.** *Such a family* $(\mathcal{L}_i)$ *is called ample. A scheme is called divisorial if it admits an ample family of invertible modules.*

Lemma (2.2.6)**.** *Let* X *be locally noetherian and let* $U \subset X$ *be an open such that, at every maximal point* x *of* $X - U$*, one has* $\dim \mathcal{O}_{X,x} \leq 1$*. Then* U *contains the maximal points of* X*.*

Proof. Put $Y = X - U$ and localize at a maximal point x of Y. The pullback Y' is the closed point of $X' = \operatorname{Spec} \mathcal{O}_{X,x}$, and U' is affine. If the dimension of X' were at least two, local cohomology would give $H^1_{Y'}(X', \mathcal{O}_{X'}) \neq 0$, while affineness of U' forces the relevant higher local cohomology to vanish, a contradiction.

Proposition (2.2.7)**.** *Every noetherian separated locally factorial scheme is divisorial. In particular every noetherian regular separated scheme is divisorial.*

Proof. After reducing to the integral case, choose a finite affine open cover (S_i). By the lemma, $S - S_i$ is purely of codimension one, hence the support of an effective divisor. Thus there are invertible sheaves $\mathcal{L}_i$ and sections f_i such that $S_{f_i} = S_i$. The family $(\mathcal{L}_i)$ is ample.

The separatedness hypothesis is essential: the affine plane with the origin doubled has only trivial invertible sheaves, and global functions cannot separate the two origins.

Lemma (2.2.8)**.** *Let* S *be divisorial, let* $(\mathcal{L}_i)$ *be an ample family, and let* $\underline{L}(S)$ *be the strictly full additive subcategory of* $\mathrm{Qcoh}(S)$ *generated by the* $\mathcal{L}_i^{\otimes -n}$*,* $(i,n) \in I \times \mathbb{N}^+$*. For* $E \in \mathrm{D}(\mathrm{Qcoh}(S))$*:*

a) $\underline{L}(S)$ *and* $\mathrm{Loclib}(S)$ *are quasi-liftable in* $\mathrm{Qcoh}(S)$ *and quasi-stable under kernels of epimorphisms.*

b) E *is* n*-pseudo-coherent, respectively pseudo-coherent, if and only if it is so relative to* $\underline{L}(S)$ *or to* $\mathrm{Loclib}(S)$*. The canonical functors*

$$\mathrm{K}^-(\underline{L}(S))/\mathrm{K}^{-,\emptyset}(\underline{L}(S)) \to \mathrm{D}(\mathrm{Qcoh}(S))$$

and

$$\mathrm{K}^-(\mathrm{Loclib}(S))/\mathrm{K}^{-,\emptyset}(\mathrm{Loclib}(S)) \to \mathrm{D}(\mathrm{Qcoh}(S))$$

are fully faithful and have essential image $\mathrm{D}^-(\mathrm{Qcoh}(S))_{\mathrm{pcoh}}$*.*

c) E *has perfect amplitude contained in* $[m,n]$ *if and only if it is isomorphic in* $\mathrm{D}(\mathrm{Qcoh}(S))$ *to a complex* F *with* $F^i = 0$ *for* $i \notin [m,n]$*, with* $F^i \in \underline{L}(S)$ *for* $i > m$*, and with* F^m *locally free of finite type. Moreover*

$$\mathrm{K}^b(\mathrm{Loclib}(S))/\mathrm{K}^{b,\emptyset}(\mathrm{Loclib}(S)) \to \mathrm{D}(\mathrm{Qcoh}(S))$$

is fully faithful with essential image $\mathrm{D}(\mathrm{Qcoh}(S))_{\mathrm{parf}}$*.*

Proof. Given an epimorphism $F \to G$ with G of finite type, take a finite type quasi-coherent submodule $F' \subset F$ still mapping epimorphically to G, then cover F' by a sum of negative powers of the ample invertible sheaves. This proves quasi-liftability. The remaining assertions follow from 1.1 applied on the Zariski site, using that objects of $\underline{L}$ and of Loclib are pseudo-coherent and using the ampleness criterion above.

Proposition (2.2.9)**.** *Assume in addition that* S *is separated or noetherian. Then:*

a) The canonical functors

$$\mathrm{K}^{-,b}(\underline{L}(S))/\mathrm{K}^{-,b,\emptyset}(\underline{L}(S)) \to \mathrm{D}^{-}(S), \qquad \mathrm{K}^{-,b}(\mathrm{Loclib}(S))/\mathrm{K}^{-,b,\emptyset}(\mathrm{Loclib}(S)) \to \mathrm{D}^{-}(S)$$

are fully faithful with essential image $\mathrm{D}^{-}(S)_{\mathrm{pcoh}}$. If S has finite cohomological dimension, the analogous bounded functors are equivalences.

b) The canonical functor

$$\mathrm{K}^{-,b}(\mathrm{Loclib}(S))/\mathrm{K}^{-,b,\emptyset}(\mathrm{Loclib}(S)) \to \mathrm{D}(S)$$

is fully faithful and has essential image $\mathrm{D}(S)_{\mathrm{parf}}$; the same explicit description of perfect amplitude as in 2.2.8 holds in $\mathrm{D}(S)$.

This is an immediate consequence of 2.2.8 together with the comparison between $\mathrm{D}(\mathrm{Qcoh}(S))$ and $\mathrm{D}(S)$ in the cases considered.

Remark (2.2.10)**.** *If $S = \operatorname{Spec} A$ is affine, the single sheaf $\mathcal{O}_S$ is ample and $\underline{L}(S) = \mathrm{Lib}(S)$. Thus $R\Gamma$ identifies the pseudo-coherent and perfect pieces of $\mathrm{D}(S)$ with the corresponding derived categories of A-modules defined by finite free and finite projective modules. In particular, a pseudo-coherent complex with bounded cohomology is locally isomorphic in $\mathrm{D}(S)$ to a strictly pseudo-coherent complex.*

3. End of the comparison with quasi-coherent complexes

The coherator, being of finite cohomological dimension, admits a right derived functor

$$\underline{R}Q : \mathrm{D}(S) \longrightarrow \mathrm{D}(\mathrm{Qcoh}(S)).$$

For $E \in \mathrm{D}(\mathrm{Qcoh}(S))$ there is a functorial canonical isomorphism

$$E \xrightarrow{\sim} \underline{R}Q(E),$$

because the terms are Q-acyclic and satisfy $E^i \xrightarrow{\sim} Q(E^i)$. For $E \in \mathrm{D}(S)$ there is a canonical morphism

$$\underline{R}Q(E) \longrightarrow E.$$

These morphisms make $\underline{R}Q$ a right adjoint to the canonical functor

$$\phi : \mathrm{D}(\mathrm{Qcoh}(S)) \longrightarrow \mathrm{D}(S).$$

Since the first morphism is an isomorphism, ϕ is fully faithful. The spectral sequence

$$R^pQ(H^q(E)) \Rightarrow R^*Q(E)$$

shows that the second morphism is an isomorphism for $E \in \mathrm{D}(S)_{\mathrm{Qcoh}}$. Hence the essential image of ϕ is $\mathrm{D}(S)_{\mathrm{Qcoh}}$.

SGA 6 – Expose III, continued

Applications: Base Change and Semi-continuity Theorems

5. Applications: base change and semi-continuity theorems

Lemma (5.1)**.** *Let $(X, \mathcal{O}_X)$ be a commutatively ringed topos, let $\mathrm{Mod}(X)$ be the category of $\mathcal{O}_X$-modules, and let $E \in \mathrm{D}^-(X)$ and $p \in \mathbb{Z}$. Consider:*

- *a) $M \mapsto H^p(E \mathbf{L}\otimes M)$ is right exact on $\mathrm{Mod}(X)$;*
- *b) the canonical map $H^p(E) \otimes M \to H^p(E \mathbf{L}\otimes M)$ is an isomorphism for all M;*
- *c) $M \mapsto H^{p+1}(E \mathbf{L}\otimes M)$ is left exact;*
- *d) the canonical map*

$$H^{p+1}(E \mathbf{L}\otimes M) \to \mathrm{Hom}(H^{-p-1}(E^\vee), M)$$

  is an isomorphism for all M.

Then $a \Leftrightarrow b \Leftrightarrow c \Leftarrow d$. If E is pseudo-coherent, all four conditions are equivalent.

The map in d) is obtained from the evaluation $E^\vee \mathbf{L}\otimes E \to \mathcal{O}_X$, hence

$$E \mathbf{L}\otimes M \to \mathrm{R}\underline{\mathrm{Hom}}(E^\vee, M),$$

followed on cohomology by

$$\mathrm{Ext}^{p+1}(E^\vee, M) \to \mathrm{Hom}(H^{-p-1}(E^\vee), M).$$

Proof. The implications $b \Rightarrow a \Leftrightarrow c \Leftarrow d$ are formal. To prove $a \Rightarrow b$, reduce by a presentation of M to the case $M = \mathcal{O}_{U,X}$. Both sides then identify with extension by zero from U, and exactness of $i_!$ gives the result. If E is pseudo-coherent, it suffices for d to test injective M. For such M the last displayed map on Ext is an isomorphism, and the assertion follows from the fact that

$$E \mathbf{L}\otimes M \to \mathrm{R}\underline{\mathrm{Hom}}(E^\vee, M)$$

is an isomorphism for pseudo-coherent E and injective M, after localization, truncation, and devissage to the case $E = \mathcal{O}_X$.

Remark (5.2)**.** *These conditions are local on the topos. If X has enough points, they may be checked stalkwise. For a complex represented by flat terms, the criterion recovers the usual EGA III criterion: right exactness is equivalent to flatness of a suitable cokernel. In particular, for a perfect complex the condition for E at degree p is dual to the corresponding condition for $E^\vee$ at degree $-p-1$.*

For a quasi-separated scheme X and a complex with quasi-coherent locally bounded cohomology, the same assertions hold after restricting to quasi-coherent modules: the quasi-coherent versions of the four conditions are equivalent in the same way, and for pseudo-coherent E all are equivalent.

Proposition (5.3)**.** *Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes, let $E \in \mathrm{D}^b(X)$ have quasi-coherent cohomology and finite flat amplitude relative to f, and let $p \in \mathbb{Z}$. Consider:*

a) $M \mapsto R^p f_*(E\,\mathbf{L}\!\otimes_Y M)$ *is right exact on* $\mathrm{Qcoh}(Y)$*;*

b) the canonical map

$$R^p f_*(E) \otimes M \to R^p f_*(E\,\mathbf{L}\!\otimes_Y M)$$

is an isomorphism for all quasi-coherent M*;*

c) $M \mapsto R^{p+1} f_*(E\,\mathbf{L}\!\otimes_Y M)$ *is left exact;*

d) the canonical map

$$R^{p+1} f_*(E\,\mathbf{L}\!\otimes_Y M) o \operatorname{Hom}(H^{-p-1}(Rf_*(E)^\vee), M)$$

is an isomorphism for all quasi-coherent M*;*

e) for every base change $Y' \to Y$*, the canonical map*

$$R^p f_*(E) \otimes_Y \mathcal{O}_{Y'} \to H^p(Rf_*(E)\,\mathbf{L}\!\otimes_Y \mathcal{O}_{Y'})$$

is an isomorphism.

Then $a \Leftrightarrow b \Leftrightarrow c \Leftrightarrow e \Leftarrow d$*. If* f *is proper,* E *is pseudo-coherent relative to* f*, and the finiteness theorem applies to* f*, these conditions are all equivalent.*

Proof. The implications involving a), b), c), and d) follow from the projection formula and Lemma 5.1. It remains to compare b) and e). This follows from the following base-change form: if X is quasi-separated and $E \in \mathrm{D}^-(X)$ has locally bounded quasi-coherent cohomology, then the condition of Lemma 5.1 a) is equivalent to requiring

$$H^p(E) \otimes_X \mathcal{O}_{X'} \to H^p(E\,\mathbf{L}\!\otimes_X \mathcal{O}_{X'})$$

to be an isomorphism for every $X' \to X$. The proof is affine: write $X = \operatorname{Spec} A$, $X' = \operatorname{Spec} A'$, and $E = F^\sim$, and reduce to

$$H^p(F) \otimes_A A' \to H^p(F\,\mathbf{L}\!\otimes_A A').$$

Conversely, for a quasi-coherent module N take $A' = D_A(N)$.

If E is pseudo-coherent, the same argument applied to $E^\vee$ gives the compatibility of

$$H^{-p-1}(E^\vee) \otimes_X \mathcal{O}_{X'} \longrightarrow H^{-p-1}((E\,\mathbf{L}\!\otimes_X \mathcal{O}_{X'})^\vee),$$

which completes the proof.

Remark (5.3.2)**.** *The proposition applies in particular when* E *is a bounded complex of quasi-coherent flat* $\mathcal{O}_X$*-modules of finite presentation relative to* f*, since such a complex has finite perfect amplitude relative to* f*.*

Definition (5.4)**.** *Let* X *be a topos,* $\mathrm{Pt}(X)$ *the set of its points,* M *an ordered set, and* $f : \mathrm{Pt}(X) \to M$*. We say that* f *is upper semi-continuous at* x *(respectively lower semi-continuous, respectively locally constant) if there exists an* x*-pointed object* U *of* X *such that, for every point* y *of* U*, one has* $f(y) \le f(x)$ *(respectively* $f(y) \ge f(x)$*, respectively* $f(y) = f(x)$*). The global notions mean the corresponding pointwise conditions.*

If $s \in H^0(X, M)$ is a section of the constant sheaf, then $x \mapsto x^*s$ is locally constant. In a locally ringed topos, a perfect complex has a locally constant rank. If P is a finite type module, then

$$x \longmapsto \mathrm{rg}_{k(x)}(P_x \otimes_{\mathcal{O}_{X,x}} k(x))$$

is upper semi-continuous, by Nakayama's lemma.

Lemma (5.5)**.** *Let $(X, \mathcal{O}_X)$ be a locally ringed topos, let $E \in \mathrm{D}^-(X)$, and let $p \in \mathbb{Z}$.*

(i) If E is perfect, then the vector spaces $H^i(E_x \, \mathbf{L}\!\otimes_{\mathcal{O}_{X,x}} k(x))$ are finite-dimensional and zero except for finitely many i, and

$$x \mapsto \sum_i (-1)^i \, \mathrm{rg}_{k(x)} \, H^i(E_x \, \mathbf{L}\!\otimes k(x))$$

is locally constant.

(ii) If E is $(p-1)$-pseudo-coherent, then $H^p(E_x \, \mathbf{L}\!\otimes k(x))$ is finite-dimensional and its dimension is upper semi-continuous. If moreover $M \mapsto H^p(E \, \mathbf{L}\!\otimes M)$ is exact, then $H^p(E)$ is locally free of finite type and the same function is locally constant.

(iii) If E is p-pseudo-coherent, respectively $(p-1)$-pseudo-coherent, and if the stalk functor $M \mapsto H^p(E_x \, \mathbf{L}\!\otimes_{\mathcal{O}_{X,x}} M)$ is right exact, respectively left exact, then this exactness holds on a suitable x-pointed neighbourhood.

Proof. For (i), use the compatibility of the rank of a perfect complex with inverse images. For (ii), after truncating and shifting, reduce to a strictly perfect complex concentrated in degrees $[-1, 1]$. The Euler characteristic identity reduces upper semi-continuity of H^0 to that of the neighbouring cohomology sheaves and to local constancy of rank. Exactness gives flatness, hence finite local freeness, of the relevant cokernels, and the exact sequence

$$0 \to H^p(E) \to Z'^p(E) \to E^{p+1} \to Z'^{p+1}(E) \to 0$$

then shows that $H^p(E)$ is locally free of finite type. Part (iii) is the same openness argument applied to the flatness of Z'^p or Z'^{p+1} at the chosen point.

Corollary (5.6)**.** *Let X be a scheme, $p \in \mathbb{Z}$, and $E \in \mathrm{D}^-(X)$ a $(p-1)$-pseudo-coherent complex. Then:*

(i) $H^p(E \, \mathbf{L}\!\otimes_X k(x))$ is finite-dimensional over $k(x)$ and

$$x \mapsto \mathrm{rg}_{k(x)} \, H^p(E \, \mathbf{L}\!\otimes_X k(x))$$

is constructible and upper semi-continuous on X.

(ii) If X is quasi-separated and E has locally bounded quasi-coherent cohomology, exactness of $M \mapsto H^p(E \, \mathbf{L}\!\otimes M)$ on quasi-coherent modules implies that the preceding function is locally constant and that $H^p(E)$ is locally free of finite type. Conversely, if X is locally noetherian and reduced and the function is locally constant, the functor is exact on quasi-coherent modules.

Proof. On each irreducible component the generic point satisfies the exactness condition, hence exactness holds on a non-empty open neighbourhood and the rank is locally constant there. This gives constructibility; the remaining assertions follow from Lemma 5.5 and from the classical EGA III criterion after localization and truncation.

Proposition (5.7)**.** *Let $f : X \to Y$ be a proper morphism of quasi-compact schemes, let $E \in \mathrm{D}^b(f)_{\mathrm{parf}}$, and let $p \in \mathbb{Z}$. Assuming the finiteness theorem applies to f – for instance if f is projective or Y is noetherian – one has:*

(i) the $H^i(Rf_*E \, \mathbf{L}{\otimes}_Y \, k(y))$ are finite-dimensional over $k(y)$ and zero except for finitely many i, and their alternating rank is locally constant on Y;

(ii) $y \mapsto \mathrm{rg}_{k(y)} \, H^p(Rf_*E \, \mathbf{L}{\otimes}_Y \, k(y))$ is constructible and upper semi-continuous;

(iii) if Y is quasi-separated and $M \mapsto R^p f_*(E \, \mathbf{L}{\otimes}_Y \, M)$ is exact on quasi-coherent modules, then $R^p f_*(E)$ is locally free of finite type and the function in (ii) is locally constant. Conversely, if Y is noetherian and reduced and the function in (ii) is locally constant, the same functor is exact.

Proof. By the finiteness theorem, Rf_*E is perfect. The first two assertions are therefore 5.5 (i) and 5.6 (i), and the third follows from 5.6 (ii) together with the projection formula

$$Rf_*E \, \mathbf{L}{\otimes}_Y \, M \xrightarrow{\sim} Rf_*(E \, \mathbf{L}{\otimes}_Y \, M).$$

Remark (5.8)**.** *For a fibre inclusion $i_y : X_y \to X$, under the usual flatness or flat-complex hypotheses, Kunneth's formula gives*

$$H^i(Rf_*E \, \mathbf{L}{\otimes}_Y \, k(y)) \xrightarrow{\sim} H^i(X_y, E_y),$$

where $E_y = \mathrm{L}i_y^*E$ or i_y^*E in the flat case. An analogous analytic theory exists for complex analytic spaces; its essential inputs are Grauert's finiteness theorem for proper maps and the analytic projection formula.

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Grothendieck Groups of Ringed Topoi

L. Illusie

1. Reminders and generalities on Grothendieck groups

1.1

Let C be a triangulated category. An application f from $\operatorname{ob} C$ to an abelian group G is called additive if

$$f(E) = f(E') + f(E'')$$

for every distinguished triangle $E' \to E \to E'' \to E'[1]$. Additive maps form an abelian group functorial in G, represented by an abelian group $k(C)$ and a universal additive map

$$\mathrm{cl}_C : \operatorname{ob} C \to k(C).$$

The group $k(C)$ is functorial in exact functors, and equivalent exact functors induce the same homomorphism.

Lemma (1.2)**.** *For $L \in C$ and $n \in \mathbb{Z}$, one has*

$$\mathrm{cl}(L[n]) = (-1)^n \,\mathrm{cl}(L).$$

If $L \simeq L' \oplus L''$, then $\mathrm{cl}(L) = \mathrm{cl}(L') + \mathrm{cl}(L'')$. If for some n the coproduct $\coprod_{i \geq 0} L[2ni]$ is representable, then $\mathrm{cl}(L) = 0$.

Proof. The first assertions follow immediately from the definitions. For the last, use

$$\coprod_{i\geq 0} L[2ni] \simeq L \oplus \big(\coprod_{i \geq 0} L[2ni]\big)[2n].$$

Lemma (1.3)**.** *If A is a thick subcategory of a triangulated category C, then the inclusion and quotient functors give an exact sequence*

$$k(A) \longrightarrow k(C) \longrightarrow k(C/A) \longrightarrow 0.$$

Lemma (1.4)**.** *Let A be an additive, respectively abelian, category, and let f be additive on* $\operatorname{ob} \mathrm{K}^b(A)$*, respectively* $\operatorname{ob} \mathrm{D}^b(A)$*. Then*

$$f(E) = \sum_i (-1)^i f(E^i), \qquad \textit{respectively} \qquad f(E) = \sum_i (-1)^i f(H^i(E)).$$

1.5

Let C be abelian and let A be a full subcategory stable under finite sums. A map $f : \operatorname{ob} A \to G$ is additive if

$$\sum_i (-1)^i f(E^i) = 0$$

for every bounded acyclic complex E of C whose components lie in A. When A is stable under kernels of epimorphisms this is equivalent to additivity on short exact sequences with terms in A. If $\mathrm{K}^{b,\emptyset}(A)$ denotes the thick subcategory of acyclic bounded complexes, then

$$\mathrm{cl}_A : \mathrm{ob}\, A \to k(\mathrm{K}^b(A)/\mathrm{K}^{b,\emptyset}(A))$$

is universal for such additive maps. We write this group as $k(A)$, or $k(A, C)$ if the ambient abelian category must be specified. In particular $k(\mathrm{D}^b(C)) = k(C)$.

Lemma (1.6)**.** *Let C be abelian and let A be a thick abelian subcategory. Let $\mathrm{D}_A(C)$ be the full subcategory of $\mathrm{D}(C)$ consisting of complexes whose cohomology objects lie in A. Then $\mathrm{D}_A(C)$ is triangulated, and the canonical functor $\mathrm{D}^b(A) \to \mathrm{D}^b_A(C)$ induces an isomorphism*

$$k(A) \xrightarrow{\sim} k(\mathrm{D}^b_A(C)).$$

Proof. Triangulatedness follows from the long exact cohomology sequence. The inverse map sends E to

$$\sum_i (-1)^i \,\mathrm{cl}_A(H^i(E)),$$

which is additive, and the assertion follows from Lemma 1.4.

Definition (1.7)**.** *For a triangulated, respectively abelian, category C, a subset $C' \subset \mathrm{ob}\, C$ is exact if, in every distinguished triangle, respectively short exact sequence, whenever two of the three terms lie in C', so does the third. It is dense if $\mathrm{ob}\, C$ is the smallest exact subset containing C'.*

Lemma (1.8)**.** *If $C' \subset \mathrm{ob}\, C$ is dense, then the classes $\mathrm{cl}_C(E)$ for $E \in C'$ generate $k(C)$.*

Proof. Let $C_0 = C'$ and define C_i inductively by closing under the third term of a triangle or short exact sequence whose other two terms lie in C_{i-1}. The subgroups generated by the classes of the C_i are all equal, and the filtration is exhaustive.

Lemma (1.9)**.** *Let C be abelian and A a full subcategory stable under finite sums, subobjects, and quotients, and suppose $\mathrm{ob}\, A$ is dense in $\mathrm{ob}\, C$. Then the homomorphism $k(A, C) \to k(C)$ defined by $E \mapsto \mathrm{cl}_C(E)$ is an isomorphism.*

2. The functors $K.$ and K' of a ringed topos

2.1

Let (X, A) be a ringed topos. We use the following notation: $\mathrm{D}^b({}_AX)_{\mathrm{coh}}$ is the category of pseudo-coherent complexes with bounded cohomology of left A-modules; $\mathrm{D}({}_AX)_{\mathrm{parf}}$ is the category of complexes of finite perfect amplitude; $\mathrm{Loclib}({}_AX)$ is the category of left A-modules locally direct factors of finite free modules; and $\mathrm{Coh}({}_AX)$ is the category of coherent left A-modules.

Definition (2.2)**.** *Unless otherwise stated,*

$$K.({}_AX) = k(\mathrm{D}^b({}_AX)_{\mathrm{coh}}), \qquad K'({}_AX) = k(\mathrm{D}({}_AX)_{\mathrm{parf}}),$$

and

$$K.({}_AX)_{\text{naive}} = k(\mathrm{Coh}({}_AX)), \qquad K'({}_AX)_{\text{naive}} = k(\mathrm{Loclib}({}_AX)).$$

Thus $K.$ is the Grothendieck group of pseudo-coherent complexes and K' the Grothendieck group of perfect complexes; the naive versions are computed directly from coherent, respectively locally free, modules.

If the final object of X is quasi-compact, the local bounded variants coincide with these groups. If A is coherent as a left A-module, the canonical map

$$K.({}_AX)_{\text{naive}} \longrightarrow K.({}_AX)$$

is an isomorphism, because coherent modules form a thick abelian subcategory and pseudo-coherent bounded complexes have coherent cohomology.

There is a natural homomorphism

$$K'({}_AX) \longrightarrow K.({}_AX)$$

induced by inclusion of perfect complexes among pseudo-coherent bounded complexes. It is not generally injective or surjective. If the final object is quasi-compact, X has enough points, and all stalk modules have finite Tor-dimension, then

$$\mathrm{D}({}_AX)_{\text{parf}} = \mathrm{D}^b({}_AX)_{\text{coh}},$$

so this map is an isomorphism; for example this holds for compact spaces ringed by regular local rings.

2.6. Pairings

Let

$$f : (X', A') \longrightarrow (X, A)$$

be a morphism of ringed topoi. For bimodules left over A' and right over $f^{-1}A$, tensor product induces

$$\mathrm{Loclib}_{A'}({}_{A'}X'_A) \times \mathrm{Loclib}({}_AX) o \, \mathrm{Loclib}({}_{A'}X')$$

and hence

$$K'_{\text{naive}}({}_{A'}X'_A)_{A'} \otimes K'_{\text{naive}}({}_AX) o K'_{\text{naive}}({}_{A'}X').$$

Derived tensor product similarly gives

$$\mathbf{L}\!\otimes_A : \mathrm{D}({}_{A'}X'_A)_{A'-\text{parf}} \times \mathrm{D}({}_AX)_{\text{parf}} o \mathrm{D}({}_{A'}X')_{\text{parf}}$$

and therefore

$$K'({}_{A'}X'_A)_{A'} \otimes K'({}_AX) o K'({}_{A'}X').$$

The naive and derived pairings are compatible with the canonical maps from naive to derived groups.

2.7. Contravariance and ring structure

For a morphism of ringed topoi $f : (X', A') \to (X, A)$, the functor $\mathrm{L}f^*$ sends perfect complexes to perfect complexes and gives

$$f^* : K'({}_AX) o K'({}_{A'}X')$$

with an analogous naive map. These maps satisfy $(gf)^* = f^*g^*$.

If $(X, \mathcal{O}_X)$ is commutatively ringed, the tensor product makes $K'(X)$ and $K'(X)_{\text{naive}}$ into commutative unital rings, with unit $\text{cl}(\mathcal{O}_X)$, and the canonical naive-to-derived map is a unital ring homomorphism. Pullback maps are ring homomorphisms.

2.8. Augmentation by rank

If $(X, \mathcal{O}_X)$ is locally ringed, rank defines an augmentation

$$\text{rg} : K'(X) \to H^0(X, \mathbb{Z}),$$

compatible with the section $i : H^0(X, \mathbb{Z}) \to K'(X)_{\text{naive}}$ defined by locally free modules of prescribed rank. These structures are preserved by inverse image. Later lambda-ring structures refine this augmented algebra structure.

2.9. Comparison of K' and the naive group

The canonical map

$$K'(X)_{\text{naive}} \to K'(X)$$

need not be an isomorphism in general. The global resolution criteria of Expose II show that it is an isomorphism in particular when:

(i) X is the Zariski topos of a separated or noetherian divisorial scheme;

(ii) X is the topos of a compact space and the sheaf of rings is soft;

(iii) X is a Stein topos with quasi-compact final object and $A = \mathcal{O}_X$.

2.10. The pairing between K' and $K.$

For a commutatively ringed topos,

$$\mathbf{L}\otimes : \mathrm{D}(X)_{\text{parf}} \times \mathrm{D}^b(X)_{\text{coh}} o \mathrm{D}^b(X)_{\text{coh}}$$

defines

$$K'(X) \otimes K.(X) \to K.(X),$$

so that $K.(X)$ is a $K'(X)$-module. If $f : X \to Y$ is a morphism, then $K.(X)$ is also a $K'(Y)$-module via pullback.

2.11. Covariance on $K.$ and the projection formula

Let $f : X \to Y$ be proper and pseudo-coherent between quasi-compact schemes. Assuming the finiteness conjecture for f – for example if Y is noetherian or f is projective – the functor Rf_* induces

$$f_* : K.(X) \to K.(Y).$$

It is $K'(Y)$-linear: for $y \in K'(Y)$ and $x \in K.(X)$,

$$f_*(yx) = yf_*(x),$$

which follows from the derived projection formula

$$Rf_*(E)\, \mathbf{L}\!\otimes\, M \xrightarrow{\sim} Rf_*(E\, \mathbf{L}\!\otimes_Y\, M).$$

For a composable proper morphism $g : Y \to Z$ satisfying the same hypotheses, $g_* f_* = (gf)_*$.

The same statements hold for proper morphisms of finite-dimensional analytic spaces by Grauert's finiteness theorem and the analytic projection formula.

2.12. Pullback on $K.$ and pushforward on K'

If $f : X \to Y$ has finite Tor-dimension, then pullback defines

$$f^* : K.(Y) o K.(X), \qquad (gf)^* = f^* g^*.$$

If, in addition, one is in the proper finite-type situation above, then Rf_* sends perfect complexes to perfect complexes and gives

$$f_* : K'(X) o K'(Y).$$

This is not usually a ring homomorphism, but it satisfies the projection formula

$$f_*(f^*(y)x) = yf_*(x)$$

for $y \in K.(Y)$, respectively $K'(Y)$, and $x \in K'(X)$. For separated divisorial schemes this construction yields the previously non-obvious homomorphism

$$f_* : K'(X)_{\text{naive}} o K'(Y)_{\text{naive}},$$

which is one of the main motivations for the finiteness and resolution formalism of the preceding exposes.

3. Complements on Grothendieck groups of schemes

Proposition (3.1.0)**.** *Let*

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{h} & Y' \end{array}$$

be a cartesian square of schemes, with Y quasi-compact and f quasi-compact and quasi-separated. Let $E \in \mathrm{D}^b(X)$ have quasi-coherent cohomology. Suppose X and Y' are Tor-independent over Y, and that either h has finite Tor-dimension or E has finite flat amplitude relative to f. Then the base-change morphism

$$\mathrm{L}h^* Rf_* E \longrightarrow Rf'_* \mathrm{L}g^* E$$

is defined and is an isomorphism.

Proof. The map is defined using the trivial duality formula for h,

$$\mathrm{Hom}(\mathrm{L}h^*L, M) \xrightarrow{\sim} \mathrm{Hom}(L, Rh_*M).$$

It is enough to define $Rf_*E \to Rf_*Rg_*\mathrm{L}g^*E$, and this is the image under Rf_* of the adjunction map $E \to Rg_*\mathrm{L}g^*E$. To prove that the map is an isomorphism, localize first on Y and Y', then use the Leray spectral sequence for an affine open cover of X to reduce to the affine case. The assertion then becomes the ring identity

$$B \,\mathbf{L}\!\otimes_A A' \xrightarrow{\sim} B'$$

together with Tor-independence, and hence

$$E \,\mathbf{L}\!\otimes_A A' \xrightarrow{\sim} E \,\mathbf{L}\!\otimes_B B'.$$

Proposition (3.1.1)**.** *In the cartesian square above, assume Y and Y' quasi-compact, X and Y' Tor-independent over Y, f proper and pseudo-coherent, and the finiteness theorem valid for f and f'. If f is perfect, respectively if h has finite Tor-dimension, then for $x \in K'(X)$, respectively $x \in K.(X)$,*

$$h^*f_*(x) = f'_*g^*(x).$$

3.2. Passage to the limit

Let $(X_i, f_{ij})_{i\in I}$ be a filtered projective system of quasi-compact and quasi-separated schemes with affine transition morphisms. Put $X = \varprojlim X_i$ and let $f_i : X \to X_i$ be the canonical morphisms.

Proposition (3.2.1)**.** *The pullback maps induce an isomorphism*

$$\varinjlim K'(X_i)_{\mathrm{naive}} \xrightarrow{\sim} K'(X)_{\mathrm{naive}}.$$

Indeed, a locally free module of finite type on X descends to some X_i, and its class in the filtered colimit is independent of the descent after increasing i.

Proposition (3.2.2)**.** *The canonical map*

$$\varinjlim K'(X_i) \longrightarrow K'(X)$$

is an isomorphism.

This follows from Deligne's localization technique, which identifies the category of perfect complexes on X with the filtered colimit of the corresponding categories on the X_i.

Proposition (3.2.3)**.** *If all X_i and X are noetherian and the transition morphisms are flat, then*

$$\varinjlim K.(X_i) \xrightarrow{\sim} K.(X).$$

The module structures over K'_{naive} are compatible with these limit isomorphisms.

3.3. Relative Grothendieck groups

Let S' be an S-scheme, let X' be an S'-scheme, and let $h : X' \to X$ be an S-morphism. We assume that inverse image on X' of a vector bundle on X is again a vector bundle on X', so that there is a canonical homomorphism

$$h^* : K^\bullet(X) \to K^\bullet(X').$$

Suppose first that $h : X' \to X$ is a closed immersion. One considers the usual exact localization diagrams and defines

$$K^\bullet_{S'/S}(X) = \ker\big(K^\bullet_{S'}(X') \to K^\bullet(X' \times_S (S - S'))\big).$$

Equivalently, this is the kernel of the natural map from $K^\bullet_{S'}(X')$ to $K^\bullet(X')$ induced by the inclusion. Pullback by h then gives a homomorphism

$$K^\bullet_S(X) \longrightarrow K^\bullet_{S'/S}(X)$$

and a commutative diagram with exact rows

$$\begin{array}{ccccccccc}
0 & \to & K^\bullet_S(X) & \to & K^\bullet(X) & \to & K^\bullet(X \times_S (S - S')) & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & K^\bullet_{S'/S}(X) & \to & K^\bullet_{S'}(X') & \to & K^\bullet(X' \times_S (S - S')) & \to & 0.
\end{array}$$

If S' is empty this recovers $K^\bullet_S(X)$, while for $X = S$ and $X' = S'$ one gets $K^\bullet_{S'/S}(S) = 0$.

Closed immersions $S'' \to S' \to S$ give transition maps

$$K^\bullet_{S'/S}(X) \to K^\bullet_{S''/S}(X),$$

so that closed immersions over S define a functor to graded abelian groups. If S' is a closed subscheme of S, $X' = X \times_S S'$, and X is regular, then

$$K^\bullet_{S'/S}(X) \longrightarrow K^\bullet_S(X)$$

is an isomorphism. The inverse is obtained from the commutative square relating $K^\bullet_S(X)$, $K^\bullet_{S'/S}(X)$, $K^\bullet(X)$, and $K^\bullet_{S'}(X')$.

For a regular scheme X and a closed subscheme $S \subset X$ we write

$$K^\bullet_S(X) = K^\bullet(X \text{ on } S).$$

If $S' \subset S$ is closed, there is a commutative diagram of exact sequences

$$\begin{array}{ccccccccc}
0 & \to & K^\bullet_{S'}(X) & \to & K^\bullet_S(X) & \to & K^\bullet_{S-S'}(X - S') & \to & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \to & K^\bullet(S') & \to & K^\bullet(S) & \to & K^\bullet(S - S') & \to & 0.
\end{array}$$

The support of an element of $K^\bullet(X)$ is the smallest closed subset Z such that the element comes from $K^\bullet_Z(X)$; the group $K^\bullet(X)$ is the union of the subgroups $K^\bullet_Z(X)$ as Z runs through closed subsets of X.

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SGA 6 – Expose V

Generalities on λ-rings

P. Berthelot

This paper contains reminders and clarifications on Grothendieck's notion of a λ-ring. It also corrects and supplements the exposition in the cited source, adding material needed later in the seminar.

1. Universal polynomials

Let $\mathcal{C}_0$ be the category of commutative rings with unit. We shall use polynomial laws with integral coefficients in finite or infinite families of indeterminates. For indeterminates $X_1, \ldots, X_n$ put

$$\sigma_i^{(n)}(X_1, \ldots, X_n) = \sum_{1 \le k_1 < \cdots < k_i \le n} X_{k_1} \cdots X_{k_i},$$

the elementary symmetric functions.

For two families X and Y, the universal polynomials P_n are defined by

$$\prod_{i,j}(1 + X_i Y_j T) = \sum_{n \ge 0} P_n(\sigma_1(X), \ldots, \sigma_n(X); \sigma_1(Y), \ldots, \sigma_n(Y)) T^n.$$

Each P_n is isobaric of weight n with respect to both families of elementary symmetric functions.

If $Y_i = X_i^s$, then

$$\sigma_i(Y_1, \ldots, Y_n) = Q_{i,s}(\sigma_1(X), \ldots, \sigma_i(X)),$$

where $Q_{i,s}$ is a universal isobaric polynomial of weight is. Similarly, if the variables Z are the products $X_i Y_j$, then

$$\sigma_k(Z) = R_{n,m,k}(\sigma_1(X), \ldots, \sigma_n(X); \sigma_1(Y), \ldots, \sigma_m(Y)),$$

with $R_{n,m,k}$ universal and isobaric of weight k. We shall also use universal polynomials $C_{n,m,k}$ governing elementary symmetric functions of sums of variables.

An augmented-ring functor is a functor F from $\mathcal{C}_0$ to commutative unital rings equipped with a natural augmentation $F \to \mathrm{id}$. We write $F(A)_+$ for the kernel of $F(A) \to A$, so that $F(A) = A \oplus F(A)_+$. The category of F-algebras consists of rings A equipped with an augmentation $F(A) \to A$, with morphisms compatible with these structures.

2. Pre-λ-rings

Let K be a commutative ring with unit. A pre-λ-ring structure on K is a homomorphism of abelian groups

$$\lambda_t : K \longrightarrow 1 + K[[t]]_+$$

where the target is made into a multiplicative group. Traditionally one writes

$$\lambda_t(x) = \sum_{i \geq 0} \lambda^i(x) t^i,$$

with $\lambda^0(x) = 1$ and $\lambda^1(x) = x$.

For a scheme X, the Grothendieck group of vector bundles $K^\bullet(X)$ has such a structure, where λ^i is induced by exterior powers. More generally the Grothendieck group of a suitable exact category has the same structure.

The universal polynomials define on $1 + K[[t]]_+$ a multiplication

$$(1 + \sum a_i t^i) \circ (1 + \sum b_j t^j) = 1 + \sum_{n \geq 1} P_n(a_1, \ldots, a_n; b_1, \ldots, b_n) t^n,$$

with unit $1 + t$, and analogous formulas define the operations λ^i. With this notation

$$\lambda_t(xy) = \lambda_t(x) \circ \lambda_t(y),$$

so that λ_t is multiplicative for this universal product.

One also defines universal polynomials L_n by

$$\lambda_t(1 + \sum x_i t^i) = 1 + \sum_{n \geq 1} L_n(x_1, \ldots, x_n) t^n.$$

Finally the Adams operations are introduced by

$$\psi_t(x) = -t \frac{d}{dt} \log \lambda_{-t}(x) = \sum_{n \geq 1} \psi^n(x) t^n.$$

They are additive endomorphisms satisfying

$$\psi^1(x) = x, \qquad \psi^m(x + y) = \psi^m(x) + \psi^m(y), \qquad \psi^m(xy) = \psi^m(x)\psi^m(y).$$

If x has dimension 1, meaning $\lambda^i(x) = 0$ for $i \geq 2$, then $\psi^m(x) = x^m$.